The Triangular Lattice and Universal Optimality

Human proofs, Lean 4 verification, numerical validation in Python and Rust, and physics from fluids to plasmas

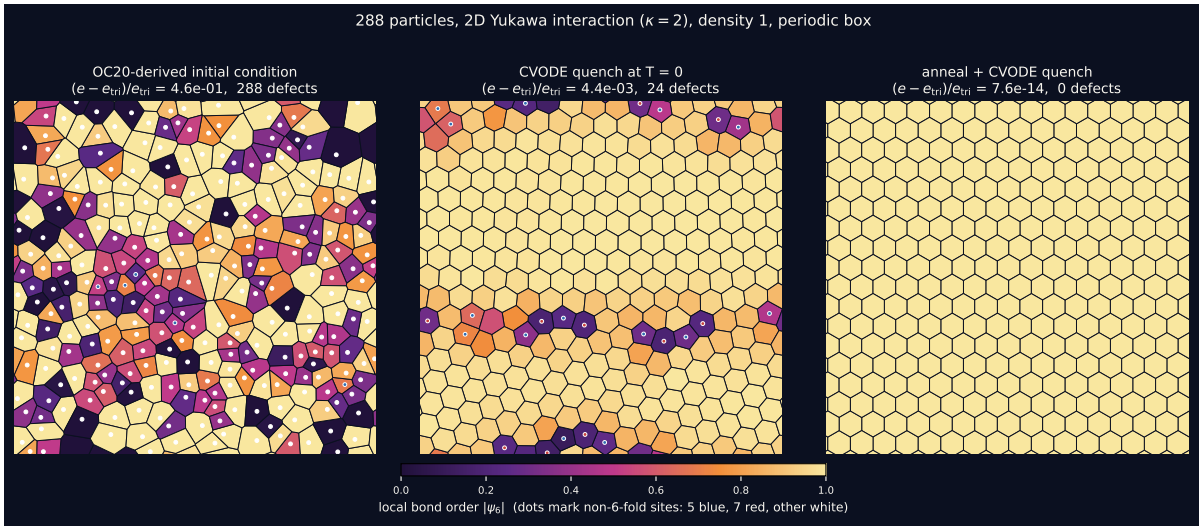
A textbook for students

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written with AI agents (Claude) under the author's direction

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288 particles, 2D Yukawa interaction (κ = 2), density 1, periodic box
OC20-derived initial condition
(e − e_tri)/e_tri = 4.6e-01, 288 defects
CVODE quench at T = 0
(e − e_tri)/e_tri = 4.4e-03, 24 defects
anneal + CVODE quench
(e − e_tri)/e_tri = 7.6e-14, 0 defects
0.0
0.2
0.4
0.6
0.8
1.0
local bond order |ψ6| (dots mark non-6-fold sites: 5 blue, 7 red, other white)

Preface

This book follows one mathematical statement from its definitions to the laboratory. The statement says that the triangular lattice has the least energy among all planar configurations of a given density, for every completely monotone repulsion. The book's line is a single one. A human proof explains *why* a statement holds. A Lean 4 proof checks that the stated proof is correct, relative to the stated definitions. A numerical experiment can only show that nothing contradicts the statement in the cases tried. A physical system shows where the mathematics applies and where it does not.

Every claim carries a tier from the SocrateAI-Scientific-Elenchus method [17]:

- **[Tier A]** kernel-checked Lean;
- **[Tier L]** cited literature;
- **[Tier C]** argument, conjecture or proposal;
- **[Tier X]** floating-point exploration, which may steer a search but never supports a claim.

Read the tier before the sentence.

Three things are *not* done here, and the book says so wherever they come up:

- the strong ("per-potential") Cohn–Kumar conjecture;
- a formal treatment of renormalized energies;
- the Riesz endpoint $s = 2$.

Renormalized energies are proved in upstream's manuscript, §8 (**[Tier L]**), but not formalised anywhere; the strong per-potential conjecture and the endpoint $s = 2$ are open. Chapter 5 explains what each would require.

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Chapter 1

Mathematical foundations

This chapter sets up everything needed to read the main theorem of the book precisely: what a configuration is, what "density one" and "energy" mean, which potentials are allowed, why Gaussians are the key case, and how a single auxiliary function gives a lower bound on the energy of *every* competitor. We give a complete human proof of that bound for periodic configurations, explain how the bound passes from Gaussians to all completely monotone potentials, and introduce just enough Lean 4 to read the formal statements. The chapter ends by reviewing a student draft, the way a referee would.

Caution : whose theorem is this?

The main theorem (Theorem 1.23 below) is *upstream's*: it was produced by an AI system and released by OpenAI [1] on 2026-09-23. It is not peer reviewed. Its Lean proof is kernel-checked and its statement was accepted by the Comparator tool against the challenge statement in our pinned build. That is **[Tier A]** for the *formal statement*, conditional on the fidelity of the Lean definitions to the intended mathematics. No human has audited that fidelity. Our own contribution, presented in Chapter 2, consists of corollaries and an audit.

Throughout, the tiers of the preface apply. In this chapter a **[Tier C]** label on a result that comes with a human proof means that a human-readable argument is given here and has not been machine-checked. It does not mean the result is doubtful.

1.1 Point configurations, density and energy

We work in the Euclidean plane $\mathbb{R}^2$ with the usual norm $|x|$. A *configuration* is a subset $C \subseteq \mathbb{R}^2$. Its points are the positions of identical particles. Because C is a set, two particles never sit at the same place.

Definition 1.1 (Local finiteness)**.** A configuration C is *locally finite* if $C \cap \overline{B}(0, R)$ is finite for every $R \in \mathbb{R}$, where $\overline{B}(0, R) = \{x : |x| \leq R\}$ is the closed disk of radius R about the origin.

For a locally finite C we write $C_R = C \cap \overline{B}(0, R)$ for the finite set of points in the closed disk and $n_C(R) = \#C_R$ for their number.

Definition 1.2 (Centred density)**.** A locally finite configuration C has *density one* if

$$\lim_{R \to \infty} \frac{n_C(R)}{\pi R^2} = 1.$$

More generally it has density ρ if the limit equals ρ.

The density is *centred*: it is measured in disks about the origin only, and the disks are *closed*. Both choices matter in principle. Centred density says nothing *uniform*: a configuration of centred density one can contain, far from the origin, unit disks holding k^2 points each (one such cluster near radius 2^k for each k), because these add only $O((\log R)^3)$ points to the disk of radius R **[Tier C]**. (For a *fixed* other centre c, the disks $\overline{B}(c,R)$ give the same limit, since $\overline{B}(0,R-|c|) \subseteq \overline{B}(c,R) \subseteq \overline{B}(0,R+|c|)$.) And a point lying exactly on the circle $|x| = R$ is counted in the closed disk but not in the open one. For locally finite configurations, the open-disk and closed-disk density and lower energy coincide **[Tier C]** (human proof in the preregistration of D4). For the formal status of this fact: **[Tier A]** kernel-checked for density one and for the lower energy, independently re-verified; other densities **[Tier C]** (details in the Editor's note).

Lean 4 : upstream's definitions of local finiteness, disk points, count and density

Quoted verbatim from `lean/ComparatorChallenges/TriangularEnergy.lean` (openai/-math `adc7f124`); the empty-set symbol is typeset through an escape because the book's listing setup has no glyph for it.

```
abbrev Plane := EuclideanSpace ℝ (Fin 2)

def LocallyFinite (C : Set Plane) : Prop :=
  ∀ R : ℝ, (C ∩ Metric.closedBall (0 : Plane) R).Finite

def diskPoints (C : Set Plane) (R : ℝ) : Finset Plane :=
  if h : (C ∩ Metric.closedBall (0 : Plane) R).Finite then h.toFinset else ∅

def diskCount (C : Set Plane) (R : ℝ) : ℕ := (diskPoints C R).card

def DensityOne (C : Set Plane) : Prop :=
  Tendsto (fun R : ℝ => (diskCount C R : ℝ) / (Real.pi * R ^ 2))
    atTop (𝓝 1)
```

Three details of the Lean text deserve attention, because a reader of the human definition could miss them.

- `diskPoints` returns the empty set when C_R is infinite. The definition is therefore total, and every theorem that needs the count to mean something must assume `LocallyFinite C`.
- `Tendsto ... atTop` $(\mathcal{N}\,1)$ is Lean's way of writing $\lim_{R\to\infty} = 1$, with R a *real* parameter. It is not a limit along integers.
- At $R = 0$ the denominator πR^2 is zero, and Lean's convention $x/0 = 0$ makes the quotient 0. This is harmless because only large R matter.

1.1.1 Pair potentials and the lower energy

A *pair potential* is written as a function g of the *squared* distance $t = |x-y|^2$. This convention, used by CKMRV [2] and by upstream [1], makes Gaussians $e^{-\alpha t}$ the basic building blocks. The Riesz potential $|x-y|^{-s}$ becomes $g(t) = t^{-s/2}$.

Definition 1.3 (Disk energy and lower energy)**.** Let $g : (0,\infty) \to [0,\infty)$ and let C be locally finite. The *disk energy* at radius R is

$$E_g(C;R) = \frac{1}{n_C(R)} \sum_{x \in C_R} \sum_{\substack{y \in C_R \\ y \neq x}} g\big(|x-y|^2\big) \;\in [0,\infty],$$

with the convention $1/0 = \infty$ in $[0,\infty]$ (and $\infty \cdot 0 = 0$). The *lower energy* of C is

$$E_g(C) = \liminf_{R\to\infty} E_g(C;R) \ \in [0,\infty].$$

Each unordered pair $\{x,y\}$ is counted twice, as (x,y) and (y,x). This is the "ordered-pair" normalisation: for a lattice it gives $\sum_{a\neq 0} g(|a|^2)$, the energy seen by one particle, without a factor $\frac{1}{2}$.

Lean 4 : upstream's disk energy and energy

Quoted verbatim from `TriangularEnergy.lean`.

```
def diskEnergy (g : ℝ → ℝ) (C : Set Plane) (R : ℝ) : ℝ≥0∞ :=
  (diskCount C R : ℝ≥0∞)⁻¹ *
    ∑ x ∈ diskPoints C R, ∑ y ∈ (diskPoints C R).erase x,
      ENNReal.ofReal (g (‖x - y‖ ^ 2))

def energy (g : ℝ → ℝ) (C : Set Plane) : ℝ≥0∞ :=
  Filter.liminf (diskEnergy g C) atTop
```

The values live in $\mathbb{R}_{\geq 0}\cup\{\infty\}$ (Lean: `ENNReal`), the extended non-negative reals $[0,\infty]$. Then every energy is defined, possibly infinite, and no summability hypothesis is needed to *state* the theorem. The function `ENNReal.ofReal` sends negative reals to 0, which is irrelevant for the non-negative potentials used here.

Caution : lower energy, not energy

$E_g(C)$ is a lim inf. CKMRV's Definition 1.3 [2] of universal optimality also requires the limit to *exist* for the optimal configuration. The Lean notion used in this book uses only the lower energy, and is therefore weaker than CKMRV's **[Tier C]**. When we say "energy" below we always mean the lower energy unless the limit is shown to exist.

Example 1.4 (Why the liminf)**.** The lim inf is the smallest value that the disk energies approach infinitely often. A competitor whose disk energies oscillate, for example because of rings of extra points at sparse radii, is judged by its *best* radii. A lower bound $\liminf \geq L$ is therefore a strong kind of lower bound: for every $\varepsilon > 0$, all sufficiently large radii have disk energy at least $L-\varepsilon$. **[Tier C]**

1.2 Lattices and the triangular lattice

A *lattice* in $\mathbb{R}^2$ is a set $\Lambda = \{jv_1 + kv_2 : (j,k)\in\mathbb{Z}^2\}$ for linearly independent v_1, v_2. Its *covolume* $|\Lambda| = |\det(v_1,v_2)|$ is the area of a fundamental parallelogram. A lattice has density $1/|\Lambda|$, so a density-one lattice has covolume one.

Definition 1.5 (The triangular lattice A)**.** Let $b = \sqrt{3}/2$ and

$$p(j,k) = \frac{1}{\sqrt{b}}\Big(j + \frac{k}{2},\ k\,b\Big), \qquad (j,k)\in\mathbb{Z}^2, \qquad A = \{p(j,k) : (j,k)\in\mathbb{Z}^2\}.$$

Lean 4 : upstream's lattice

Quoted verbatim from `TriangularEnergy.lean`.

```
def b : ℝ := Real.sqrt 3 / 2
```

```
def triangularPoint (jk : ℤ × ℤ) : Plane :=
  (Real.sqrt b)⁻¹ •
    (EuclideanSpace.single 0 ((jk.1 : ℝ) + (jk.2 : ℝ) / 2) +
      EuclideanSpace.single 1 ((jk.2 : ℝ) * b))

def A : Set Plane := Set.range triangularPoint

/-- The full nonzero-lattice energy, allowing a divergent series. -/
def latticeEnergy (g : ℝ → ℝ) : ℝ≥0∞ :=
  ∑' a : {x : Plane // x ∈ A ∧ x ≠ 0}, ENNReal.ofReal (g (‖a.val‖ ^ 2))
```

The basis vectors are $v_1 = p(1,0) = b^{-1/2}(1,0)$ and $v_2 = p(0,1) = b^{-1/2}(\frac{1}{2}, b)$.

Proposition 1.6 (Covolume and Gram form)**.** *$|\det(v_1, v_2)| = 1$, so A has covolume one. The Gram matrix is*

$$G = \begin{pmatrix} v_1 \cdot v_1 & v_1 \cdot v_2 \\ v_2 \cdot v_1 & v_2 \cdot v_2 \end{pmatrix} = \frac{1}{b}\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix},$$

and consequently

$$|p(j,k)|^2 = \frac{1}{b}\,(j^2 + jk + k^2) = \frac{2}{\sqrt{3}}\,(j^2 + jk + k^2). \tag{1.1}$$

Human proof. $\det(v_1, v_2) = b^{-1} \det \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & b \end{pmatrix} = b^{-1} \cdot b = 1$. Next, $v_1 \cdot v_1 = b^{-1}$, $v_1 \cdot v_2 = b^{-1} \cdot \frac{1}{2}$ and $v_2 \cdot v_2 = b^{-1}(\frac{1}{4} + b^2) = b^{-1}(\frac{1}{4} + \frac{3}{4}) = b^{-1}$. Finally $|jv_1 + kv_2|^2 = (j,k)\,G\,(j,k)^{\mathsf{T}} = b^{-1}(j^2 + jk + k^2)$. **[Tier C]** for the covolume and the Gram matrix (human proof). The norm identity (1.1) is also kernel-checked **[Tier A]**. □

Lean 4 : the norm identity (our file `formal_cert/Triangular/RieszFinite.lean`)

```
/-- The squared norm of a lattice point. -/
lemma norm_sq_triangularPoint (j k : ℤ) :
    ‖triangularPoint (j, k)‖ ^ 2 = (1 / b) * ((j : ℝ) ^ 2 + (j : ℝ) * k + (k : ℝ) ^ 2)
    := by
```

All six nearest neighbours of the origin are at squared distance $1/b = 2/\sqrt{3}$. The minimal distance of the density-one triangular lattice is therefore $(2/\sqrt{3})^{1/2} \approx 1.0746$ **[Tier C]**. In the physics chapters the same lattice appears with spacing $a = \sqrt{2/\sqrt{3}}$.

1.2.1 The dual lattice

Definition 1.7 (Dual lattice)**.** The dual of a lattice Λ is $\Lambda^* = \{w \in \mathbb{R}^2 : w \cdot \lambda \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$. Its Gram matrix in the dual basis is G^{-1}, and $|\Lambda^*| = 1/|\Lambda|$.

Proposition 1.8 (A^* is a rotated copy of A)**.** *Let J be the rotation by $90°$. Then $A^* = JA$. In particular A^* is isometric to A and has covolume one.*

Human proof. $Jv_1 = b^{-1/2}(0,1)$ and $Jv_2 = b^{-1/2}(-b, \frac{1}{2})$. Compute $Jv_1 \cdot v_1 = 0$, $Jv_1 \cdot v_2 = b^{-1}b = 1$, $Jv_2 \cdot v_1 = -b^{-1}b = -1$ and $Jv_2 \cdot v_2 = b^{-1}(-\frac{b}{2} + \frac{b}{2}) = 0$. All four are integers, so $JA \subseteq A^*$. Both lattices have covolume 1: J preserves area, and $|A^*| = 1/|A| = 1$. A sublattice of the same covolume is the whole lattice, so $JA = A^*$. As a cross-check, $G^{-1} = b^{-1}\begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$ (note $\det G = b^{-2} \cdot \frac{3}{4} = 1$), whose quadratic form $b^{-1}(j^2 - jk + k^2)$ becomes $b^{-1}(j^2 + jk + k^2)$ under $k \mapsto -k$. **[Tier C]** □

1.2.2 Lattice energy as an Epstein-type sum

For a lattice the natural energy is the energy seen from one lattice point:

$$\mathcal{E}_g(A) = \sum_{a \in A \setminus \{0\}} g\big(|a|^2\big) = \sum_{(j,k) \neq (0,0)} g\Big(\tfrac{2}{\sqrt{3}}(j^2 + jk + k^2)\Big) \in [0, \infty],$$

which is upstream's `latticeEnergy`. Group the terms by the value $n = j^2 + jk + k^2$, and let $r(n)$ be the number of (j,k) with that value. Then $\mathcal{E}_g(A) = \sum_{n \geq 1} r(n)\, g(2n/\sqrt{3})$. For the Riesz potential $g(t) = t^{-s/2}$ this is

$$\mathcal{E}_{t^{-s/2}}(A) = b^{s/2} \sum_{(j,k) \neq 0} (j^2 + jk + k^2)^{-s/2},$$

an Epstein zeta function of the form $j^2 + jk + k^2$ evaluated at $s/2$. It is finite exactly when $s > 2$. Finiteness for $s > 2$ is kernel-checked as `latticeEnergy_riesz_lt_top` **[Tier A]**. Divergence for $s \leq 2$ follows by comparing the sum with $\int_{|x| \geq 1} |x|^{-s}\, dx$ **[Tier C]**.

Example 1.9 (The first shells)**.** The smallest values of $j^2 + jk + k^2$ and their multiplicities can be checked by hand **[Tier C]**.

- $n = 1$: $(\pm 1, 0)$, $(0, \pm 1)$, $\pm(1, -1)$, so $r(1) = 6$.
- $n = 3$: $\pm(1, 1)$, $\pm(2, -1)$, $\pm(1, -2)$, so $r(3) = 6$.
- $n = 4$: twice the $n = 1$ vectors, so $r(4) = 6$.
- $n = 7$: the twelve vectors $\pm(2, 1)$, $\pm(1, 2)$, $\pm(3, -1)$, $\pm(3, -2)$, $\pm(1, -3)$, $\pm(2, -3)$, so $r(7) = 12$.
- $n = 2, 5, 6$ do not occur.

The script below cross-checks these counts **[Tier X]**. Running it printed the line shown.

```
from collections import Counter

M = 20
count = Counter(j * j + j * k + k * k
                for j in range(-M, M + 1) for k in range(-M, M + 1))
print([(n, count[n]) for n in range(1, 14) if count[n] > 0])
```

```
[(1, 6), (3, 6), (4, 6), (7, 12), (9, 6), (12, 6), (13, 12)]
```

For the Gaussian $g(t) = e^{-\pi t}$ (that is, $\alpha = 1$ in the normalisation of §1.5), the shell contributions $r(n)e^{-\pi n/b}$ decay very fast. The following script was run and printed the output shown **[Tier X]**.

```
import math

b = math.sqrt(3) / 2
shells = [(1, 6), (3, 6), (4, 6), (7, 12)]
for n, m in shells:
    print(n, m, m * math.exp(-math.pi * n / b))
print(sum(m * math.exp(-math.pi * n / b) for n, m in shells))
```

```
1 6 0.15947960085851695
3 6 0.0001126711998980545 8
4 6 2.9947929979986557e-06
7 12 1.1247564017693356e-10
0.15959526696388865
```

So $\sum_{a \in A \setminus 0} e^{-\pi |a|^2} \approx 0.1595953$, and the first shell alone carries more than 99.9% of it **[Tier X]**. Compare this with the full truncated theta sum in Exercise 1.13.

1.3 Completely monotone potentials

Definition 1.10 (Completely monotone; admissible potential)**.** A function $g : (0, \infty) \to \mathbb{R}$ is *completely monotone* if it is C^∞ on $(0, \infty)$ and

$$(-1)^r g^{(r)}(t) \;\geq\; 0 \qquad \text{for all } r = 0, 1, 2, \ldots \text{ and all } t > 0.$$

Upstream calls g *admissible* if it is C^∞ on $(0, \infty)$, non-negative there, and completely monotone. The non-negativity is the case $r = 0$.

Lean 4 : upstream's admissible potentials

Quoted verbatim from `TriangularEnergy.lean`.

```
def AdmissiblePotential (g : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧
  (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
  ∀ (r : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r g t
```

Note three things.

1. The inequalities are *weak*: $g \equiv 1$ is admissible, and so is $g \equiv 0$. A definition with strictly alternating signs would be a different (smaller) class.
2. The smoothness is C^∞ (`ContDiffOn` with smoothness index $\top$). It is *not* real-analyticity. Analyticity is a consequence, through Bernstein's theorem below, but it is not part of the definition.
3. Only $t > 0$ matters. The value of g at 0 (or at negative t) never enters the energy, because distinct points have $|x - y|^2 > 0$.

Example 1.11 (Gaussians)**.** For $\alpha \geq 0$, $g(t) = e^{-\alpha t}$ satisfies $(-1)^r g^{(r)}(t) = \alpha^r e^{-\alpha t} \geq 0$. So it is admissible **[Tier C]**. In space, $g(|x|^2) = e^{-\alpha|x|^2}$ is a Gaussian. The case $\alpha = 0$ is the constant potential.

Example 1.12 (Riesz potentials)**.** For $s > 0$, $g(t) = t^{-s/2}$ has $(-1)^r g^{(r)}(t) = \frac{s}{2}(\frac{s}{2} + 1) \cdots (\frac{s}{2} + r - 1)\, t^{-s/2-r} > 0$. It is admissible, and this is kernel-checked as `TriangularRiesz.riesz_admissible` **[Tier A]**. The Lean proof rewrites the r-th derivative with `Real.iter_deriv_rpow_const`, converts the falling factorial at $-s/2$ into a rising factorial at $s/2$ (`ascPochhammer_eval_neg_eq_descPochhammer`), and uses positivity of the rising factorial (`ascPochhammer_pos`). At $s = 0$ it is the constant 1, which is admissible. For $s < 0$ it is not: for instance $t \mapsto t$ (the case $s = -2$) has first derivative $1 > 0$. That case is kernel-checked as `riesz_neg_two_not_admissible` (statement: not `AdmissiblePotential (TriangularRiesz.riesz (-2))`) **[Tier A]**.

Example 1.13 (Yukawa in the variable $t = r^2$)**.** The screened Coulomb (Yukawa) potential $V(r) = e^{-\kappa r}/r$, $\kappa > 0$, becomes $g(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$. We claim it is completely monotone **[Tier C]**. The argument uses the classical integral

$$\frac{e^{-\kappa\sqrt{t}}}{\sqrt{t}} = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-tu}\, u^{-1/2} e^{-\kappa^2/(4u)}\, du \qquad (t > 0). \tag{1.2}$$

It follows from $\int_0^\infty u^{-1/2} e^{-tu - a/u}\, du = \sqrt{\pi/t}\, e^{-2\sqrt{at}}$ with $a = \kappa^2/4$. Formula (1.2) writes g as a positive mixture of exponentials e^{-tu}. Differentiating under the integral (dominated convergence, since $u^r e^{-tu}$ is integrable against the weight for $t > 0$) gives $(-1)^r g^{(r)}(t) = \pi^{-1/2} \int_0^\infty u^r e^{-tu} u^{-1/2} e^{-\kappa^2/(4u)}\, du \geq 0$. A numerical spot check of (1.2) at $\kappa = 2$ ran as follows **[Tier X]**:

```python
import math
from scipy.integrate import quad

kappa = 2.0
for t in (0.25, 1.0, 4.0):
    lhs = math.exp(-kappa * math.sqrt(t)) / math.sqrt(t)
    integrand = lambda u: u ** -0.5 * math.exp(-kappa ** 2 / (4 * u)) * math.exp(-t * u)
    val, err = quad(integrand, 0, math.inf)
    print(f"t={t:4.2f}  lhs={lhs:.12f}  rhs={val / math.sqrt(math.pi):.12f}")
```

```
t=0.25  lhs=0.735758882343  rhs=0.735758882343
t=1.00  lhs=0.135335283237  rhs=0.135335283237
t=4.00  lhs=0.009157819444  rhs=0.009157819444
```

The spot check steers but does not prove. The proof is the integral identity. No Lean proof of Yukawa admissibility exists in this book's files.

Two closure properties are useful **[Tier C]**. Positive combinations of completely monotone functions are completely monotone, which is obvious from the definition. Products of completely monotone functions are completely monotone, because Leibniz's rule gives $(-1)^r (fg)^{(r)} = \sum_i \binom{r}{i} [(-1)^i f^{(i)}] [(-1)^{r-i} g^{(r-i)}] \geq 0$.

Theorem 1.14 (Bernstein's theorem)**.** *A function $g : (0,\infty) \to \mathbb{R}$ is completely monotone if and only if there is a non-negative Borel measure μ on $[0,\infty)$ such that*

$$g(t) = \int_{[0,\infty)} e^{-\alpha t}\, d\mu(\alpha) \qquad \text{for all } t > 0,$$

the integral being finite for every $t > 0$. The measure μ is then unique. ***[Tier L]*** *(classical result of Bernstein, in the form attributed by upstream [1] to Bernstein–Hausdorff–Widder. Not proved in this book.)*

Two features of the statement matter for energy problems.

- **The measure lives on $[0,\infty)$, closed at 0.** An atom $\mu(\{0\}) = c > 0$ contributes the constant c to g. For instance $g(t) = 1 + t^{-2}$ is completely monotone with $\mu = \delta_0 +$ (a density on $(0,\infty)$).
- **A constant potential has infinite energy at density one.** If $g \equiv 1$ then $E_1(C;R) = n_C(R) - 1$ (each of the $n_C(R)$ points sees $n_C(R) - 1$ others), and $n_C(R) \to \infty$ at density one. Hence $E_1(C) = \infty$ for every density-one C **[Tier C]** (the human proof is this one line). Upstream proves the same fact in Lean as `energy_one` (file `OAI/Analysis/Triangular/Energy/Potentials.lean`), with the formal statement `energy (fun _ => 1) C = ∞` under `DensityOne C`. It is part of the build whose main theorem is kernel-checked. The lattice energy of the constant is also ∞, so the atom at 0 never breaks the inequality: both sides are ∞.

1.4 Fourier analysis and Poisson summation

We use the normalisation of upstream [1]:

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x)\, e^{-2\pi i\, x\cdot\xi}\, dx.$$

With this choice $\widehat{\widehat{f}}(x) = f(-x)$ for Schwartz f (Fourier inversion, **[Tier L]**, classical), and no factors of 2π appear in the Poisson summation formula.

Proposition 1.15 (Gaussians). *For $\alpha > 0$ let $G_\alpha(x) = e^{-\pi\alpha|x|^2}$ on $\mathbb{R}^2$. Then*

$$\widehat{G_\alpha}(\xi) = \frac{1}{\alpha}\, e^{-\pi|\xi|^2/\alpha}.$$

In particular $\widehat{G_\alpha} > 0$, and $G_1 = \widehat{G_1}$.

Human proof. In one variable, $\int_{\mathbb{R}} e^{-\pi\alpha x^2} e^{-2\pi i x\xi}\, dx = \alpha^{-1/2} e^{-\pi\xi^2/\alpha}$. This is the standard computation: complete the square and shift the contour, or check that both sides solve the same first-order ODE in ξ with the same value at $\xi = 0$. In two variables the integral factorises into a product of two such integrals, giving $\alpha^{-1/2} \cdot \alpha^{-1/2} = \alpha^{-1}$. **[Tier C]** □

Theorem 1.16 (Poisson summation for a lattice). *Let $\Lambda \subset \mathbb{R}^2$ be a lattice of covolume $|\Lambda|$ and f a Schwartz function on $\mathbb{R}^2$. For every $v \in \mathbb{R}^2$,*

$$\sum_{\lambda\in\Lambda} f(\lambda + v) = \frac{1}{|\Lambda|} \sum_{w\in\Lambda^*} \widehat{f}(w)\, e^{2\pi i\, w\cdot v},$$

both sides converging absolutely. ***[Tier L]*** *(classical. The sketch below is* ***[Tier C]****.)*

Sketch for Schwartz functions. Let $F(v) = \sum_\lambda f(\lambda + v)$. Since f and all its derivatives decay faster than any power, the series and all its term-by-term derivatives converge uniformly on compact sets. So F is C^∞, and it is Λ-periodic. The characters of $\mathbb{R}^2/\Lambda$ are $e^{2\pi i w\cdot v}$ with $w \in \Lambda^*$. The Fourier coefficient of F is

$$c_w = \frac{1}{|\Lambda|}\int_D F(v) e^{-2\pi i w\cdot v}\, dv = \frac{1}{|\Lambda|}\int_{\mathbb{R}^2} f(x) e^{-2\pi i w\cdot x}\, dx = \frac{\widehat{f}(w)}{|\Lambda|},$$

where D is a fundamental domain. We unfold the integral over D and the sum over λ into one integral over $\mathbb{R}^2$, using $e^{-2\pi i w\cdot\lambda} = 1$. A smooth periodic function has rapidly decaying Fourier coefficients and equals its absolutely convergent Fourier series everywhere. This gives the formula. □

Example 1.17 (Theta functions of A). Apply Theorem 1.16 to $f = G_\alpha$, $\Lambda = A$ and $v = 0$. Proposition 1.15 gives $\widehat{G_\alpha}$. By Proposition 1.8, $|w|$ runs over the same values on A^* as $|a|$ on A, with the same multiplicities. Writing $\theta_A(\alpha) = \sum_{a\in A} e^{-\pi\alpha|a|^2}$ we get the functional equation

$$\theta_A(\alpha) = \frac{1}{\alpha}\,\theta_A(1/\alpha) \qquad (\alpha > 0).$$

[Tier C] (it follows from the two propositions and the theorem). Exercise 1.13 checks it in floating point.

1.5 The linear-programming bound

The idea goes back to Cohn and Elkies for sphere packing and to Cohn and Kumar [3] for energy. It turns the problem "bound the energy of every configuration" into the problem "find one good auxiliary function". We state it for the Gaussian $g_\alpha(t) = e^{-\pi\alpha t}$, so that $g_\alpha(|x|^2) = G_\alpha(x)$.

Definition 1.18 (Auxiliary function). Let $\alpha > 0$. A radial Schwartz function $f : \mathbb{R}^2 \to \mathbb{R}$ is an *auxiliary function for G_α* if

(i) $f(x) \le G_\alpha(x)$ for all $x \neq 0$ (f is a *minorant* of the potential), and

(ii) $\widehat{f}(\xi) \ge 0$ for all ξ.

It is *sharp for* A if in addition

(iii) $f(a) = G_\alpha(a)$ for all $a \in A \setminus \{0\}$, and

(iv) $\widehat{f}(w) = 0$ for all $w \in A^* \setminus \{0\}$.

Since f is radial and real, $\widehat{f}$ is real and radial, so (ii) makes sense. Conditions (iii) and (iv) are the "root conditions". In (iii), since $G_\alpha - f \geq 0$, every point of $A \setminus 0$ is a minimum of $G_\alpha - f$.

1.5.1 Periodic configurations: a complete proof

Definition 1.19 (Periodic configuration and its energy)**.** Let Λ be a lattice and $v_1, \dots, v_N \in \mathbb{R}^2$ with $v_i - v_j \notin \Lambda$ for $i \neq j$. The configuration $C = \bigcup_{i=1}^{N} (\Lambda + v_i)$ is Λ*-periodic with* N *points per period*. Its density is $N/|\Lambda|$. Its *periodic energy* is

$$E_g^{\mathrm{per}}(C) = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{\substack{\lambda \in \Lambda \\ \lambda + v_j - v_i \neq 0}} g\big(|\lambda + v_j - v_i|^2\big),$$

the average over the N cosets of the energy seen by one particle.

The condition $\lambda + v_j - v_i \neq 0$ excludes exactly one term: $i = j$ and $\lambda = 0$. For $i \neq j$ the vector $\lambda + v_j - v_i$ is never zero, because the cosets are distinct.

Theorem 1.20 (LP bound, periodic case)**.** *Let $\alpha > 0$, let f be an auxiliary function for G_α, and let C be a Λ-periodic configuration of density one ($|\Lambda| = N$). Then*

$$E_{g_\alpha}^{\mathrm{per}}(C) \;\geq\; \widehat{f}(0) - f(0).$$

If f is sharp for A, then $E_{g_\alpha}^{\mathrm{per}}(A) = \widehat{f}(0) - f(0) = \sum_{a \in A \setminus 0} e^{-\pi\alpha|a|^2}$. Hence A minimises the periodic Gaussian energy among periodic density-one configurations. ***[Tier C]*** *(complete human proof below.)*

Human proof. All series below converge absolutely, because G_α and f are Schwartz.

Step 1 (minorant). By (i), each term of E^{per} is at least the corresponding value of f:

$$N\, E_{g_\alpha}^{\mathrm{per}}(C) = \sum_{i,j} \sum_{\lambda + v_j - v_i \neq 0} G_\alpha(\lambda + v_j - v_i) \;\geq\; \sum_{i,j} \sum_{\lambda + v_j - v_i \neq 0} f(\lambda + v_j - v_i).$$

Step 2 (put back the diagonal). The excluded terms are exactly the N terms with $i = j$ and $\lambda = 0$, each equal to $f(0)$:

$$\sum_{i,j} \sum_{\lambda + v_j - v_i \neq 0} f(\lambda + v_j - v_i) = \sum_{i,j} \sum_{\lambda \in \Lambda} f(\lambda + v_j - v_i) \;-\; N f(0).$$

Step 3 (Poisson). By Theorem 1.16 with $v = v_j - v_i$ and $|\Lambda| = N$,

$$\sum_{i,j} \sum_{\lambda} f(\lambda + v_j - v_i) = \frac{1}{N} \sum_{w \in \Lambda^*} \widehat{f}(w) \sum_{i,j} e^{2\pi i\, w \cdot (v_j - v_i)} = \frac{1}{N} \sum_{w \in \Lambda^*} \widehat{f}(w) \Big| \sum_{j=1}^{N} e^{2\pi i\, w \cdot v_j} \Big|^2.$$

The interchange of the finite sum over (i, j) with the sum over w is harmless.

Step 4 (positivity). Each term on the right is ≥ 0 by (ii). Keeping only $w = 0$, where the inner sum equals N, gives a lower bound of $\frac{1}{N}\widehat{f}(0) N^2 = N\widehat{f}(0)$.

Combining the four steps, $N\, E^{\mathrm{per}} \geq N\widehat{f}(0) - N f(0)$. Divide by N.

Equality on A. Take $\Lambda = A$, $N = 1$, $v_1 = 0$. In Step 1, (iii) makes every term an equality. In Step 4, (iv) makes every dropped term zero. So both inequalities are equalities: $\sum_{a \neq 0} G_\alpha(a) = \widehat{f}(0) - f(0)$. Since $G_\alpha(a) = e^{-\pi\alpha|a|^2}$, this is the stated value. □

Remark 1.21 (Other densities). The same proof at density $\rho = N/|\Lambda|$ gives $E^{\mathrm{per}} \geq \rho\, \widehat{f}(0) - f(0)$. The upstream theorem is stated at density one. Chapter 2 transfers it to every density by scaling. **[Tier C]**

1.5.2 From the periodic energy to the lower energy

Theorem 1.20 uses E^{per}, not the disk energy of Definition 1.3. These are different definitions, and we connect them explicitly rather than silently identifying them.

Proposition 1.22 (Disk energy of a periodic configuration)**.** *Let C be periodic as in Definition 1.19 and $\alpha > 0$. Then $\lim_{R\to\infty} E_{g_\alpha}(C;R) = E^{\mathrm{per}}_{g_\alpha}(C)$. In particular the lower energy $E_{g_\alpha}(C)$ equals the periodic energy, and the limit exists.* ***[Tier C]***

Human proof. For $x \in C$ let $e(x) = \sum_{y\in C,\, y\neq x} G_\alpha(x-y)$. It is constant on each coset $\Lambda + v_i$, say equal to e_i, and $E^{\mathrm{per}} = \frac{1}{N}\sum_i e_i$. Let $n_i(R)$ be the number of points of $\Lambda + v_i$ in $\overline{B}(0,R)$. Lattice point counting, done by comparing the disk with the union of fundamental parallelograms centred at the points it contains, gives $n_i(R)/(\pi R^2) \to 1/|\Lambda|$. Hence $n_i(R)/n_C(R) \to 1/N$.

Upper bound. Dropping the restriction $y \in C_R$ only adds non-negative terms, so $E(C;R) \le \frac{1}{n_C(R)}\sum_{x\in C_R} e(x) = \sum_i \frac{n_i(R)}{n_C(R)} e_i \to E^{\mathrm{per}}$.

Lower bound. Let $\tau(L) = \sup_{u\in\mathbb{R}^2} \sum_{\lambda\in\Lambda,\,|\lambda+u|>L} G_\alpha(\lambda+u)$. This tends to 0 as $L\to\infty$ by Gaussian decay. If $x \in C_{R-L}$, every $y \in C$ with $|x-y| \le L$ lies in C_R, so the inner sum at x is at least $e(x) - N\tau(L)$. Discarding the points of $C_R \setminus C_{R-L}$, which only lowers the sum,

$$E(C;R) \;\ge\; \frac{1}{n_C(R)} \sum_{x\in C_{R-L}} \big(e(x) - N\tau(L)\big) \;\xrightarrow[R\to\infty]{}\; E^{\mathrm{per}} - N\tau(L).$$

The limit uses $n_i(R-L)/n_C(R) \to 1/N$. Letting $L\to\infty$ shows that the lim inf is at least E^{per}. □

Together, Theorem 1.20 and Proposition 1.22 prove: *for every periodic density-one configuration C and every $\alpha > 0$ for which a sharp auxiliary function exists, $E_{g_\alpha}(C) \ge \sum_{a\in A\setminus 0} e^{-\pi\alpha|a|^2} = E_{g_\alpha}(A)$* **[Tier C]**.

1.5.3 What upstream adds

Two things are missing from the argument above. Upstream [1] supplies both.

1. **Existence of sharp auxiliary functions for every $\alpha > 0$.** This is the hard part. In dimension two no construction by modular forms is known; that is the method of CKMRV [2] in dimensions 8 and 24. Upstream instead builds f_α from a sine product with double zeros encoding the periodic contact set. Multiplying by double-pole and simple-pole terms gives entire functions with prescribed values and slopes. Compactly supported spectral measures, Gaussian damping, and finite reference lists handled by positive quadrature complete the construction. The quadrature rests on certified matrix and polynomial inequalities, checked with python-flint ball arithmetic. **[Tier L]** [1]

2. **Arbitrary competitors.** The competitor need not be periodic. Upstream compares sums over disk points with area integrals, with a uniform error, and obtains $E_{g_\alpha}(C) \ge \widehat{f_\alpha}(0) - f_\alpha(0)$ for every locally finite configuration of centred density one. **[Tier L]** [1]. The full theorem, including this step, is kernel-checked in Lean **[Tier A]** (Theorem 1.23).

1.6 From Gaussians to all completely monotone potentials

Suppose we know, for every $\alpha > 0$ and every admissible competitor C,

$$E_{g_\alpha}(C) \;\ge\; L(\alpha) := \sum_{a\in A\setminus 0} e^{-\pi\alpha|a|^2}. \tag{1.3}$$

How do we get the bound for a general admissible g? Here is the human argument in outline **[Tier C]**.

Step 1 (Bernstein). Write $g(t) = \int_{[0,\infty)} e^{-\beta t}\, d\mu(\beta)$ as in Theorem 1.14. In the normalisation of (1.3) this reads $e^{-\beta t} = g_{\beta/\pi}(t)$.

Step 2 (finite sums commute with the integral). For fixed R the disk energy is a *finite* sum of non-negative terms, so

$$E_g(C;R) = \int_{[0,\infty)} E_{g_{\beta/\pi}}(C;R)\, d\mu(\beta).$$

Step 3 (Fatou, not linearity). The lim inf is not linear: in general $\liminf(u_R + w_R) \neq \liminf u_R + \liminf w_R$. The correct tool is Fatou's lemma, which gives exactly the inequality we need. Fatou is a statement about *sequences*, while our lim inf is over a real parameter $R \to \infty$. So first choose radii $R_n \to \infty$ with $E_g(C;R_n) \to E_g(C)$, and then apply Fatou along (R_n):

$$E_g(C) = \lim_n \int E_{g_{\beta/\pi}}(C;R_n)\, d\mu \;\geq\; \int \liminf_n E_{g_{\beta/\pi}}(C;R_n)\, d\mu \;\geq\; \int E_{g_{\beta/\pi}}(C)\, d\mu(\beta).$$

The last step holds because a lim inf along a subsequence of radii is at least the lim inf over all radii.

Step 4 (the Gaussian bound, then Tonelli). On $(0,\infty)$ use (1.3). On the lattice side, Tonelli's theorem (non-negative terms) swaps sum and integral:

$$\int_{(0,\infty)} L(\beta/\pi)\, d\mu(\beta) = \sum_{a\in A\setminus 0} \int_{(0,\infty)} e^{-\beta|a|^2} d\mu(\beta).$$

Step 5 (the atom at 0). If $\mu(\{0\}) = c > 0$ then $g \geq c$, so $E_g(C) \geq c\,E_1(C) = \infty$ by the constant-potential fact of §1.3, and the bound is trivial. If $\mu(\{0\}) = 0$, Steps 3–4 give $E_g(C) \geq \sum_{a\neq 0} g(|a|^2) = \mathcal{E}_g(A)$.

Step 6 (attainment). A is itself locally finite and of density one (kernel-checked in our file `AnyDensity.lean` as `A_lf` and `A_d1`, which are used in the proof of the audited `TriangularDensity.attained` **[Tier A]**), so the bound applies to it. For $C = A$, each point sees at most the full lattice sum, so $E_g(A;R) \leq \mathcal{E}_g(A)$ for every R (the upper-bound step of Proposition 1.22). Then $\limsup \leq \mathcal{E}_g(A) \leq \liminf$, the second inequality being the bound just proved for $C = A$.

1.6.1 How upstream completes it

Upstream completes the passage with lower-limit, endpoint and infinite-energy arguments (upstream manuscript, §5) **[Tier L]** [1]. Its Lean proof follows a route that avoids Bernstein's theorem on the open interval and its endpoint difficulties at $t \to 0$. We read the following off upstream's Lean files; the description of the route is ours **[Tier C]**. The proof of `universal_energy_minimum` in `Universal.lean` calls `universal_from_gaussian_bounds`, which is built from the lemmas below. So they lie in the dependency cone of the audited theorem, whose axiom footprint is the allowlist (§1.7). We did not print their axioms separately.

- *Shift.* For $\varepsilon > 0$ the shifted function $g(\varepsilon + t)$ is a mixture $\int_{[0,1]} v^t\, d\nu(v)$ of *power potentials* v^t. Here ν is a finite measure on $[0,1]$ of mass $g(\varepsilon)$, obtained by upstream's own finite-difference argument (Lean: `CM.shifted_mixture`). With $v = e^{-\pi\alpha}$ one has $v^t = e^{-\pi\alpha t}$: $v \in (0,1)$ are Gaussians, $v = 1$ is the constant, and $v = 0$ is the zero potential.

- *Gaussian to power bounds.* The case split $v = 0$ / $v = 1$ / $0 < v < 1$ is the following lemma.

- *Removing the shift.* Since g is antitone, $E_{g(\varepsilon+\cdot)}(C) \le E_g(C)$ (`shifted_energy_le`). The lattice energy is the supremum of the shifted ones over $\varepsilon = 1/(n+1)$ (`latticeEnergy_shift_iSup`). Fatou and Tonelli enter through `shifted_mixture_energy`.

Lean 4 : upstream, `OAI/Analysis/Triangular/Energy/Potentials.lean` (verbatim)

```
lemma power_bounds_from_gaussians {C : Set Plane} (hD : DensityOne C)
    (hgauss : ∀ α : ℝ, 0<α → latticeEnergy (gaussPot α)≤energy (gaussPot α) C)
    (v : I) : latticeEnergy (powerPotential v)≤energy (powerPotential v) C := by
  by_cases h0 : (v:ℝ)=0
  · have hl : latticeEnergy (powerPotential v)=0 := by
```

(. . .) and, in the case $v = 1$:

```
  by_cases h1 : (v:ℝ)=1
  · have he : powerPotential v=(fun _ => (1:ℝ)) := by
      funext t
      simp only [powerPotential,h1,Real.one_rpow]
    rw [he,energy_one hD]
    exact le_top
```

So the atom at $\alpha = 0$ ($v = 1$) is handled exactly as in Step 5, by `energy_one`.

1.7 A Lean 4 primer for this book

You do not need to write Lean to use this book, but you must be able to *read* a statement and know what a "kernel-checked" label does and does not certify.

1.7.1 Reading a theorem statement

Lean 4 : the main theorem as stated in the challenge file (verbatim)

```
/-- Universal energy minimality for smooth completely monotone potentials, with
    attainment. -/
theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := by
  sorry
```

Read it left to right.

- `(g : ℝ → ℝ) (C : Set Plane)` are the objects: a potential and a configuration.
- `(hg : ...) (hC : ...) (hd : ...)` are *hypotheses*. Each is a named proof that the stated property holds. To apply the theorem you must supply all three.
- After the final colon comes the *conclusion*. Here it is a conjunction ($\wedge$) of the bound and the attainment.
- `:= by sorry` is the proof. In the *challenge* file it is `sorry` *by design*. The proved version is `OAI/Analysis/Triangular/Energy/Universal.lean`. The Comparator tool certifies that the two statements agree, and the build of the proved file is kernel-checked.

In human language:

Theorem 1.23 (Upstream's main theorem [1]). *Let g be admissible (Definition 1.10) and $C \subseteq \mathbb{R}^2$ locally finite with centred density one. Then*

$$\mathcal{E}_g(A) \;\le\; E_g(C) \qquad \text{and} \qquad \mathcal{E}_g(A) = E_g(A),$$

with values in $[0,\infty]$, and E_g the lower energy over closed centred disks. ***[Tier A]*** *for the Lean statement* ***`OAI.AtomicTriangular.universal_energy_minimum`*** *(kernel-checked, conditional on the fidelity of the definitions). The proof is AI-generated and not peer reviewed.*

1.7.2 Corollaries are short

Once the main theorem is available, a corollary only needs to check the hypothesis. Here is the Riesz case, quoted verbatim from our file `formal_cert/Triangular/Riesz.lean`:

Lean 4 : `formal_cert/Triangular/Riesz.lean` (verbatim excerpt)

```
/-- Riesz potential in the variable `t = r^2`. -/
noncomputable def riesz (s : ℝ) : ℝ → ℝ := fun t => t ^ (-(s / 2))

theorem riesz_admissible {s : ℝ} (hs : 0 < s) : AdmissiblePotential (riesz s) := by
  refine ⟨?_, ?_, ?_⟩
  · intro x hx
    exact (Real.contDiffAt_rpow_const_of_ne (ne_of_gt hx)).contDiffWithinAt
  · intro t ht
    exact (Real.rpow_pos_of_pos ht _).le
```

```
...
```

```
theorem triangular_riesz_optimal {s : ℝ} (hs : 0 < s) (C : Set Plane) (hC :
    LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy (riesz s) ≤ energy (riesz s) C :=
  (universal_energy_minimum (riesz s) C (riesz_admissible hs) hC hd).1
```

The anonymous constructor `⟨?_, ?_, ?_⟩` splits `AdmissiblePotential` into its three conjuncts: smoothness, non-negativity and complete monotonicity. Each bullet $(\cdot)$ proves one of them. The last line applies the main theorem and takes the first component `.1` of the conjunction. The docstring in the file also remarks that for $s \le 2$ the lattice energy is ∞, so the corollary is then vacuous. That remark is a comment, not a theorem **[Tier C]**. The *finite* side ($s > 2$) is the kernel-checked `latticeEnergy_riesz_lt_top` **[Tier A]**.

1.7.3 `sorry`, axioms and the allowlist

Lean accepts a file containing `sorry`. It prints a warning and *exits with code* 0; the recorded compile below shows exactly this (return code 0 with a `sorry` warning). Behind the scenes, `sorry` is the axiom `sorryAx`, which proves anything. A user can also declare a new `axiom`, for example one asserting `False`, and prove anything from it; again the compiler exits 0. The exit code is therefore *not* a proof check. What is checked is the list of axioms a declaration depends on, printed by `##print axioms`. This book accepts exactly the allowlist

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\},$$

the standard axioms of Lean's classical mathematics. Anything else, in particular `sorryAx`, rejects the result.

Our audit file `formal_cert/Triangular/Axioms.lean` prints the axioms of 30 declarations (22 for D1–D3, 8 for D4) and of two planted controls. The controls, quoted verbatim, are:

```
theorem ctl_sorry : TriangularRiesz.riesz 1 1 = 0 := by sorry
axiom ctl_cheat : False
theorem ctl_axiom : TriangularRiesz.riesz 1 1 = 0 := ctl_cheat.elim
#print axioms ctl_sorry
#print axioms ctl_axiom
```

Both controls "prove" the false statement $1^{-1/2} = 0$. The recorded compile of this file (`results/discovery/D1/compile_Axioms.json`, return code 0) contains, among its output lines:

```
'TriangularRiesz.triangular_riesz_optimal' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'OAI.AtomicTriangular.universal_energy_minimum' depends on axioms: [propext, Classical.choice,
    Quot.sound]
.../formal_cert/Triangular/Axioms.lean:42:8: warning: declaration uses 'sorry'
'ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
'ctl_axiom' depends on axioms: [ctl_cheat, propext, Classical.choice, Quot.sound]
```

(The path in the warning line is shortened here.) The return code is 0 even though two false theorems were "proved". The audit catches them, because their footprints contain `sorryAx` and `ctl_cheat`. This is the positive and negative control of the audit: it must flag the planted cheats, and it does.

1.7.4 What "kernel-checked" attaches to

Following the Elenchus method [17], the label **[Tier A]** attaches to the *formal statement*, never to its gloss in English. Lean certifies that `universal_energy_minimum`, *as written with upstream's definitions*, follows from the allowlisted axioms. It does not certify that `DensityOne` is the density a physicist means, or that `energy` (a lim inf over closed centred disks) is the energy CKMRV mean. Those are questions of *statement adequacy*, and this chapter's §1.1 raises several of them: centred disks, closed disks, lim inf, and the empty-set convention of `diskPoints`. No human has audited statement adequacy for these files. The ledger records 6 open flags. So the correct phrase is "kernel-checked, conditional on the fidelity of the definitions". Avoid "proved true", "certified physics" or "100% verified".

1.8 Reviewing a draft

A student draft titled *"Universal Optimality of the Planar Triangular Lattice: A Formal Proof Architecture"* (header: "Statement-Adequacy Review Draft") summarised the material of this chapter. Reviewing it is a good exercise. Before reading the tables, try to find what is wrong with each quoted claim yourself.

	Draft says	Correct statement	Why
1	"derivatives strictly alternate in sign"	$(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 0$, $t > 0$ (weak inequalities), with g C^∞ on $(0, \infty)$.	Constants are admissible. The Lean definition `AdmissiblePotential` uses $0 \leq$, not $0 <$.
2	Bernstein's representation is "strictly sufficient"; the passage from Gaussians to mixtures is not justified	$g(t) = \int_{[0,\infty)} e^{-\alpha t} d\mu(\alpha)$ with $\mu \geq 0$, and μ may charge $\alpha = 0$. Gaussian bounds are *sufficient*. The passage needs Tonelli, Fatou along a sequence of radii, and the atom at 0 treated separately.	lim inf is not linear (§1.6). The constant part has infinite energy at density one (`energy_one`).
3	Cohn–Elkies-type lemma by "majorization"	$f \leq G_\alpha$ (a *minorant*) and $\widehat{f} \geq 0$ give $E(C) \geq \widehat{f}(0) - f(0)$ at density one. Root conditions on $A \setminus 0$ and $A^* \setminus 0$ give equality on A by Poisson summation. Here $G_\alpha = e^{-\pi\alpha\lvert x\rvert^2}$.	Theorem 1.20. A majorant gives no lower bound. The draft also omits the bound itself, the density, and the normalisation.
4	auxiliary functions from "integral transforms of weakly holomorphic modular forms adapted to hexagonal symmetry"	Upstream uses sine-product cardinal interpolation, Gaussian damping and certified finite quadrature.	Modular forms are the CKMRV method in dimensions 8 and 24 [2]. No modular-form construction is known in dimension 2.

	Draft says	Correct statement	Why
5	"We demonstrate" the theorem; author line "Statement-Adequacy Review Draft"	The main theorem is upstream's [1]: AI-generated, kernel-checked, Comparator-accepted, not peer reviewed. Our contribution is corollaries and an audit.	Attribution, and honesty about the review status.
6	Theorem 2 (arbitrary densities) without the hypotheses of local finiteness and density ρ	For $\rho > 0$ and C locally finite with density ρ (`DensityRho`): $E_g(C) \geq \mathcal{E}_{g(\rho^{-1}\cdot)}(A)$, attained by $\rho^{-1/2}A$. The energy is the lower energy.	Without local finiteness the counts are meaningless. Lean: `universal_any_density` (D2, see Chapter 2).
7	Theorem 3 (non-uniqueness), with no qualifier on the class of configurations	Non-uniqueness holds among *all* locally finite configurations: $A \setminus \{0\}$ is optimal and is not an isometric image of A. Uniqueness among *periodic* configurations is open in 2D and not claimed.	$A \setminus \{0\}$ is not periodic. Lean: `universal_optimum_not_unique`. Upstream: "It identifies the minimum value, not all minimizers."
8	(missing)	E_g is the *lower* energy (lim inf). CKMRV Def. 1.3 also requires the limit to exist. Disks are closed and centred.	The Lean notion is weaker than CKMRV's. Open vs closed disks: **[Tier A]** kernel-checked for density one and for the lower energy, independently re-verified; other densities **[Tier C]** (details in the Editor's note).

The tiers in the tables are those of the sources cited in each row. Rows 1, 6 and 7 point to kernel-checked Lean declarations **[Tier A]**. Row 3 rests on Theorem 1.20 **[Tier C]** for periodic competitors and on upstream **[Tier L]** [1] for arbitrary ones. Row 4 is **[Tier L]** [1, 2]. Row 8 is

a statement-adequacy remark **[Tier C]**. Row 2 combines **[Tier L]** (Bernstein) and **[Tier C]** (the Fatou argument).

1.8.1 The corrected statement on one page

Definition 1.24 (Setting)**.** A configuration $C \subseteq \mathbb{R}^2$ is locally finite. It has density one if $\#(C \cap \overline{B}(0,R))/(\pi R^2) \to 1$ (closed, centred disks). A potential g of the squared distance is admissible if it is C^∞ on $(0,\infty)$ and $(-1)^r g^{(r)} \geq 0$ there for all $r \geq 0$. The lower energy is $E_g(C) = \liminf_{R\to\infty} \frac{1}{\#C_R} \sum_{x\neq y\in C_R} g(|x-y|^2) \in [0,\infty]$.

Lemma 1.25 (LP bound, density one)**.** *Let $\alpha > 0$ and $G_\alpha(x) = e^{-\pi\alpha|x|^2}$, $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x\cdot\xi}dx$. If f is radial Schwartz with $f \leq G_\alpha$ on $\mathbb{R}^2 \setminus 0$ and $\widehat{f} \geq 0$, then every density-one configuration C satisfies $E_{e^{-\pi\alpha t}}(C) \geq \widehat{f}(0) - f(0)$. This is **[Tier C]** for periodic C (Theorem 1.20 with Proposition 1.22) and **[Tier L]** [1] for arbitrary locally finite C. If moreover $f = G_\alpha$ on $A \setminus 0$ and $\widehat{f} = 0$ on $A^* \setminus 0$, then $\widehat{f}(0) - f(0) = \sum_{a\in A\setminus 0} e^{-\pi\alpha|a|^2} = E_{e^{-\pi\alpha t}}(A)$ **[Tier C]**.*

Theorem 1.26 (Upstream [1]; AI-generated, not peer reviewed)**.** *For every admissible g and every locally finite C of density one, $\sum_{a\in A\setminus 0} g(|a|^2) \leq E_g(C)$, and equality holds for $C = A$. The values are in $[0,\infty]$. The theorem identifies the minimum value, not all minimisers. **[Tier A]** for the Lean statement `OAI.AtomicTriangular.universal_energy_minimum`, conditional on the fidelity of the definitions. The strong (per-potential) Cohn–Kumar conjecture is* not *claimed and remains open.*

1.9 Exercises

Difficulty: (E) easy, (M) medium, (H) hard.

Exercise 1.1 ((E))**.** Show that $\mathbb{Z}^2$ and A both have density one in the sense of Definition 1.2. You may use the lattice point count $n(R) = \pi R^2/|\Lambda| + O(R)$. Why does Definition 1.2 need local finiteness?

Exercise 1.2 ((E))**.** Verify the Gram matrix of Proposition 1.6 and find all (j,k) with $j^2 + jk + k^2 = 13$. Check that there are twelve.

Exercise 1.3 ((E))**.** Which of these are admissible in the sense of Definition 1.10? (a) $e^{-t} + 3$; (b) t^{-1}; (c) e^{-t^2}; (d) $1/(1+t)$; (e) $\sqrt{t}$. For (c), compute $g''(t)$ near $t = 0$.

Exercise 1.4 ((E))**.** Show that $E_1(C;R) = n_C(R) - 1$ for the constant potential $g \equiv 1$, and deduce that $E_1(C) = \infty$ at density one. Find the corresponding upstream Lean lemma and compare its hypotheses with yours.

Exercise 1.5 ((M))**.** Prove that a product of two completely monotone functions is completely monotone. Use this and (1.2) to give two different proofs that $e^{-\kappa\sqrt{t}}/\sqrt{t}$ is completely monotone. (For the first proof you also need: $e^{-\kappa\sqrt{t}}$ is completely monotone. Prove it from (1.2) with an extra differentiation in κ, or via the composition rule.)

Exercise 1.6 ((M))**.** Prove Proposition 1.15 in one variable by showing that $\phi(\xi) = \int e^{-\pi\alpha x^2} e^{-2\pi i x\xi}dx$ satisfies $\phi'(\xi) = -(2\pi\xi/\alpha)\phi(\xi)$.

Exercise 1.7 ((M))**.** Repeat the proof of Theorem 1.20 at density $\rho = N/|\Lambda|$ and show that the bound becomes $\rho\widehat{f}(0) - f(0)$. Where exactly does the density enter?

Exercise 1.8 ((M))**.** Give an example of real sequences with $\liminf(u_n + w_n) > \liminf u_n + \liminf w_n$. Explain why Step 3 of §1.6 would be wrong if it claimed equality, and why the inequality that Fatou gives is the one needed.

Exercise 1.9 ((M)). Let $C = A \setminus \{0\}$. Show directly that C has density one, and that $E_g(C) = E_g(A)$ for g Gaussian. Why is C not periodic? Which Lean declaration states the general version?

Exercise 1.10 ((H)). Complete the lattice point counting step of Proposition 1.22: show that $n_i(R) = \pi R^2/|\Lambda| + O(R)$ for a translate $\Lambda + v_i$, uniformly in v_i.

Exercise 1.11 ((H)). Suppose a radial Schwartz f satisfies (i)–(iv) of Definition 1.18. Show that $\nabla(G_\alpha - f) = 0$ at every $a \in A \setminus 0$. In one radial variable, explain why the roots of $G_\alpha - f$ at $|a|$ have even order. Why does this suggest a construction from a product with double zeros?

Exercise 1.12 ((H)). The draft review (§1.8) lists eight problems. Find a *ninth* adequacy question about upstream's definitions that the table does not cover. Hint: think about `diskPoints` on a configuration that is not locally finite, or about centred versus uncentred disks. State it precisely and say which tier an answer could reach.

Exercise 1.13 ((M, Python)). *Poisson summation and the theta comparison.* The script below computes truncated theta sums $\theta_A(\alpha)$, the Poisson-dual expression $\theta_A(1/\alpha)/\alpha$ (Example 1.17), and $\theta_{\mathbb{Z}^2}(\alpha)$ for the square lattice, which also has density one. It was run and printed the output below **[Tier X]**.

```python
import math

B = math.sqrt(3) / 2  # b = sqrt(3)/2, as in upstream


def norm_sq(j: int, k: int) -> float:
    """|p(j,k)|^2 = (1/b)(j^2 + jk + k^2) for the covolume-one triangular lattice A."""
    return (j * j + j * k + k * k) / B


def theta_A(alpha: float, M: int = 40) -> float:
    """Truncated theta sum over A: sum of exp(-pi*alpha*|a|^2), |j|,|k| <= M."""
    return sum(math.exp(-math.pi * alpha * norm_sq(j, k))
               for j in range(-M, M + 1) for k in range(-M, M + 1))


def theta_Z2(alpha: float, M: int = 40) -> float:
    """Theta sum over the square lattice Z^2 (also covolume one)."""
    s = sum(math.exp(-math.pi * alpha * n * n) for n in range(-M, M + 1))
    return s * s


print("alpha   theta_A(alpha)      theta_A(1/alpha)/alpha   theta_Z2(alpha)")
for alpha in (0.5, 1.0, 2.0):
    lhs = theta_A(alpha)
    rhs = theta_A(1.0 / alpha) / alpha
    print(f"{alpha:4.1f}  {lhs:.15f}  {rhs:.15f}  {theta_Z2(alpha):.15f}")
```

```
alpha   theta_A(alpha)      theta_A(1/alpha)/alpha   theta_Z2(alpha)
 0.5  2.008477918597919  2.008477918597920  2.014967440690170
 1.0  1.159595266963928  1.159595266963928  1.180340599016097
 2.0  1.004238959298960  1.004238959298960  1.007483720345085
```

(a) Explain why columns 2 and 3 agree, citing the two propositions and the theorem used. Why do they agree only to about 10^{-15}? (b) Compute $\theta_A(1) - 1$ and compare with Example 1.9. (c) The Gaussian energies at $\alpha = 1$ are $\theta - 1$: about 0.15960 for A and 0.18034 for $\mathbb{Z}^2$ **[Tier X]**. This is consistent with the main theorem, but it supports nothing (Tier X). Which theorem in the literature compares theta functions of all lattices of covolume one? (See [7].) (d) Change `M` to 5 and explain why the output barely changes.

Exercise 1.14 ((E, Python))**.** Modify the shell-count script of Example 1.9 to list the values $n \le 50$ represented by $j^2 + jk + k^2$, and the corresponding $r(n)$. Conjecture which primes p occur, then check your conjecture by hand for $p \le 13$. Report only what your run prints.

Exercise 1.15 ((H))**.** Using Theorem 1.20, show that among *lattices* of covolume one, the triangular lattice minimises $\sum_{\lambda \neq 0} e^{-\pi \alpha |\lambda|^2}$ for every α for which a sharp auxiliary function exists. Explain why this lattice statement is much weaker than Theorem 1.23.

Chapter 2

The theorems, side by side: human proof and Lean 4

This chapter states every result of the book twice. First comes the mathematics: an informal statement, a precise statement and a human proof that you can check with pen and paper. Then comes the Lean 4 text, quoted verbatim from the files in `formal_cert/Triangular/`, together with a short paragraph explaining how the formal proof follows the human one. Where a listing is shortened, the omission is marked by a comment `- ...`; nothing else in a listing is changed.

How the tiers are used in this chapter.

- A Lean theorem that compiled under the pinned toolchain (Lean 4.34.1, Mathlib `d13f23b7`, openai/math `adc7f124`), with an axiom footprint inside {`propext`, `Classical.choice`, `Quot.sound`}, is **[Tier A]**. Tier A attaches to the *formal statement*. It means "kernel-checked, conditional on the fidelity of the definitions". It does not mean "true of the physical world", and it does not mean that the Lean definitions say what the English says: no human has audited the adequacy of the statements, and the claims ledger carries six open adequacy flags.
- Every result below except the upstream theorem is a corollary of the upstream theorem `OAI.AtomicTriangular.universal_energy_minimum` [1]. That theorem is AI-generated (OpenAI release). We kernel-checked it and Comparator accepted it under the pinned toolchain. It has not been peer reviewed.
- A human proof written out in this book is an argument, **[Tier C]**, until it is matched by a kernel-checked Lean proof. When it is matched, the *Lean statement* is Tier A; the human proof stays an explanation of it.
- Statements taken from the upstream manuscript or the published literature are **[Tier L]**.

2.1 The objects, as upstream defines them

We write $\mathbb{P} = \mathbb{R}^2$ with the Euclidean norm, and $B_R = \{x : \|x\| \le R\}$ for the *closed* disk of radius R centred at the origin. All definitions in this section are upstream's [1]. They are taken from the challenge file `lean/ComparatorChallenges/TriangularEnergy.lean` (openai/math `adc7f124`, sha256 `af51979c...0616`; we checked the full hash, `af51979c7baecd3e9852637bbba29eff96cbb09cec84e15eff7d686f1a770616`, with `sha256sum` on our pinned copy).

Definition 2.1 (Configurations, counts, density)**.** A *configuration* is a set $C \subseteq \mathbb{P}$ (no multiplicities). It is *locally finite* if $C \cap B_R$ is finite for every R. Its *disk count* is $N_R(C) = \#(C \cap B_R)$;

Result	Main Lean name (file)	Tier
Upstream	`universal_energy_minimum` (upstream)	A (AI-generated, not peer reviewed)
D1	`triangular_riesz_optimal` (`Riesz.lean`)	A
D1b	`latticeEnergy_riesz_lt_top` (`RieszFinite.lean`)	A
D2	`triangular_optimal_any_density` (`AnyDensity.lean`)	A
Controls	`riesz_neg_two_not_admissible`, ... (`Controls.lean`)	A
D3	`universal_optimum_not_unique` (`UniversalOptimality.lean`)	A
D4	`energy_eq_openEnergy`, `densityOne_iff_open` (`OpenDisk.lean`, `OpenDiskAny.lean`)	A

Table 2.1: The results of this chapter. "A" always means: kernel-checked, conditional on the fidelity of the definitions.

by convention the count is 0 if $C \cap B_R$ is infinite. C has *density one* if $N_R(C)/(\pi R^2) \to 1$ as $R \to \infty$, and density ρ if the limit is ρ.

Definition 2.2 (Admissible potentials)**.** $g : \mathbb{R} \to \mathbb{R}$ is *admissible* if g is C^∞ on $(0, \infty)$, $g(t) \geq 0$ for $t > 0$, and $(-1)^r g^{(r)}(t) \geq 0$ for every $r \in \mathbb{N}$ and every $t > 0$. The inequalities are *weak*, and the smoothness is C^∞, not analyticity. The variable t is the *squared* distance: a pair at distance r interacts through $g(r^2)$.

Definition 2.3 (Disk energy, lower energy, lattice energy)**.** With values in $[0, \infty]$,

$$E_g(C; R) = \frac{1}{N_R(C)} \sum_{x \in C \cap B_R} \sum_{\substack{y \in C \cap B_R \\ y \neq x}} g\big(\|x - y\|^2\big), \qquad E_g(C) = \liminf_{R \to \infty} E_g(C; R).$$

(Each term is truncated at 0 by `ENNReal.ofReal`, and $0^{-1} = \infty$, $\infty \cdot 0 = 0$ in $[0, \infty]$, so an empty disk has energy 0.) $E_g(C)$ is the *lower* energy: it is a lim inf, and nothing requires the limit to exist. Let $b = \sqrt{3}/2$ and

$$p(j, k) = b^{-1/2}\Big(j + \tfrac{k}{2},\ kb\Big), \qquad A = \{p(j, k) : (j, k) \in \mathbb{Z}^2\}.$$

A is the triangular lattice of covolume one. Its *lattice energy* is $\mathcal{E}_g(A) = \sum_{a \in A \setminus \{0\}} g(\|a\|^2) \in [0, \infty]$; the sum may be ∞.

Lean 4 (upstream challenge file, verbatim)

```
-- lean/ComparatorChallenges/TriangularEnergy.lean, lines 11-36 (namespace OAI.
    TriangularUniversal)
abbrev Plane := EuclideanSpace ℝ (Fin 2)

def LocallyFinite (C : Set Plane) : Prop :=
  ∀ R : ℝ, (C ∩ Metric.closedBall (0 : Plane) R).Finite

def diskPoints (C : Set Plane) (R : ℝ) : Finset Plane :=
  if h : (C ∩ Metric.closedBall (0 : Plane) R).Finite then h.toFinset else ∅

def diskCount (C : Set Plane) (R : ℝ) : ℕ := (diskPoints C R).card

def DensityOne (C : Set Plane) : Prop :=
  Tendsto (fun R : ℝ => (diskCount C R : ℝ) / (Real.pi * R ^ 2))
    atTop (𝓝 1)

def AdmissiblePotential (g : ℝ → ℝ) : Prop :=
```

```
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧
  (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
  ∀ (r : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r g t

def diskEnergy (g : ℝ → ℝ) (C : Set Plane) (R : ℝ) : ℝ≥0∞ :=
  (diskCount C R : ℝ≥0∞)⁻¹ *
    ∑ x ∈ diskPoints C R, ∑ y ∈ (diskPoints C R).erase x,
      ENNReal.ofReal (g (‖x - y‖ ^ 2))

def energy (g : ℝ → ℝ) (C : Set Plane) : ℝ≥0∞ :=
  Filter.liminf (diskEnergy g C) atTop
-- ... lines 40-63 (namespace OAI.AtomicTriangular)
def b : ℝ := Real.sqrt 3 / 2

def triangularPoint (jk : ℤ × ℤ) : Plane :=
  (Real.sqrt b)⁻¹ •
    (EuclideanSpace.single 0 ((jk.1 : ℝ) + (jk.2 : ℝ) / 2) +
      EuclideanSpace.single 1 ((jk.2 : ℝ) * b))

def A : Set Plane := Set.range triangularPoint
-- ...
def latticeEnergy (g : ℝ → ℝ) : ℝ≥0∞ :=
  ∑' a : {x : Plane // x ∈ A ∧ x ≠ 0}, ENNReal.ofReal (g (‖a.val‖ ^ 2))
```

Two details of the formal definitions are worth noticing before any proof. First, `diskPoints` returns the empty set when $C \cap B_R$ is infinite, so every useful statement carries the hypothesis `LocallyFinite C`. Second, `iteratedDeriv` is the iterated *global* derivative of $g : \mathbb{R} \to \mathbb{R}$; at a point $t > 0$ it agrees with the classical derivative because g is smooth on the open set $(0, \infty)$ around t (this is item "Potential class" of the statement-fidelity audit, `results/discovery/statement_fidelity.md`) **[Tier C]**.

2.2 Result 0: the upstream theorem

Informal statement. Among all locally finite planar configurations of density one, the triangular lattice has the least lower energy, simultaneously for every admissible potential.

Theorem 2.4 (Universal energy minimum [1])**.** *Let g be admissible and C locally finite of density one. Then*

$$\mathcal{E}_g(A) \le E_g(C) \qquad \text{and} \qquad \mathcal{E}_g(A) = E_g(A).$$

Status: **[Tier A]** for the Lean statement below (AI-generated proof, kernel-checked and Comparator-accepted by us under the pinned toolchain, footprint {`propext`, `Classical.choice`, `Quot.sound`}; not peer reviewed); **[Tier L]** for the manuscript's proof.

What is claimed, and what is not.

- The theorem is an *energy comparison*: it identifies the minimum *value* $\mathcal{E}_g(A)$ of the lower energy. In upstream's words, "It identifies the minimum value, not all minimizers." **[Tier L]** Section 2.6 shows, in Lean, that the minimizer is indeed not unique.
- Upstream does *not* claim the per-potential form of the Cohn–Kumar conjecture, in which each potential has a sharp auxiliary function of its own. That strong form is **open**. **[Tier L]**
- The competitor C need not be periodic; it only has to be locally finite with density one.

- Energies are lower energies over *closed* disks centred at 0. Section 2.7 shows that open disks give the same density and the same lower energy.

- The Riesz endpoint $s = 2$ and renormalized (Coulomb-type) energies are not covered. Renormalized energies are treated only in the upstream manuscript (its Section 8), and not in Lean anywhere. **[Tier L]**

2.2.1 Proof architecture (Tier L)

We give only the architecture of upstream's proof (upstream manuscript, Section 1), not its details. Everything in this subsection is **[Tier L]**.

Step 1: Gaussians. Use the normalisation $\hat{f}(\xi) = \int f(x)e^{-2\pi i x\cdot\xi}\,dx$, the Gaussian $G_\alpha(x) = e^{-\pi\alpha\|x\|^2}$ ($\alpha > 0$) and the dual lattice A^*. For every $\alpha > 0$ upstream constructs a radial Schwartz function f_α with

$$f_\alpha \le G_\alpha, \qquad \hat{f}_\alpha \ge 0, \qquad f_\alpha(a) = G_\alpha(a)\ (a \in A \setminus \{0\}), \qquad \hat{f}_\alpha(w) = 0\ (w \in A^* \setminus \{0\}).$$

So f_α is a *minorant* of the potential (not a majorant), with nonnegative Fourier transform.

Step 2: the linear-programming bound. For every competitor C of centred density one,

$$E_{G_\alpha}(C) \;\ge\; \hat{f}_\alpha(0) - f_\alpha(0).$$

This is a Cohn–Elkies-type bound [3, 2]. The competitor does not need to be periodic; upstream compares the disk sums with the area of the disk and controls the error uniformly.

Step 3: equality on the lattice. Poisson summation on A (covolume one) reads $\sum_{a\in A} f_\alpha(a) = \sum_{w\in A^*} \hat{f}_\alpha(w)$. The interpolation conditions turn it into $f_\alpha(0) + \sum_{a\neq 0} G_\alpha(a) = \hat{f}_\alpha(0)$. Hence

$$\hat{f}_\alpha(0) - f_\alpha(0) = \sum_{a\in A\setminus\{0\}} e^{-\pi\alpha\|a\|^2} = \mathcal{E}_{G_\alpha}(A),$$

and the bound of Step 2 is attained by A.

Step 4: the construction of f_α. A sine product with double zeros encodes the periodic contact set. Multiplying it by double-pole and simple-pole terms gives entire functions with prescribed values and slopes. Compactly supported spectral measures and Gaussian damping follow. Finite reference lists are handled by positive quadrature, with certified matrix and polynomial inequalities (the verification scripts use `python-flint` ball arithmetic). The construction does *not* use modular forms; modular forms are the method of [2] in dimensions 8 and 24.

Step 5: from Gaussians to all admissible g. An admissible g is a positive mixture of Gaussians in $t = \|x\|^2$: $g(t) = \int_{[0,\infty)} e^{-\pi\alpha t}\,d\mu(\alpha)$, i.e. $g(\|x\|^2) = \int G_\alpha(x)\,d\mu(\alpha)$ in the normalisation of Step 1 (Bernstein–Hausdorff–Widder, after rescaling α by π; upstream gives its own finite-difference proof). The measure may charge $\alpha = 0$, i.e. contain a constant, whose energy is infinite at density one (upstream lemma `energy_one`). Integrating the Gaussian bounds needs monotone convergence and Tonelli, and, because lim inf is not linear, a separate lower-limit argument; the endpoint and infinite-energy cases are treated in upstream's Section 5.

Lean 4 (upstream proof, top level, verbatim)

The challenge file states the theorem with `sorry` by design; the proof lives in `OAI/Analysis/Triangular/Energy/Universal.lean`, and Comparator certifies that the two statements agree. We quote the 26-line proof file from our copy of the upstream release `fd4aeeb2` (the triangular files are unchanged from `adc7f124`).

```
-- ComparatorChallenges/TriangularEnergy.lean, lines 65-69
/-- Universal energy minimality for smooth completely monotone potentials, with
    attainment. -/
theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := by
  sorry

-- OAI/Analysis/Triangular/Energy/Universal.lean, lines 7-22
namespace AtomicTriangular
/-- The triangular lattice minimizes every nonnegative completely monotone energy. -/
theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (_hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := by
  have hb (α : ℝ) (hα : 0<α) :
      TriangularUniversal.latticeEnergy (TriangularUniversal.gaussPot α)≤
        TriangularUniversal.energy (TriangularUniversal.gaussPot α) C := by
    rw [←latticeEnergy_eq_support]
    exact gaussian_energy_bound hd α hα
  have hu := TriangularUniversal.universal_from_gaussian_bounds hd hb hg
  constructor
  · rw [latticeEnergy_eq_support]
    exact hu.1
  · rw [latticeEnergy_eq_support,A_eq_support]
    exact hu.2
```

How the Lean proof mirrors the architecture. The top-level proof has exactly two moves. The local fact `hb` is Steps 1–4: a lower bound for every Gaussian potential `gaussPot Îś`, obtained from `gaussian_energy_bound`. The call `universal_from_gaussian_bounds` is Step 5: it passes from Gaussian bounds to every admissible g and also yields attainment. The rewrites `latticeEnergy_eq_support` and `A_eq_support` translate between the challenge file's description of the lattice and the one used inside the library. Notice that the hypothesis `_hC : LocallyFinite C` is unused in the proof (the leading underscore says so): local finiteness is part of the *statement* that Comparator checks, and it matters for the meaning of `diskPoints`, but the bound itself does not need it **[Tier C]** (the hypothesis is unused in upstream's proof term; no theorem without it is stated).

2.3 Result D1: Riesz potentials

Informal statement. For every $s > 0$ the triangular lattice minimizes the Riesz energy $\sum |x - y|^{-s}$ among locally finite configurations of density one. The statement has content only for $s > 2$.

The Riesz kernel at distance r is r^{-s}, so in the variable $t = r^2$ the potential is

$$g_s(t) = t^{-s/2}.$$

In Lean this is `TriangularRiesz.riesz s := fun t => t ^ (-(s / 2))`, and the control `riesz_sq` checks $g_s(r^2) = r^{-s}$ for $r > 0$ **[Tier A]**.

Lemma 2.5 (Admissibility)**.** *For every $s > 0$, g_s is admissible.*

Human proof. Write $\sigma = s/2 > 0$. We check the three conditions of Definition 2.2.
Smoothness. On $(0, \infty)$, $t \mapsto t^{-\sigma} = e^{-\sigma \log t}$ is a composition of smooth functions.
Positivity. For $t > 0$, $t^{-\sigma} > 0$.
Complete monotonicity. Differentiating r times,

$$g_s^{(r)}(t) = (-\sigma)(-\sigma - 1) \cdots (-\sigma - r + 1)\, t^{-\sigma - r}.$$

The coefficient is the *descending* Pochhammer symbol $(x)_r^{\downarrow} = x(x-1)\cdots(x-r+1)$ at $x = -\sigma$. Pulling a factor -1 out of each of the r factors gives

$$(-\sigma)_r^{\downarrow} = (-1)^r\, \sigma(\sigma+1)\cdots(\sigma+r-1) = (-1)^r (\sigma)_r^{\uparrow},$$

with the *ascending* Pochhammer symbol $(\sigma)_r^{\uparrow}$. Therefore

$$(-1)^r g_s^{(r)}(t) = (-1)^r (-1)^r (\sigma)_r^{\uparrow}\, t^{-\sigma - r} = (\sigma)_r^{\uparrow}\, t^{-\sigma - r}.$$

Each factor $\sigma + i$ $(0 \le i < r)$ is positive because $\sigma > 0$, so $(\sigma)_r^{\uparrow} > 0$, and $t^{-\sigma - r} > 0$. The product is positive, in particular ≥ 0. □

Theorem 2.6 (D1, Riesz optimality)**.** *For every $s > 0$ and every locally finite C of density one, $\sum_{a \in A \setminus \{0\}} \|a\|^{-s} \le E_{g_s}(C)$ in $[0, \infty]$.*

Human proof. Apply Theorem 2.4 with $g = g_s$, which is admissible by Lemma 2.5, and keep the first conjunct. □

Status: **[Tier A]** for the Lean statement (conditional on upstream and on the fidelity of the definitions). The upstream manuscript already says informally that the theorem "includes every inverse power $g(t) = t^{-p}$ with $p > 0$"; D1 is therefore the kernel-checked form of a stated special case, not a new mathematical claim.

Caution (informative only for $s > 2$)

For $0 < s \le 2$ the left-hand side is ∞, so Theorem 2.6 only says $E_{g_s}(C) = \infty$. The reason is a counting argument **[Tier C]**: the lattice has about $2\pi k$ points with $k \le \|a\| < k + 1$, each contributing about k^{-s}, and $\sum_k k^{1-s} = \infty$ for $s \le 2$ (Exercise 2.2). We have *not* proved this divergence in Lean. For $s > 2$ the left-hand side is finite (D1b, §2.4), so the theorem is not the trivial $\infty \le \infty$. The endpoint $s = 2$ is open.

Lean 4 (`formal_cert/Triangular/Riesz.lean`, verbatim)

```
/-- Riesz potential in the variable `t = r^2`. -/
noncomputable def riesz (s : ℝ) : ℝ → ℝ := fun t => t ^ (-(s / 2))

theorem riesz_admissible {s : ℝ} (hs : 0 < s) : AdmissiblePotential (riesz s) := by
  refine ⟨?_, ?_, ?_⟩
  · intro x hx
    exact (Real.contDiffAt_rpow_const_of_ne (ne_of_gt hx)).contDiffWithinAt
  · intro t ht
    exact (Real.rpow_pos_of_pos ht _).le
  · intro r t ht
    have ha : 0 < s / 2 := by positivity
    rw [iteratedDeriv_eq_iterate]
    unfold riesz
    rw [Real.iter_deriv_rpow_const]
    have hasc := ascPochhammer_eval_neg_eq_descPochhammer (R := ℝ) (-(s / 2)) r
```

```
    rw [neg_neg] at hasc
    have hdesc : (descPochhammer ℝ r).eval (-(s / 2)) = (-1) ^ r * (ascPochhammer ℝ r).
    eval (s / 2) := by
      have h1 : ((-1 : ℝ) ^ r) * ((-1) ^ r) = 1 := by rw [← mul_pow]; norm_num
      calc (descPochhammer ℝ r).eval (-(s / 2))
          = ((-1) ^ r * (-1) ^ r) * (descPochhammer ℝ r).eval (-(s / 2)) := by rw [h1,
    one_mul]
        _ = (-1) ^ r * (ascPochhammer ℝ r).eval (s / 2) := by rw [hasc]; ring
    rw [hdesc]
    have hpos := ascPochhammer_pos r (s / 2) ha
    have hpow := Real.rpow_pos_of_pos ht (-(s / 2) - r)
    have h1 : ((-1 : ℝ) ^ r) * ((-1) ^ r) = 1 := by rw [← mul_pow]; norm_num
    calc (0 : ℝ) ≤ (ascPochhammer ℝ r).eval (s / 2) * t ^ (-(s / 2) - r) := by
    positivity
      _ = (-1) ^ r * ((-1) ^ r * (ascPochhammer ℝ r).eval (s / 2) * t ^ (-(s / 2) - r))
    := by
        rw [← mul_assoc, ← mul_assoc, h1, one_mul]

theorem triangular_riesz_optimal {s : ℝ} (hs : 0 < s) (C : Set Plane) (hC :
    LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy (riesz s) ≤ energy (riesz s) C :=
  (universal_energy_minimum (riesz s) C (riesz_admissible hs) hC hd).1
```

How the Lean proof mirrors the human proof. The anonymous constructor `â§Ĺ?_, ?_, ?_â§Ľ` opens the three conditions in the order of the human proof. Smoothness is the Mathlib fact that $x \mapsto x^p$ is smooth away from 0 (`Real.contDiffAt_rpow_const_of_ne`); positivity is `Real.rpow_pos_of_pos`. For complete monotonicity, `iteratedDeriv_eq_iterate` turns the r-th derivative into the r-fold iterate of `deriv`, and `Real.iter_deriv_rpow_const` is the displayed formula with the descending Pochhammer polynomial evaluated at $-\sigma$. The identity $(-\sigma)^{\downarrow}_r = (-1)^r(\sigma)^{\uparrow}_r$ is `ascPochhammer_eval_neg_eq_descPochhammer` read backwards; it is packaged as `hdesc`, using $(-1)^r(-1)^r = 1$ (`h1`). Positivity of the ascending symbol at $\sigma > 0$ is `ascPochhammer_pos`, and the final `calc` cancels the two signs, exactly as in the last display of the human proof. The optimality theorem is one line: the first projection `.1` of the upstream theorem.

Lean 4 (negative control, `Controls.lean`, verbatim)

```
theorem riesz_neg_two_not_admissible : ¬ AdmissiblePotential (TriangularRiesz.riesz (-2)
    ) := by
  intro h
  have h3 := h.2.2 1 1 one_pos
  have hfun : TriangularRiesz.riesz (-2) = fun t => t := by
    funext t
    unfold TriangularRiesz.riesz
    norm_num
  rw [hfun] at h3
  simp at h3 <;> linarith
```

The hypothesis $s > 0$ cannot be dropped: for $s = -2$ the "potential" is $g(t) = t$, whose first derivative is $+1$, so the condition $(-1)^1 g'(1) \geq 0$ fails. The control picks exactly that instance (`h.2.2 1 1`: $r = 1$, $t = 1$). It shows that `AdmissiblePotential` is not vacuous **[Tier A]**.

2.4 Result D1b: the Riesz lattice energy is finite for $s > 2$

Informal statement. For $s > 2$ the Epstein-type sum $\sum_{a \in A \setminus \{0\}} \|a\|^{-s}$ converges.

Lemma 2.7 (Norm identity)**.** $\|p(j,k)\|^2 = \dfrac{1}{b}\,(j^2 + jk + k^2)$ *for all* $(j,k) \in \mathbb{Z}^2$.

Human proof. $\|p(j,k)\|^2 = b^{-1}\big((j+k/2)^2 + k^2 b^2\big) = b^{-1}\big(j^2 + jk + \tfrac{1}{4}k^2 + \tfrac{3}{4}k^2\big) = b^{-1}(j^2 + jk + k^2)$, using $b^2 = 3/4$. □

Theorem 2.8 (D1b)**.** *For every* $s > 2$, $\mathcal{E}_{g_s}(A) = \sum_{a \in A \setminus \{0\}} \|a\|^{-s} < \infty$.

Human proof. Let $\sigma = s/2 > 1$ and $Q(j,k) = j^2 + jk + k^2$.

Step 1 (summability over $\mathbb{Z}^2$). Since $jk \geq -\frac{1}{2}(j^2 + k^2)$, we have $Q(j,k) \geq \frac{1}{2}(j^2 + k^2)$, so Q vanishes only at $(0,0)$ and $Q^{-\sigma} \leq 2^{\sigma}(j^2 + k^2)^{-\sigma}$ elsewhere. The $8n$ integer points with $\max(|j|, |k|) = n \geq 1$ satisfy $j^2 + k^2 \geq n^2$, so

$$\sum_{(j,k) \neq (0,0)} Q(j,k)^{-\sigma} \leq 2^{\sigma} \sum_{n \geq 1} 8n \cdot n^{-2\sigma} < \infty \qquad (2\sigma - 1 > 1).$$

Step 2 (scaling). By Lemma 2.7, $\|p(j,k)\|^{-s} = (Q/b)^{-\sigma} = b^{\sigma} Q^{-\sigma}$, a constant multiple of a summable family.

Step 3 (reindexing). Each $a \in A \setminus \{0\}$ is $p(j,k)$ for some index; choose one, $\iota(a)$. Then $p(\iota(a)) = a$, so ι is injective (if $\iota(a) = \iota(a')$ then $a = p(\iota(a)) = p(\iota(a')) = a'$). A nonnegative summable family stays summable when restricted along an injective map, so $\sum_a \|a\|^{-s} = \sum_a b^{\sigma} Q(\iota(a))^{-\sigma} < \infty$. □

Lean 4 (`formal_cert/Triangular/RieszFinite.lean`, **verbatim, shortened**)

```
lemma norm_sq_triangularPoint (j k : ℤ) :
    ‖triangularPoint (j, k)‖ ^ 2 = (1 / b) * ((j : ℝ) ^ 2 + (j : ℝ) * k + (k : ℝ) ^ 2)
    := by
  rw [EuclideanSpace.real_norm_sq_eq]
  -- ...
  field_simp
  rw [hsb, hb2]
  ring

/-- A chosen index of a lattice point. -/
noncomputable def toIdx (a : {x : Plane // x ∈ A ∧ x ≠ 0}) : ℤ × ℤ := Classical.choose
     a.2.1

lemma toIdx_spec (a : {x : Plane // x ∈ A ∧ x ≠ 0}) : triangularPoint (toIdx a) = a.val
     :=
  Classical.choose_spec a.2.1

lemma toIdx_injective : Function.Injective toIdx := by
  intro a a' h
  apply Subtype.ext
  rw [← toIdx_spec a, ← toIdx_spec a', h]

/-- **D1b.** For every `s > 2` the Riesz `s`-energy of the density-one triangular
    lattice is finite. -/
theorem latticeEnergy_riesz_lt_top {s : ℝ} (hs : 2 < s) : latticeEnergy (riesz s) < ⊤
    := by
  have hσ : 1 < s / 2 := by linarith
  set Q : ℤ × ℤ → ℝ := fun jk => (jk.1 : ℝ) ^ 2 + (jk.1 : ℝ) * jk.2 + (jk.2 : ℝ) ^ 2
    with hQ
  -- ...
```

```
-- Σ Q^(-σ) over ℤ² from openai/math, via (j,k) ↦ (j,-k)
have hQsum : Summable (fun jk : ℤ × ℤ => Q jk ^ (-(s / 2))) := by
  have hn := LatticeSummability.summable_normForm_neg_rpow hσ
  -- ...
have hh : Summable (fun jk : ℤ × ℤ => ((1 / b) * Q jk) ^ (-(s / 2))) := by
  -- ...
  exact hQsum.mul_left _
-- ...
have hfsum : Summable f := by
  have := hh.comp_injective toIdx_injective
  -- ...
rw [← ENNReal.ofReal_tsum_of_nonneg hf0 hfsum]
exact ENNReal.ofReal_lt_top
```

(The elided lines include an equivalence of $\mathbb{Z} \times \mathbb{Z}$ written with a Unicode arrow that this book's listing style cannot typeset; read the file for the full text.)

How the Lean proof mirrors the human proof. The three steps appear in the same order. Step 1 is not reproved. The openai/math library already contains the lemma

`LatticeSummability.summable_normForm_neg_rpow`,

summability of $(x^2 - xy + y^2)^{-\sigma}$ for $\sigma > 1$, and the file transports it to Q by the substitution $(j,k) \mapsto (j,-k)$, which turns $j^2 - jk + k^2$ into $j^2 + jk + k^2$. (In Lean's real powers $0^{-\sigma} = 0$, so the term at $(0,0)$ does no harm and the sum may run over all of $\mathbb{Z}^2$.) Step 2 is `hh`, a constant multiple via `Real.mul_rpow`. Step 3 is `toIdx`: an index chosen with `Classical.choose`, whose injectivity is proved exactly as in the human proof, from $p \circ \iota = \mathrm{id}$, without ever proving that p itself is injective. `Summable.comp_injective` restricts the summable family, and `ENNReal.ofReal_tsum_of_nonneg` moves the real sum into $[0,\infty]$, where a real number is $< \infty$.

2.5 Result D2: every density

Informal statement. At every density $\rho > 0$ the triangular lattice rescaled to density ρ, $\rho^{-1/2}A$, is optimal for every admissible potential, and it attains the bound.

Theorem 2.9 (D2). *Let $\rho > 0$, h admissible and C locally finite of density ρ. Then*

$$\sum_{a \in A \setminus \{0\}} h\Big(\frac{\|a\|^2}{\rho}\Big) \le E_h(C), \qquad \text{and} \qquad E_h(\rho^{-1/2}A) = \sum_{a \in A \setminus \{0\}} h\Big(\frac{\|a\|^2}{\rho}\Big).$$

Consequently $\rho^{-1/2}A$ is locally finite, has density ρ, and $E_h(\rho^{-1/2}A) \le E_h(C)$.

The proof consists of four scaling lemmas. Throughout, $c > 0$ and $cC = \{cx : x \in C\}$.

Lemma 2.10 (Scaling counts). *If C is locally finite, so is cC, and $N_R(cC) = N_{R/c}(C)$.*

Human proof. $cx \in B_R \iff \|x\| \le R/c$, so $cC \cap B_R = c\,(C \cap B_{R/c})$, the image of a finite set under the injective map $x \mapsto cx$. It is finite and has the same number of elements. □

Lemma 2.11 (Scaling density). *If C is locally finite with density ρ, then $\sqrt{\rho}\,C$ has density one.*

Human proof. With $c = \sqrt{\rho}$ and Lemma 2.10, for $R > 0$

$$\frac{N_R(cC)}{\pi R^2} = \frac{N_{R/c}(C)}{\pi R^2} = \frac{1}{\rho} \cdot \frac{N_{R/c}(C)}{\pi (R/c)^2} \xrightarrow[R \to \infty]{} \frac{1}{\rho} \cdot \rho = 1,$$

because $R/c \to \infty$ when $R \to \infty$. □

Lemma 2.12 (Scaling energies)**.** *If C is locally finite, $E_g(cC; R) = E_{g(c^2\,\cdot)}(C; R/c)$ for every R, and $E_g(cC) = E_{g(c^2\,\cdot)}(C)$.*

Human proof. The disk sets correspond under $x \mapsto cx$ (Lemma 2.10), the counts agree, and $g(\|cx - cy\|^2) = g(c^2\|x - y\|^2)$, which proves the first identity term by term. For the second, put $F(R) = E_{g(c^2\,\cdot)}(C; R)$. The map $R \mapsto R/c$ is an increasing bijection of $\mathbb{R}$, so it maps the filter of neighbourhoods of $+\infty$ onto itself, and $\liminf_{R\to\infty} F(R/c) = \liminf_{R\to\infty} F(R)$. □

Lemma 2.13 (Rescaled potentials)**.** *If g is admissible and $c > 0$, then $t \mapsto g(ct)$ is admissible.*

Human proof. Smoothness and positivity are preserved because $t > 0 \Rightarrow ct > 0$. By the chain rule and induction on r, $\frac{d^r}{dt^r} g(ct) = c^r g^{(r)}(ct)$, so $(-1)^r \frac{d^r}{dt^r} g(ct) = c^r \cdot (-1)^r g^{(r)}(ct) \geq 0$. □

Human proof. [Human proof of Theorem 2.9] *Bound.* Let $c = \sqrt{\rho}$ and $h_\rho(t) = h(t/\rho)$, admissible by Lemma 2.13. The configuration cC is locally finite with density one (Lemmas 2.10 and 2.11). Theorem 2.4 gives $\mathcal{E}_{h_\rho}(A) \leq E_{h_\rho}(cC)$. By Lemma 2.12, $E_{h_\rho}(cC) = E_{h_\rho(c^2\,\cdot)}(C)$, and $h_\rho(c^2 t) = h(\rho t/\rho) = h(t)$. So $\mathcal{E}_{h_\rho}(A) \leq E_h(C)$, which is the bound.

Attainment. By Lemma 2.12 with $c = \rho^{-1/2}$, $E_h(\rho^{-1/2}A) = E_{h(\rho^{-1}\,\cdot)}(A) = E_{h_\rho}(A)$, and the second conjunct of Theorem 2.4 for h_ρ gives $E_{h_\rho}(A) = \mathcal{E}_{h_\rho}(A)$.

Density of $\rho^{-1/2}A$. With $c = \rho^{-1/2}$, $N_R(cA)/(\pi R^2) = \rho \cdot N_{R\sqrt{\rho}}(A)/(\pi(R\sqrt{\rho})^2) \to \rho$. □

Lean 4 (`formal_cert/Triangular/AnyDensity.lean`, **verbatim, shortened**)

```
/-- ‘C‘ has density ‘ρ‘. -/
def DensityRho (ρ : ℝ) (C : Set Plane) : Prop :=
  Tendsto (fun R : ℝ => (diskCount C R : ℝ) / (Real.pi * R ^ 2)) atTop (𝓝 ρ)

lemma admissible_comp_mul {g : ℝ → ℝ} (hg : AdmissiblePotential g) {c : ℝ} (hc : 0 < c
    ) :
    AdmissiblePotential (fun t => g (c * t)) := by
  -- ...
lemma diskCount_smul {c : ℝ} (hc : 0 < c) {C : Set Plane} (hC : LocallyFinite C) (R : ℝ
    ) :
    diskCount (c • C) R = diskCount C (R / c) := by
  -- ...
lemma density_smul {ρ : ℝ} (hρ : 0 < ρ) {C : Set Plane} (hC : LocallyFinite C) (hd :
    DensityRho ρ C) :
    DensityOne (Real.sqrt ρ • C) := by
  -- ...
lemma diskEnergy_smul {c : ℝ} (hc : 0 < c) (g : ℝ → ℝ) {C : Set Plane} (hC :
    LocallyFinite C) (R : ℝ) :
    diskEnergy g (c • C) R = diskEnergy (fun t => g (c ^ 2 * t)) C (R / c) := by
  -- ... (liminf_comp_div is quoted in full below)
lemma energy_smul {c : ℝ} (hc : 0 < c) (g : ℝ → ℝ) {C : Set Plane} (hC : LocallyFinite
     C) :
    energy g (c • C) = energy (fun t => g (c ^ 2 * t)) C := by
  -- ...

/-- **D2.** Universal optimality at every density ‘ρ > 0‘. -/
theorem universal_any_density (h : ℝ → ℝ) (hh : AdmissiblePotential h) {ρ : ℝ} (hρ : 0
     < ρ) (C : Set Plane)
    (hC : LocallyFinite C) (hd : DensityRho ρ C) :
    latticeEnergy (fun t => h (ρ^(-1) * t)) ≤ energy h C := by
  have hs : 0 < Real.sqrt ρ := Real.sqrt_pos.2 hρ
  have hg := admissible_comp_mul hh (inv_pos.2 hρ)
  have key := (universal_energy_minimum (fun t => h (ρ^(-1) * t)) (Real.sqrt ρ • C) hg (
    locallyFinite_smul hs hC)
    (density_smul hρ hC hd)).1
  rw [energy_smul hs _ hC] at key
```

```
  have e : (fun t => (fun t => h (ρ⁻¹ * t)) (Real.sqrt ρ ^ 2 * t)) = h := by
    funext t
    simp only
    rw [Real.sq_sqrt hρ.le, ← mul_assoc, inv_mul_cancel₀ hρ.ne', one_mul]
  rw [e] at key
  exact key

theorem attained (h : ℝ → ℝ) (hh : AdmissiblePotential h) {ρ : ℝ} (hρ : 0 < ρ) :
    energy h ((Real.sqrt ρ)⁻¹ • A) = latticeEnergy (fun t => h (ρ⁻¹ * t)) := by
  have hs : 0 < Real.sqrt ρ := Real.sqrt_pos.2 hρ
  rw [energy_smul (inv_pos.2 hs) h A_lf, inv_pow, Real.sq_sqrt hρ.le]
  exact ((universal_energy_minimum _ A (admissible_comp_mul hh (inv_pos.2 hρ)) A_lf A_d1
    ).2).symm

/-- **D2, optimality form.** At every density `ρ > 0` the rescaled lattice `ρ^{-1/2} A`
    has density `ρ`, and its energy is
`≤` that of every locally finite density-`ρ` configuration. -/
theorem triangular_optimal_any_density (h : ℝ → ℝ) (hh : AdmissiblePotential h) {ρ : ℝ
    } (hρ : 0 < ρ) :
    LocallyFinite ((Real.sqrt ρ)⁻¹ • A) ∧ DensityRho ρ ((Real.sqrt ρ)⁻¹ • A) ∧
      ∀ C : Set Plane, LocallyFinite C → DensityRho ρ C → energy h ((Real.sqrt ρ)⁻¹ •
    A) ≤ energy h C :=
  ⟨locallyFinite_smul (inv_pos.2 (Real.sqrt_pos.2 hρ)) A_lf, scaled_density hρ, fun C hC
     hd => by
    rw [attained h hh hρ]; exact universal_any_density h hh hρ C hC hd⟩
```

The reparametrisation of the lim inf (Lemma 2.12, second part) is the following, verbatim:

```
lemma liminf_comp_div {c : ℝ} (hc : 0 < c) (F : ℝ → ENNReal) :
    liminf (fun R => F (R / c)) atTop = liminf F atTop := by
  have hmap : map (fun R : ℝ => R / c) atTop = atTop := by
    have h1 := (OrderIso.mulLeft₀ c⁻¹ (inv_pos.2 hc)).map_atTop
    have h2 : ((OrderIso.mulLeft₀ c⁻¹ (inv_pos.2 hc)) : ℝ → ℝ) = fun R => R / c := by
      funext R
      exact (div_eq_inv_mul R c).symm
    rw [← h2]
    exact h1
  rw [show (fun R => F (R / c)) = F ∘ (fun R : ℝ => R / c) from rfl]
  unfold Filter.liminf
  rw [← Filter.map_map, hmap]
```

How the Lean proof mirrors the human proof. The four lemmas appear under their own names: `diskCount_smul` (counts), `density_smul` (density), `diskEnergy_smul` together with `liminf_comp_div` and `energy_smul` (energies), and `admissible_comp_mul` (potentials; its proof is the induction $\frac{d^r}{dt^r} g(ct) = c^r g^{(r)}(ct)$, stated as `iterate_deriv_comp_mul`). The main proof is the human "Bound" paragraph line by line: `key` is the upstream bound for h_ρ and $\sqrt{\rho}\,C$; `energy_smul` rewrites the energy of $\sqrt{\rho}\,C$; `e` is the computation $h_\rho(c^2 t) = h(t)$. The lim inf lemma shows a typical formalisation move: "an increasing bijection maps $+\infty$ to $+\infty$" becomes `OrderIso.map_atTop`, and the lim inf of a composition becomes the lim inf along the image filter (`Filter.map_map`).

2.5.1 Why the factor ρ^{-1} is needed: a negative control

It is tempting to state D2 as "the density-one lattice energy $\mathcal{E}_h(A)$ is a lower bound at every density". That is false. At density $\rho < 1$ the points are farther apart, so repulsive energies are smaller.

Proposition 2.14 (`unscaled_false`). *It is not true that $\mathcal{E}_{g_4}(A) \le E_{g_4}(C)$ for every locally finite C of density $1/4$.*

Human proof. Take $C = 2A = (1/4)^{-1/2}A$, which has density $1/4$ by Theorem 2.9. By attainment, $E_{g_4}(2A) = \sum_{a\neq 0}(4\|a\|^2)^{-2} = \frac{1}{16}\sum_{a\neq 0}\|a\|^{-4}$. The sum is finite (D1b with $s = 4$) and positive, so $E_{g_4}(2A) < \mathcal{E}_{g_4}(A)$. □

In Lean the strict inequality is proved termwise rather than by the factor $1/16$: every term $((1/4)^{-1}\|a\|^2)^{-2}$ is $\le$, and the term at $p(1,0)$ is $<$, the corresponding term $(\|a\|^2)^{-2}$; since the right-hand sum is finite (`latticeEnergy_riesz_lt_top`), `ENNReal.tsum_lt_tsum` gives a strict inequality of the sums.

Lean 4 (`AnyDensity.lean`, verbatim, shortened)

```
theorem unscaled_false :
    ¬ (∀ C : Set Plane, LocallyFinite C → DensityRho (1 / 4) C →
        latticeEnergy (TriangularRiesz.riesz 4) ≤ energy (TriangularRiesz.riesz 4) C) :=
     by
  intro H
  have hρ : (0 : ℝ) < 1 / 4 := by norm_num
  have key := H _ (locallyFinite_smul (inv_pos.2 (Real.sqrt_pos.2 hρ)) A_lf) (
    scaled_density hρ)
  rw [attained _ (TriangularRiesz.riesz_admissible (by norm_num)) hρ] at key
  -- key : Σ ofReal(t^{-2}) ≤ Σ ofReal((4t)^{-2}); but (4t)^{-2} < t^{-2} termwise and
    the sum is finite
  have hfin := TriangularRiesz.latticeEnergy_riesz_lt_top (s := 4) (by norm_num)
  -- ...
  have strict := ENNReal.tsum_lt_tsum hfin2 hle (hlt ⟨_, hpt⟩)
  exact absurd key (not_le.2 strict)
```

For the Riesz family the correct bound has a closed form. Since $g_s(t/\rho) = \rho^{s/2}g_s(t)$,

$$\mathcal{E}_{g_s(\rho^{-1}\cdot)}(A) = \rho^{s/2}\,\mathcal{E}_{g_s}(A), \qquad \text{so} \qquad \rho^{s/2}\sum_{a\in A\setminus\{0\}}\|a\|^{-s} \le E_{g_s}(C)$$

for every locally finite C of density ρ **[Tier A]**. In Lean these are `riesz_lattice_homogeneous` and `riesz_any_density_homogeneous` in `Controls.lean`. At $\rho = 1/4$, $s = 4$ the factor is $1/16$, as in the proof above. The control `rho_one` checks that at $\rho = 1$ D2 gives back the density-one statement, and `densityRho_one_iff` that `DensityRho 1` *is* `DensityOne` (by `Iff.rfl`) **[Tier A]**.

2.6 Result D3: universal optimality, and a second minimizer

2.6.1 The definition

Definition 2.15 (Universally optimal, lower-energy form). A configuration C is *universally optimal at density ρ* if C is locally finite, has density ρ, and $E_g(C) \le E_g(C')$ for every admissible g and every locally finite C' of density ρ.

This transcribes Definition 1.3 of [2] into upstream's notions of density and lower energy. It is *weaker* than CKMRV's notion: CKMRV also require the energy of C itself to exist as a limit, while here only the lim inf enters **[Tier C]**. A Lean theorem about `UniversallyOptimal` therefore says nothing about whether the limit exists.

Theorem 2.16 (`triangular_universally_optimal`). *For every $\rho > 0$, $\rho^{-1/2}A$ is universally optimal at density ρ.*

Human proof. This is the last sentence of Theorem 2.9, read against Definition 2.15. □

Lean 4 (`formal_cert/Triangular/UniversalOptimality.lean`, **verbatim**)

```
/-- CKMRV Definition 1.3, in upstream's density and lower-energy notions. -/
def UniversallyOptimal (ρ : ℝ) (C : Set Plane) : Prop :=
  LocallyFinite C ∧ DensityRho ρ C ∧
    ∀ g : ℝ → ℝ, AdmissiblePotential g → ∀ C' : Set Plane, LocallyFinite C' →
    DensityRho ρ C' →
      energy g C ≤ energy g C'

/-- **Universal optimality of the triangular lattice at every density** (conditional on
    upstream). -/
theorem triangular_universally_optimal {ρ : ℝ} (hρ : 0 < ρ) :
    UniversallyOptimal ρ ((Real.sqrt ρ)⁻¹ • A) := by
  refine ⟨locallyFinite_smul (inv_pos.2 (Real.sqrt_pos.2 hρ)) A_lf, scaled_density hρ, ?
    _⟩
  intro g hg C' hC' hd
  exact (triangular_optimal_any_density g hg hρ).2.2 C' hC' hd
```

2.6.2 The vacancy theorem

Let $V = A \setminus \{0\}$, the lattice with one point removed (a *vacancy*).

Theorem 2.17 (Vacancy). *(i) V is locally finite and has density one.*

(ii) $E_g(V) = \mathcal{E}_g(A)$ for every admissible g.

(iii) There is no isometry f of the plane with $f(A) = V$.

Hence V is universally optimal at density one but is not an isometric copy of A: universal optima, in the lower-energy sense, are not unique up to isometry among all locally finite configurations.

Human proof. Write $N = N_R(A)$, and assume $R \geq 0$, so that $0 \in A \cap B_R$ and $N \geq 1$.

(i) Counts and density. $V \subseteq A$, so V is locally finite, and $V \cap B_R = (A \cap B_R) \setminus \{0\}$ has $N - 1$ elements. Hence

$$\frac{N_R(V)}{\pi R^2} = \frac{N_R(A)}{\pi R^2} - \frac{1}{\pi R^2} \longrightarrow 1 - 0 = 1.$$

(ii), upper bound. Denote by $S_V(R)$ and $S_A(R)$ the two double sums in the disk energies. Every ordered pair (x, y) of distinct points of $V \cap B_R$ is also a pair of $A \cap B_R$, and all terms are ≥ 0, so $S_V(R) \leq S_A(R)$. Since $N \to \infty$ (density one forces $N \sim \pi R^2$), for large R we have $N \geq 2$ and

$$E_g(V; R) = \frac{S_V(R)}{N - 1} \leq \frac{N}{N - 1} \cdot \frac{S_A(R)}{N} = \kappa(R)\, E_g(A; R), \qquad \kappa(R) = \frac{N}{N - 1} \longrightarrow 1.$$

Now we take lower limits. If $L = \liminf_R E_g(A; R) = \infty$ there is nothing to prove. Otherwise choose $R_n \to \infty$ with $E_g(A; R_n) \to L$. Then $\kappa(R_n) E_g(A; R_n) \to 1 \cdot L$, so

$$E_g(V) = \liminf_R E_g(V; R) \leq \liminf_n \kappa(R_n) E_g(A; R_n) = L = E_g(A) = \mathcal{E}_g(A),$$

the last equality by the second conjunct of Theorem 2.4.

(ii), lower bound. By (i), V is an admissible competitor: locally finite of density one. The first conjunct of Theorem 2.4 gives $\mathcal{E}_g(A) \leq E_g(V)$. (Notice that this direction uses the universal theorem itself; we do not compute $E_g(V)$ from below by hand.)

(iii) No isometry. Suppose f is an isometry of $\mathbb{R}^2$ with $f(A) = V$. An isometry of a strictly convex real normed space, in particular of the Euclidean plane, is affine, so

$$f(2a - b) = 2f(a) - f(b) \qquad \text{for all } a, b.$$

The points $p(1,0)$ and $p(2,0)$ are nonzero (their squared norms are $1/b$ and $4/b$ by Lemma 2.7), so they lie in $V = f(A)$: $p(1,0) = f(a)$ and $p(2,0) = f(b)$ with $a, b \in A$. The lattice is closed under integer combinations, so $2a - b \in A$, and

$$f(2a - b) = 2p(1,0) - p(2,0) = p(0,0) = 0.$$

So $0 \in f(A) = V$, which contradicts $0 \notin V$. In words: $p(1,0)$ is the midpoint of 0 and $p(2,0)$; an affine bijection preserves midpoints and the lattice is closed under "reflect b through a", so the image of A must contain the point that the vacancy has removed.

Conclusion. By (i) and (ii), for every admissible g and every competitor C', $E_g(V) = \mathcal{E}_g(A) \le E_g(C')$; so V is universally optimal at density one. With (iii) this is the non-uniqueness statement. □

Lean 4 (`UniversalOptimality.lean`, verbatim, shortened)

```
def vacancy : Set Plane := A \ {0}

lemma diskCount_vacancy {R : ℝ} (hR : 0 ≤ R) : diskCount vacancy R = diskCount A R - 1
    := by
  -- ...
/-- The correction factor `N / (N - 1)`. -/
noncomputable def corr (R : ℝ) : ENNReal :=
  ENNReal.ofReal ((diskCount A R : ℝ) / ((diskCount A R : ℝ) - 1))

lemma corr_tendsto : Tendsto corr atTop (𝓝 1) := by
  -- ...
lemma diskEnergy_vacancy_le (g : ℝ → ℝ) :
    ∀ᶠ R in atTop, diskEnergy g vacancy R ≤ corr R * diskEnergy g A R := by
  -- ...

/-- **The vacancy configuration is also a minimiser**, for every admissible `g`. -/
theorem vacancy_energy (g : ℝ → ℝ) (hg : AdmissiblePotential g) : energy g vacancy =
    latticeEnergy g := by
  apply le_antisymm
  · have hA := (universal_energy_minimum g A hg A_lf A_d1).2
    have hc := corr_tendsto
    calc energy g vacancy = liminf (diskEnergy g vacancy) atTop := rfl
      _ ≤ liminf (corr * diskEnergy g A) atTop := liminf_le_liminf (
    diskEnergy_vacancy_le g)
      _ ≤ limsup corr atTop * liminf (diskEnergy g A) atTop := by
          apply ENNReal.liminf_mul_le <;> left <;> rw [hc.limsup_eq] <;> simp
      _ = latticeEnergy g := by
          rw [hc.limsup_eq, one_mul, hA]
          rfl
  · exact (universal_energy_minimum g vacancy hg vacancy_lf vacancy_density).1

theorem vacancy_not_isometric_image : ¬ ∃ f : Plane → Plane, Isometry f ∧ f '' A =
    vacancy := by
  rintro ⟨f, hf, hfA⟩
  set φ := hf.affineIsometryOfStrictConvexSpace
  have hφ : ∀ p, φ p = f p := fun p => rfl
  have key : ∀ a b : Plane, f ((2 : ℝ) • a - b) = (2 : ℝ) • f a - f b := by
    -- ...
  have hp : triangularPoint (1, 0) ∈ vacancy := by
    -- ...
```

```
  have hq : triangularPoint (2, 0) ∈ vacancy := by
    -- ...
  rw [← hfA] at hp hq
  obtain ⟨a, ha, hfa⟩ := hp
  obtain ⟨b, hb, hfb⟩ := hq
  have h0 : f ((2 : ℝ) • a - b) = 0 := by
    rw [key, hfa, hfb, ← triangularPoint_comb]
    convert triangularPoint_zero using 2 <;> norm_num
  have hmem : (0 : Plane) ∈ f '' A := ⟨_, A_comb ha hb, h0⟩
  rw [hfA] at hmem
  exact hmem.2 rfl

/-- **Non-uniqueness.** There is a universal optimum at density one that is not an
    isometric copy of the triangular
lattice: the minimiser is not determined, only the minimum value. -/
theorem universal_optimum_not_unique :
    ∃ C : Set Plane, UniversallyOptimal 1 C ∧ ¬ ∃ f : Plane → Plane, Isometry f ∧ f ''
     A = C :=
  ⟨vacancy, vacancy_universally_optimal, vacancy_not_isometric_image⟩
```

(Typographical note: in the file, the quantifier of `diskEnergy_vacancy_le` is Lean's "eventually" binder, written with a superscript letter f after ∀; the listing shows it as $\forall^{\mathsf{f}}$.)

How the Lean proof mirrors the human proof. `diskCount_vacancy` is the count $N-1$; `vacancy_density` is the limit $1-0$. The factor κ is `corr`, and `corr_tendsto` is $\kappa \to 1$. `diskEnergy_vacancy_le` is the inequality $E_g(V;R) \le \kappa(R) E_g(A;R)$ for all large R, built from `pair_sum_le` ($S_V \le S_A$, by dropping terms) and `coeff_eq` ($\frac{1}{N-1} = \frac{N}{N-1} \cdot \frac{1}{N}$ in $[0,\infty]$). The upper bound in `vacancy_energy` is a three-step `calc`. The human proof's subsequence argument is replaced by the library inequality `ENNReal.liminf_mul_le`, $\liminf(uv) \le \limsup u \cdot \liminf v$ in $[0,\infty]$, whose side conditions ($\limsup u \neq 0$ or $\liminf v \neq \infty$, and the symmetric one) are discharged because $\limsup \kappa = 1$. The lower bound is literally "apply the upstream theorem to V". In `vacancy_not_isometric_image`, `Isometry.affineIsometryOfStrictConvexSpace` is the statement "isometries are affine", `key` is $f(2a-b) = 2f(a) - f(b)$, `triangularPoint_comb` and `A_comb` say that the lattice is closed under $(a,b) \mapsto 2a-b$, and the last three lines derive $0 \in V$ and the contradiction.

2.6.3 Interpretation and what remains open

- **[Tier C]** The lower energy cannot see a single missing point, and the same argument applies to any finite set of missing points (Exercise 2.7). So any statement of the form "the minimizer is the triangular lattice" is *false* among all locally finite configurations. This is the remark after CKMRV Theorem 1.4 ("removing a single particle changes neither the density nor the energy") [2] **[Tier L]**, here in Lean **[Tier A]** (for the lower-energy notion).

- **[Tier C]** The vacancy is not periodic. The meaningful uniqueness question is therefore uniqueness among *periodic* configurations, the form CKMRV prove in dimensions 8 and 24. In dimension 2 it is **open**, and nothing in this book addresses it.

- **[Tier C]** For physics (Chapter 4), the theorem says that the triangular lattice has the least energy per particle in the bulk; it does not say that a physical system will be found in that state, or that defects cost energy per particle. A single vacancy changes the total pair energy of a large disk by an amount that stays bounded (for $s > 2$ Riesz potentials, by the argument of Exercise 2.8), and the energy per particle does not see it.

2.7 Result D4: open versus closed disks

Upstream measures densities and energies on *closed* disks B_R. A reader might worry that open disks $B_R^\circ = \{x : \|x\| < R\}$ give a different theory: on the boundary circle a lattice can carry many points. We now prove that, for locally finite configurations, nothing changes.

Notation. For locally finite C write $N_R^\circ(C) = \#(C \cap B_R^\circ)$ and define $E_g^\circ(C;R)$ and $E_g^\circ(C) = \liminf_R E_g^\circ(C;R)$ exactly as in Definition 2.3, with B_R° in place of B_R. C has *open density* ρ if $N_R^\circ(C)/(\pi R^2) \to \rho$.

Informal statement. For a locally finite configuration, open and closed disks give the same density and the same lower energy. Hence upstream's theorem holds verbatim with open disks.

Theorem 2.18 (D4)**.** *Let C be locally finite, $g : \mathbb{R} \to \mathbb{R}$ any function and $\rho \in \mathbb{R}$. Then*

(i) C has density ρ if and only if it has open density ρ;

(ii) $E_g^\circ(C) = E_g(C)$.

Lean covers (i) for $\rho = 1$ only (`densityOne_iff_open`) and (ii) (`energy_eq_openEnergy`); `OpenDiskAny.lean` proves both without the local-finiteness hypothesis, and `ctl_open` and `ctl_closed` show the identity is not definitional (a point on the unit circle). Case $\rho \neq 1$: **[Tier C]** (the same argument applies, not formalised).

Status: **[Tier C]** for the human proof below. Lean status of D4: **[Tier A]** kernel-checked for density one and for the lower energy, independently re-verified; other densities **[Tier C]** (details in the Editor's note).

The disk energy at radius R depends on R only through the finite set $C \cap B_R$ (respectively $C \cap B_R^\circ$): the count is its cardinality and the double sum runs over it. The whole proof rests on comparing these sets.

Lemma 2.19 (Comparing point sets)**.** *Let C be locally finite and $R \in \mathbb{R}$. There are radii $\varphi(R) \in [R-1, R]$ and $\psi(R) \in [R, R+1]$ with*

$$C \cap B_R^\circ = C \cap B_{\varphi(R)} \qquad \text{and} \qquad C \cap B_R = C \cap B_{\psi(R)}^\circ.$$

Human proof. First identity. The set $F = C \cap B_R^\circ$ is finite, since it lies in $C \cap B_R$. Let $m = \max\{\|x\| : x \in F\}$ if $F \neq \emptyset$, and $m = -\infty$ otherwise. Every $x \in F$ has $\|x\| < R$, so $m < R$. Put $\varphi(R) = \max(m, R - \frac{1}{2})$, so $R - 1 \le \varphi(R) < R$. If $x \in C$ and $\|x\| \le \varphi(R)$, then $\|x\| < R$ and $x \in F$. Conversely, if $x \in F$ then $\|x\| \le m \le \varphi(R)$. So $F = C \cap B_{\varphi(R)}$.

Second identity. The set $G = \{x \in C : R < \|x\| \le R+1\}$ is finite, since it lies in $C \cap B_{R+1}$. Let $m' = \min\{\|x\| : x \in G\}$ if $G \neq \emptyset$, and $m' = +\infty$ otherwise; then $m' > R$. Put $\psi(R) = \min(m', R + \frac{1}{2})$, so $R < \psi(R) \le R+1$. If $\|x\| \le R$ then $\|x\| < \psi(R)$. Conversely, let $x \in C$ with $\|x\| < \psi(R) \le R + \frac{1}{2}$. If we had $\|x\| > R$, then $x \in G$ and $\|x\| \ge m' \ge \psi(R)$, a contradiction; so $\|x\| \le R$. Hence $C \cap B_R = C \cap B_{\psi(R)}^\circ$. □

Finiteness is used twice, to make the maximum and the minimum exist. Without it the lemma fails (Exercise 2.9).

Lemma 2.20 (Lower limits under reparametrisation)**.** *Let $u : \mathbb{R} \to [0, \infty]$ and $\varphi : \mathbb{R} \to \mathbb{R}$ with $\varphi(R) \to \infty$ as $R \to \infty$. Then $\liminf_{R\to\infty} u(\varphi(R)) \ge \liminf_{R\to\infty} u(R)$.*

Human proof. Recall $\liminf_R u(R) = \sup_X \inf_{x \ge X} u(x)$. Fix X and put $m_X = \inf_{x \ge X} u(x)$. Since $\varphi(R) \to \infty$ there is Y with $\varphi(R) \ge X$ for all $R \ge Y$, hence $u(\varphi(R)) \ge m_X$ for all $R \ge Y$. Therefore $\inf_{R \ge Y} u(\varphi(R)) \ge m_X$, and so $\liminf_R u(\varphi(R)) \ge m_X$. Taking the supremum over X gives the claim. □

Human proof. [Human proof of Theorem 2.18] *(ii) Energies.* Put $u(R) = E_g(C;R)$ and $u^\circ(R) = E_g^\circ(C;R)$. Because the disk energy depends only on the point set, Lemma 2.19 gives

$$u^\circ(R) = u(\varphi(R)), \qquad u(R) = u^\circ(\psi(R)) \qquad \text{for every } R.$$

Since $\varphi(R) \geq R-1$ and $\psi(R) \geq R$, both reparametrisations tend to ∞. Lemma 2.20 applied twice gives

$$\begin{aligned} E_g^\circ(C) = \liminf_{R\to\infty} u(\varphi(R)) &\geq \liminf_{R\to\infty} u(R) = E_g(C), \\ E_g(C) = \liminf_{R\to\infty} u^\circ(\psi(R)) &\geq \liminf_{R\to\infty} u^\circ(R) = E_g^\circ(C). \end{aligned}$$

Hence $E_g^\circ(C) = E_g(C)$.

(i) Densities. Since $B_{R-1} \subseteq B_R^\circ \subseteq B_R \subseteq B_{R+1}^\circ$,

$$N_{R-1}(C) \leq N_R^\circ(C) \leq N_R(C) \leq N_{R+1}^\circ(C) \qquad (R \geq 1).$$

Suppose C has density ρ. For $R > 1$,

$$\frac{N_{R-1}(C)}{\pi R^2} = \frac{N_{R-1}(C)}{\pi(R-1)^2} \cdot \frac{(R-1)^2}{R^2} \longrightarrow \rho \cdot 1 = \rho,$$

and $N_R(C)/(\pi R^2) \to \rho$. The middle term $N_R^\circ(C)/(\pi R^2)$ is squeezed between them, so C has open density ρ. Conversely, suppose C has open density ρ. Then $N_R^\circ(C) \leq N_R(C) \leq N_{R+1}^\circ(C)$ and $N_{R+1}^\circ(C)/(\pi R^2) = \frac{N_{R+1}^\circ(C)}{\pi(R+1)^2} \cdot \frac{(R+1)^2}{R^2} \to \rho$; the same squeeze gives density ρ. □

Corollary 2.21 (Upstream with open disks)**.** *Let g be admissible and C locally finite with open density one. Then $\mathcal{E}_g(A) \leq E_g^\circ(C)$ and $\mathcal{E}_g(A) = E_g^\circ(A)$.*

Human proof. By Theorem 2.18(i), C has density one, so Theorem 2.4 gives $\mathcal{E}_g(A) \leq E_g(C) = E_g^\circ(C)$, the equality by Theorem 2.18(ii). The lattice A is locally finite, so $\mathcal{E}_g(A) = E_g(A) = E_g^\circ(A)$. □

Status: **[Tier C]** for the corollary and its proof. Lean status of D4: **[Tier A]** kernel-checked for density one and for the lower energy, independently re-verified; other densities **[Tier C]** (details in the Editor's note).

Remark 2.22. The preregistration of D4 (`results/discovery/D4/preregistration.json`) was committed before the Lean file was written. It names the hypothesis, the kill criterion ("the liminf comparison for ENNReal-valued step functions on atTop (real index) cannot be closed in-session"), the controls (axiom audit with planted `sorry`/axiom; non-vacuity at $C = A$) and a prior probability of about 65% that the Lean proof would go through. The proof went through: the statements below were kernel-checked, with the footprints recorded in §2.8.

Lean 4 (`formal_cert/Triangular/OpenDisk.lean`, verbatim, proofs elided)

```
/-- **D4, energy.** The lower energy is the same with open and with closed disks. -/
theorem energy_eq_openEnergy (g : ℝ → ℝ) {C : Set Plane} (hC : LocallyFinite C) :
    energy g C = openEnergy g C := by
  -- ...
/-- **D4, density.** Centred density one is the same with open and with closed disks. -/
theorem densityOne_iff_open {C : Set Plane} (hC : LocallyFinite C) : DensityOne C ↔
    DensityOneOpen C := by
  -- ...
/-- **D4.** Upstream's universal optimality theorem, verbatim with open disks. -/
theorem universal_energy_minimum_open (g : ℝ → ℝ) (C : Set Plane) (hg :
```

```
    AdmissiblePotential g)
    (hC : LocallyFinite C) (hd : DensityOneOpen C) :
    latticeEnergy g ≤ openEnergy g C ∧ latticeEnergy g = openEnergy g A := by
  -- ...
theorem A_densityOneOpen : DensityOneOpen A :=
  -- ...
```

Lean 4 (formal_cert/Triangular/OpenDiskAny.lean, verbatim, proofs elided)

```
theorem energy_eq_openEnergy_any (g : ℝ → ℝ) (C : Set Plane) : energy g C = openEnergy
    g C := by
  -- ...
theorem densityOne_iff_open_any (C : Set Plane) : DensityOne C ↔ DensityOneOpen C := by
  -- ...
lemma ctl_open : openDiskCount {e} 1 = 0 := by
  -- ...
lemma ctl_closed : diskCount {e} 1 = 1 := by
  -- ...
```

Here `e` is `EuclideanSpace.single 0 1`, a point of norm one.

2.8 The axiom audit and the controls

A Lean file that "compiles" has not thereby been proved. Two failure modes exit with code 0:

- a proof containing `sorry` compiles with a *warning*, not an error;
- a declared `axiom` can prove anything, including `False`, and compiles silently.

Our gate therefore reads the axiom footprint of every theorem with `##print axioms` and accepts a declaration only if its footprint is contained in the whitelist {`propext`, `Classical.choice`, `Quot.sound`}; the presence of `sorryAx` is a rejection. To show that the gate can fail, the audit file plants one `sorry` and one false axiom, and the gate must flag both.

Lean 4 (formal_cert/Triangular/Axioms.lean, verbatim, shortened)

```
import Triangular.Riesz
import Triangular.RieszFinite
import Triangular.AnyDensity
import Triangular.Controls
import Triangular.UniversalOptimality
import Triangular.OpenDisk
import Triangular.OpenDiskAny

/-! Axiom audit of the D1 corollary, with planted controls. -/

#print axioms TriangularRiesz.riesz_admissible
#print axioms TriangularRiesz.triangular_riesz_optimal
#print axioms OAI.AtomicTriangular.universal_energy_minimum
-- ... (22 audited declarations for D1-D3; 30 in total)
#print axioms TriangularUniversalOptimality.universal_optimum_not_unique
#print axioms TriangularOpenDisk.energy_eq_openEnergy
#print axioms TriangularOpenDisk.densityOne_iff_open
#print axioms TriangularOpenDisk.universal_energy_minimum_open
#print axioms TriangularOpenDisk.A_densityOneOpen
#print axioms TriangularOpenDiskAny.energy_eq_openEnergy_any
#print axioms TriangularOpenDiskAny.densityOne_iff_open_any
```

```
#print axioms TriangularOpenDiskAny.ctl_open
#print axioms TriangularOpenDiskAny.ctl_closed

theorem ctl_sorry : TriangularRiesz.riesz 1 1 = 0 := by sorry
axiom ctl_cheat : False
theorem ctl_axiom : TriangularRiesz.riesz 1 1 = 0 := ctl_cheat.elim
#print axioms ctl_sorry
#print axioms ctl_axiom
```

Note that the two control theorems state something *false*: $g_1(1) = 1^{-1/2} = 1 \neq 0$. Both nevertheless compile. This is the point of the controls.

Numerics (recorded compiler output, not a number but a log)

The following is the output stored in `results/discovery/D1/compile_Axioms.json` (fields `rc` $= 0$, `seconds` $= 15.4$) from compiling `Axioms.lean` with the pinned toolchain. We did not rerun the compile for this book (the machine is shared); the log is quoted, shortened, verbatim.

```
'TriangularRiesz.riesz_admissible' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'TriangularRiesz.triangular_riesz_optimal' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'OAI.AtomicTriangular.universal_energy_minimum' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'TriangularRiesz.norm_sq_triangularPoint' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'TriangularRiesz.latticeEnergy_riesz_lt_top' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'OAI.SevenEighths.LatticeSummability.summable_normForm_neg_rpow' depends on axioms: [
    propext,
 Classical.choice,
 Quot.sound]
'TriangularDensity.universal_any_density' depends on axioms: [propext, Classical.choice,
     Quot.sound]
...
'TriangularUniversalOptimality.vacancy_not_isometric_image' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'TriangularUniversalOptimality.vacancy_universally_optimal' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'TriangularUniversalOptimality.universal_optimum_not_unique' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'TriangularOpenDisk.energy_eq_openEnergy' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'TriangularOpenDisk.densityOne_iff_open' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'TriangularOpenDisk.universal_energy_minimum_open' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'TriangularOpenDisk.A_densityOneOpen' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'TriangularOpenDiskAny.energy_eq_openEnergy_any' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'TriangularOpenDiskAny.densityOne_iff_open_any' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'TriangularOpenDiskAny.ctl_open' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'TriangularOpenDiskAny.ctl_closed' depends on axioms: [propext, Classical.choice, Quot.
    sound]
.../formal_cert/Triangular/Axioms.lean:42:8: warning: declaration uses ‘sorry‘
'ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
```

```
'ctl_axiom' depends on axioms: [ctl_cheat, propext, Classical.choice, Quot.sound]
```

How to read the log.

- The return code is 0 although the file contains a `sorry` and a false axiom. An exit code is not a proof gate.
- All 30 audited declarations (22 for D1–D3, 8 for D4), including upstream's `universal_energy_minimum`, report the footprint [`propext`, `Classical.choice`, `Quot.sound`]: the three standard axioms of Lean's classical mathematics, which Mathlib uses everywhere. This is *not* "zero axioms". **[Tier A]**
- `ctl_sorry` reports `sorryAx` (and the compiler warns, at line 42), so the gate rejects it. `ctl_axiom` reports `ctl_cheat`, outside the whitelist, so the gate rejects it. Both controls are flagged, which shows that the gate is able to fail.
- A clean footprint certifies that the proofs are complete relative to the definitions. It does not certify that the definitions mean what the English says (statement adequacy), which is a separate human audit and has not been done.

The other controls, collected. Every positive result in this chapter is accompanied by a control showing that its hypotheses or conclusions are not vacuous **[Tier A]**:

- `riesz_neg_two_not_admissible`: the admissibility hypothesis can fail (§2.3);
- `riesz_sq`: the Lean potential is the Riesz kernel in the distance;
- `smoothness_order_is_infty`: upstream's smoothness order is C^∞, not analytic;
- `scaled_density`, `attained`: the hypothesis `DensityRho ÏĄ` is satisfiable, and the bound of D2 is attained, so D2 is an optimality statement, not just an inequality (§2.5);
- `unscaled_false`: the naive unscaled statement is false (§2.5);
- `latticeEnergy_riesz_lt_top`: D1 is not $\infty \le \infty$ for $s > 2$ (§2.4);
- `rho_one`, `densityRho_one_iff`: D2 specialises correctly to $\rho = 1$;
- `vacancy_not_isometric_image`: the second minimizer is genuinely different (§2.6).

2.9 Exercises

Exercises marked **(Lean)** give a Lean statement with a `sorry` placeholder for you to replace by a proof. These statements were written for this book and have *not* been compiled by us; part of the exercise is to make them typecheck (they assume `open OAI.AtomicTriangular Filter Topology` and the namespaces of the files above). A solution counts only if it passes the axiom gate of §2.8: compile, then `##print axioms`, with footprint inside the whitelist and no `sorryAx`.

Exercise 2.1 (Pen and paper)**.** Using Lemma 2.7, find the smallest nonzero value of $\|a\|^2$ for $a \in A$ and all lattice points that attain it. Show that there are exactly six, and that the minimal distance is $(2/\sqrt{3})^{1/2}$.

Exercise 2.2 (Pen and paper)**.** Show that $\sum_{a\in A\setminus\{0\}} \|a\|^{-s} = \infty$ for $0 < s \le 2$. *Hint:* bound the number of lattice points in the annulus $k \le \|a\| < k+1$ from below by a constant times k, for large k, using the density of A.

Exercise 2.3 (Pen and paper)**.** Prove $(-x)_r^{\downarrow} = (-1)^r (x)_r^{\uparrow}$ by induction on r, and deduce that $(-1)^r \frac{d^r}{dt^r} t^{-\sigma} > 0$ for $\sigma > 0$, $t > 0$. What goes wrong for $\sigma < 0$? Compare with `riesz_neg_two_not_admissible`.

Exercise 2.4 (Lean)**.** Prove that $t \mapsto e^{-t}$ is admissible.

```
theorem exp_neg_admissible : AdmissiblePotential (fun t => Real.exp (-t)) := by
  sorry
```

Hint: the r-th derivative is $(-1)^r e^{-t}$; Mathlib's `iteratedDeriv` lemmas for compositions with $t \mapsto -t$ may help, or prove $\mathtt{deriv}^{[r]}$ by induction as in `iterate_deriv_comp_mul`.

Exercise 2.5 (Pen and paper, then Lean)**.** Show that the constant $g \equiv 1$ is admissible and that $\mathcal{E}_1(A) = \infty$. Then show directly from the definition that $E_1(C) = \infty$ for every locally finite C of density one ($E_1(C; R) = N_R(C) - 1$). Explain why this is consistent with Theorem 2.4, and why the measure in the Gaussian-mixture representation may charge $\alpha = 0$.

```
theorem const_admissible : AdmissiblePotential (fun _ => (1 : ℝ)) := by
  sorry
```

Exercise 2.6 (Pen and paper)**.** Prove the homogeneity identity $\mathcal{E}_{g_s(\rho^{-1}\cdot)}(A) = \rho^{s/2}\mathcal{E}_{g_s}(A)$ and deduce, for $s > 2$, that the minimal Riesz energy at density ρ is a strictly increasing function of ρ. Why does Proposition 2.14 follow?

Exercise 2.7 (Pen and paper)**.** Let $F \subseteq A$ be finite. Adapt the proof of Theorem 2.17 to show that $A \setminus F$ has density one and $E_g(A \setminus F) = \mathcal{E}_g(A)$ for every admissible g. Is $A \setminus F$ ever an isometric image of A when $F \neq \emptyset$?

Exercise 2.8 (Pen and paper)**.** Let $s > 2$ and $x \notin A$. Show that $A \cup \{x\}$ has density one and that $E_{g_s}(A \cup \{x\}) = \mathcal{E}_{g_s}(A)$. *Hint:* the extra pairs contribute $2\sum_{a\in A\cap B_R} \|x - a\|^{-s}$, which stays bounded as $R \to \infty$; compare with D1b. For $s \le 2$ the conclusion still holds, trivially: both sides are ∞. Which step of the finiteness argument above fails there?

Exercise 2.9 (Pen and paper)**.** Let $C = \{(1 - \frac{1}{n}, 0) : n \ge 1\} \cup \{(2, 0)\}$. Show that C is not locally finite, and that there is no $r < 1$ with $C \cap B_1^\circ = C \cap B_r$. Which line of the proof of Lemma 2.19 fails? What does the Lean convention "`diskPoints` is empty when the set is infinite" give for $N_R(C)$, $R \ge 1$?

Exercise 2.10 (Lean)**.** Prove that disk counts are monotone in the radius for locally finite configurations.

```
theorem diskCount_mono {C : Set Plane} (hC : LocallyFinite C) {R R' : ℝ} (h : R ≤ R') :
    diskCount C R ≤ diskCount C R' := by
  sorry
```

Then show by an example that the hypothesis `hC` cannot be removed. (Proved in `OpenDisk.lean` as `diskCount_mono`; use it as the solution key.)

Exercise 2.11 (Lean)**.** Formalise Lemma 2.20.

```
theorem liminf_le_liminf_of_tendsto (u : ℝ → ENNReal) (φ : ℝ → ℝ)
    (hφ : Tendsto φ atTop atTop) :
    liminf u atTop ≤ liminf (u ∘ φ) atTop := by
  sorry
```

Hint: lim inf along `atTop` of $u \circ \varphi$ is lim inf u along the filter `map ÏĘ atTop`, and `map ÏĘ atTop âĽď atTop`. (Proved in `OpenDisk.lean` as `liminf_le_liminf_comp`; use it as the solution key.)

Exercise 2.12 (Lean)**.** Generalise the vacancy theorem to every density: show that $(\rho^{-1/2}A)\setminus\{0\}$ is universally optimal at density ρ.

```
theorem vacancy_rho_universally_optimal {ρ : ℝ} (hρ : 0 < ρ) :
    TriangularUniversalOptimality.UniversallyOptimal ρ (((Real.sqrt ρ)⁻¹ • A) \ {0}) := by
  sorry
```

Hint: either redo the proof of `vacancy_energy` at density ρ, or show $((\rho^{-1/2})A) \setminus \{0\} = \rho^{-1/2}(A \setminus \{0\})$ and use the scaling lemmas of `AnyDensity.lean`.

Exercise 2.13 (Pen and paper)**.** Let $u, v : \mathbb{R} \to [0,\infty]$ with $u(R) \to 1$. Prove $\liminf uv = \liminf v$ (both inequalities), handling the case $\liminf v = \infty$. Where does the proof of Theorem 2.17 need only one of the two inequalities?

Exercise 2.14 (Discussion)**.** Definition 2.15 uses the lower energy. Write down the stronger CKMRV-style definition in which the limit $\lim_R E_g(C;R)$ must exist. Does the vacancy configuration satisfy it? (You may use, without proof, that $\lim_R E_g(A;R)$ exists for the potentials in question; state clearly that this is an assumption.) Which of the Lean theorems of this chapter would need new proofs under the stronger definition?

Exercise 2.15 (Pen and paper)**.** Show that for the triangular lattice the sequence $N_R(A) - N_R^\circ(A)$ is unbounded as $R \to \infty$ (the boundary circle can carry many lattice points), and nevertheless both $N_R(A)/(\pi R^2)$ and $N_R^\circ(A)/(\pi R^2)$ tend to 1. Why does this not contradict Theorem 2.18? *Hint:* the number of representations of an integer by $j^2 + jk + k^2$ is unbounded.

Chapter 3

Numerical validation in Python and Rust (Tier X)

Chapters 1 and 2 stated upstream's theorem (AI-generated, not peer reviewed; kernel-checked in Lean, conditional on the fidelity of the definitions) that the triangular lattice A has the least lower energy among all locally finite configurations of density one, for every admissible potential g. This chapter does something much weaker, and the weakness is the subject of its first section. We put a few hundred particles in a periodic box, let them interact through a screened Coulomb (Yukawa) law, let the energy run downhill, and look at where it ends. The code is the Rust crate `qf-yukawa-crystal` in rusty-SUNDIALS [18] (PR #72), and a short Python companion written for this book. Every number in the chapter is **[Tier X]** unless a different tier is shown.

All Rust code below is quoted verbatim from the files `crates/qf-yukawa-crystal/src/{lib.rs,jellium.rs,main.rs}` on the branch `physics/jellium-wigner-2026-10-11` of rusty-SUNDIALS, pinned to commit `b7e4380` of that branch, with the line numbers given in each listing title. Every listing and every line number in this chapter was re-checked against `git show b7e4380:<path>`. Lines left out are marked `// [...]` (an insertion of ours, not part of the file).

3.1 What numerics can and cannot do

3.1.1 The gap between the theorem and the computer

The theorem of Chapter 2 is about objects that a computer cannot hold.

- *Infinite configurations.* C is an arbitrary locally finite subset of $\mathbb{R}^2$, and its lower energy is a lim inf over disks of radius $R \to \infty$. A simulation holds finitely many particles, here $N = 288$.
- *All competitors.* The theorem quantifies over every C. A simulation visits the few end states that a particular algorithm reaches from a few hundred starting points.
- *Exact potentials.* The theorem uses g exactly. The code truncates the potential at a cutoff r_c and modifies it near r_c (Section 3.2). The truncated potential is *not* completely monotone, so the theorem does not even apply to it.
- *Exact arithmetic.* The code uses IEEE double precision, with relative rounding $\approx 1.1\times10^{-16}$ per operation and sums of tens of thousands of terms.

A periodic box is itself a competitor of a special kind. A configuration with N particles in a periodic rectangle of area N is, after periodic extension, a locally finite configuration of density one in the plane. Its energy per particle in the box (with the minimum-image rule and r_c less

than half the box) equals one half of the lower energy E_g of Chapter 1 for the periodic extension (Chapter 1 counts ordered pairs; this chapter, like the physics literature, counts each pair once), for the truncated potential. So the box computes honest energies of honest competitors, but only of *periodic* ones, with one fixed period cell, and only for the truncated potential.

Caution : Tier X never supports a claim

In the Elenchus scheme [17] a floating-point experiment is **[Tier X]**: it may steer a search, suggest a conjecture or expose a bug, but it is never evidence for a theorem. "No relaxation went below the lattice" is consistent with Chapter 2. It does not support it: the theorem is already kernel-checked (conditional on the fidelity of its definitions), and a few hundred periodic states say nothing about the infinitely many non-periodic ones.

What, then, is the point? There are four honest uses.

1. *Testing code against a theorem.* If a relaxation ever ended below the lattice energy, the first suspect would be the code, the second the truncation, and only a distant third the mathematics. A theorem is a strong test oracle for numerical software.

2. *Exploring the landscape.* The theorem identifies the minimum *value*, not the local minima, their depths or basins. Numerics can explore those.

3. *Checking statement adequacy.* If the Lean definitions of Chapter 1 described the wrong object, the energies computed from the same formulas would disagree with physical intuition or with the literature (for example the Madelung constants of Section 3.5).

4. *Connecting to physics* (Chapter 4): what finite systems at finite temperature actually do.

3.1.2 Preregistration and commit order

An experiment whose success criterion is chosen after the results are seen can always be declared a success. The study in this chapter was therefore *preregistered*: its predictions, tolerances, potentials, box, initial conditions and analysis were written into `results/preregistration.json` and committed (rusty-SUNDIALS commit `636c514`) *before* the simulation code (`968b0f2`) and before the results (`1f46e70`). The order of commits is the evidence: anyone can check with `git log` that the criteria predate the code. The three predictions read, verbatim:

> **P1** `no_undercut`: "Over all runs (T = 0 quenches and annealed runs, all initial conditions), no relaxed configuration has e < e_tri - 1e-9 * |e_tri| (kappa = 2). A violation would contradict the theorem (or reveal a truncation artefact, to be checked at kappa = 1 and with larger r_c)."
>
> **P2** `annealed_reaches_lattice`: "At least one annealed run per initial-condition family reaches the perfect lattice: zero defects and |e - e_tri| < 1e-8 |e_tri|. Expected but not guaranteed (annealing is heuristic); failure is reported, not hidden."
>
> **P3** `quench_glassy`: "T = 0 quenches from disordered starts mostly end in defected local minima (e > e_tri, defects > 0). This is a property of gradient flow, not evidence against the theorem."

P1 is the only prediction connected to the theorem. P2 is a prediction about an algorithm (annealing), and P3 about another (gradient descent). A failed P2 is reported as failed (Section 3.4); it says nothing about the theorem.

The P1 wording contains a hedge worth noticing: the potential is truncated, so a genuine undercut *could* be a truncation artefact. In the preregistration this is not an escape hatch but a test plan, fixed in advance: an undercut would have to be re-examined at another κ and a larger r_c.

3.1.3 Controls must be shown to fire

A check that cannot fail carries no information. Before any production number is reported, every observable is run on inputs for which the right answer is known, in both directions.

- A *positive control* is an input on which the check must pass: the perfect lattice must stay put, a slightly perturbed lattice must relax back to it.
- A *negative control* is an input on which the check must fire: the square lattice must be detected as defected and as a saddle, and a box that cannot hold the triangular lattice must produce defected states above e_{tri}.

If the defect counter reported zero defects for the square lattice, every "zero defects" in the results would be worthless. The same goes for the Hessian test and for the energy comparison. This is the numerical twin of the negative controls of the Lean work (`unscaled_false`, `riesz_neg_two_not_admissible`): an auditor that accepts everything is not an auditor.

3.2 The model

3.2.1 The Yukawa potential and complete monotonicity

Two particles at distance r interact through

$$V(r) = \frac{e^{-\kappa r}}{r}, \qquad \kappa > 0. \tag{3.1}$$

In the language of Chapter 1, the potential is $g(t) = V(\sqrt{t}) = e^{-\kappa\sqrt{t}}\, t^{-1/2}$ as a function of the squared distance $t = r^2$.

Proposition 3.1. *For every $\kappa \geq 0$, $g(t) = e^{-\kappa\sqrt{t}}\, t^{-1/2}$ is completely monotone on $(0, \infty)$.*

Human proof. **[Tier C]** (a standard argument, not formalised here). The function $t^{-1/2}$ is completely monotone: $(-1)^r \frac{d^r}{dt^r} t^{-1/2} = \frac{1}{2} \cdot \frac{3}{2} \cdots \frac{2r-1}{2}\, t^{-1/2-r} \geq 0$; this is the $s = 1$ case of `riesz_admissible`. The function $h(t) = \kappa\sqrt{t}$ has a completely monotone derivative, and the composition e^{-h} of the completely monotone $u \mapsto e^{-u}$ with such an h is completely monotone (differentiate e^{-h} repeatedly: every term is a product of e^{-h} and derivatives of $-h$, each with the right sign). The product of two completely monotone functions is completely monotone by the Leibniz rule, since every term of $(-1)^r (fg)^{(r)} = \sum_k \binom{r}{k} (-1)^k f^{(k)}\, (-1)^{r-k} g^{(r-k)}$ is non-negative. □

So the theorem predicts that no density-one configuration has lower energy per particle than A for the exact Yukawa potential. The values $\kappa = 2$ (main) and $\kappa = 1$ (check) are used throughout.

3.2.2 Shifted-force cutoff

The sum of V over a lattice converges quickly, but a simulation needs a finite neighbour list. We cut at $r_c = 8$ and *shift the force*:

$$V_{\mathrm{sf}}(r) = \begin{cases} V(r) - V(r_c) - (r - r_c)\, V'(r_c), & r < r_c, \\ 0, & r \geq r_c. \end{cases} \tag{3.2}$$

Then $V_{\mathrm{sf}}(r_c) = 0$ and $V'_{\mathrm{sf}}(r_c^-) = V'(r_c) - V'(r_c) = 0$: energy and force are continuous at r_c. The second derivative is *not* shifted, $V''_{\mathrm{sf}} = V''$ for $r < r_c$, so the Hessian jumps by $V''(r_c)$ when a pair crosses r_c. The derivatives are

$$V'(r) = -\frac{e^{-\kappa r}(\kappa r + 1)}{r^2}, \qquad V''(r) = \frac{e^{-\kappa r}(\kappa^2 r^2 + 2\kappa r + 2)}{r^3}. \tag{3.3}$$

In the code (`lib.rs`, lines 42–63):

lib.rs, lines 42–63

```
    fn v(&self, r: f64) -> f64 {
        (-self.kappa * r).exp() / r
    }
    fn dv(&self, r: f64) -> f64 {
        -(-self.kappa * r).exp() * (self.kappa * r + 1.0) / (r * r)
    }
    fn d2v(&self, r: f64) -> f64 {
        let k = self.kappa;
        (-k * r).exp() * (k * k * r * r + 2.0 * k * r + 2.0) / (r * r * r)
    }
    /// Shifted-force pair potential and its first two derivatives (zero for `r >= rc`).
    pub fn pair(&self, r: f64) -> (f64, f64, f64) {
        if r >= self.rc {
            return (0.0, 0.0, 0.0);
        }
        let (vc, dvc) = (self.v(self.rc), self.dv(self.rc));
        (
            self.v(r) - vc - (r - self.rc) * dvc,
            self.dv(r) - dvc,
            self.d2v(r),
        )
    }
```

The `refs` command records $V(r_c) = e^{-\kappa r_c}/r_c$ as `truncation_v_rc`: $e^{-16}/8 \approx 1.41 \times 10^{-8}$ at $\kappa = 2$ and $e^{-8}/8 \approx 4.19 \times 10^{-5}$ at $\kappa = 1$.[1] This is the pointwise value of V at the cut, *not* the change in the energy. The linear shift $-V(r_c) - (r - r_c)V'(r_c)$ acts on every pair inside r_c, and there are many such pairs (about $\pi r_c^2 \approx 200$ neighbours per particle), so the energy per particle moves much more. For the triangular lattice, a scratch sum of the *unshifted* V out to $r = 60$ gives 0.3935428489 at $\kappa = 2$ and 1.6253213396 at $\kappa = 1$, against the shifted-force values 0.3935333668 and 1.6076092172 of Table 3.1: differences 9.5×10^{-6} and 1.77×10^{-2}.[2] At $\kappa = 1$ the shift is larger than $e_{\mathrm{sq}} - e_{\mathrm{tri}}$ itself. It is almost the same for the two lattices, though: the unshifted square-minus-triangular differences are 9.53×10^{-3} ($\kappa = 1$) and 7.43×10^{-3} ($\kappa = 2$), close to the shifted ones. Section 3.6 shows the same effect when r_c goes from 6 to 8 at $\kappa = 2$: the shifted-force lattice energy moves by 3.0×10^{-4}, while $V(6)$ is about 10^{-6}. What saves the study is that the lattice reference is computed with the same V_{sf}: every energy comparison in this chapter is between equally truncated energies.

Remark 3.2. V_{sf} is not completely monotone (it vanishes on $[r_c, \infty)$, is positive just below, and its second derivative jumps), so P1 is a prediction for a potential the theorem does not cover. The modification is not small at $\kappa = 1$, only nearly constant per particle, which suggests (**[Tier C]**) that it preserves the ordering of configurations.

3.2.3 The periodic box at density one

At density one the triangular lattice has spacing a with $a^2\sqrt{3}/2 = 1$, i.e.

$$a = \sqrt{2/\sqrt{3}} \approx 1.074570. \tag{3.4}$$

[1] `python -I -c "import math; print(math.exp(-16)/8, math.exp(-8)/8)"`, output `1.406689683990739e-08 4.193282848781398e-05`.

[2] Scratch double loop over $|i|, |j| \le 200$ (not shipped), $\frac{1}{2}\sum_{0<|v|<60} e^{-\kappa|v|}/|v|$. Output (triangular, square): $\kappa = 1$: `1.6253213396062602`, `1.6348554953171857`; $\kappa = 2$: `0.3935428489323898`, `0.40097424876505755`.

The box is $L_x \times L_y = 16a \times 9\sqrt{3}\,a \approx 17.1931 \times 16.7509$.[3] It holds 16 sites per row and 18 rows (rows are $a\sqrt{3}/2$ apart, and $18 \cdot a\sqrt{3}/2 = 9\sqrt{3}\,a$), so $N = 288$ and the area is $16 \cdot 9\sqrt{3}\,a^2 = 288$: density one, and the triangular lattice fits periodically. Such a box is called *commensurate* with A. A square box of side $\sqrt{289} = 17$ holds the square lattice ($N = 289$) but not the triangular one; it is used as a negative control.

Displacements use the minimum-image convention: `delta` (`lib.rs`, lines 65–72) replaces each coordinate difference d_x by `dx -= lx * (dx / lx).round()`, i.e. $d_x - L_x \operatorname{round}(d_x/L_x) \in [-L_x/2, L_x/2]$, and likewise for d_y. The minimum image is the correct nearest copy as long as $r_c \le \min(L_x, L_y)/2$, which holds here ($8 < 8.375$). Under this condition each pair contributes at most once, and the energy of the box is

$$E(x) = \sum_{1 \le i < j \le N} V_{\mathrm{sf}}\big(|x_i - x_j|_{\mathrm{mi}}\big), \qquad e(x) = E(x)/N. \tag{3.5}$$

The minimum-image map jumps when a coordinate difference crosses $\pm L/2$, but there $|x_i - x_j|_{\mathrm{mi}} \ge L/2 > r_c$ and $V_{\mathrm{sf}} = 0$, so E is continuously differentiable everywhere and smooth away from the pair distances r_c.

The energy per particle of the infinite lattice with basis b_1, b_2 is

$$e_{\mathrm{lat}} = \frac{1}{2} \sum_{v \in \Lambda \setminus \{0\}} V_{\mathrm{sf}}(|v|), \tag{3.6}$$

which `lattice_energy` (`lib.rs`, lines 224–239) computes by a direct double loop. For the commensurate box the two must agree, and the unit test `periodic_lattice_energy_equals_direct_lattice_sum` checks this to 10^{-12} relative; the study's references agree to about 10^{-14} (Table 3.1).

3.2.4 Force and Hessian

Write $d = x_i - x_j$ (minimum image), $r = |d|$, $u = d/r$, and $\varphi(d) = V_{\mathrm{sf}}(|d|)$.

Proposition 3.3. *For $0 < r < r_c$,*

$$\nabla_d \varphi = V'_{\mathrm{sf}}(r)\, u, \qquad \nabla_d^2 \varphi = K(d) := V''(r)\, uu^{\mathsf{T}} + \frac{V'_{\mathrm{sf}}(r)}{r}\big(I - uu^{\mathsf{T}}\big). \tag{3.7}$$

Consequently the force on particle i is $F_i = -\partial E/\partial x_i = -\sum_{j \ne i} V'_{\mathrm{sf}}(r_{ij})\, u_{ij}$, and the Hessian $H = \nabla^2 E \in \mathbb{R}^{2N \times 2N}$ is the sum over pairs $i < j$ with $r_{ij} < r_c$ of the 2×2 blocks

$$H_{ii} \mathrel{+}= K, \qquad H_{jj} \mathrel{+}= K, \qquad H_{ij} \mathrel{-}= K, \qquad H_{ji} \mathrel{-}= K. \tag{3.8}$$

Human proof. $\partial r/\partial d_a = d_a/r = u_a$, so $\partial_a \varphi = V'_{\mathrm{sf}}(r) u_a$. Differentiating again, $\partial_b u_a = (\delta_{ab} - u_a u_b)/r$, hence $\partial_b \partial_a \varphi = V''_{\mathrm{sf}}(r) u_a u_b + V'_{\mathrm{sf}}(r)(\delta_{ab} - u_a u_b)/r$, and $V''_{\mathrm{sf}} = V''$ below r_c. Since $d = x_i - x_j$, $\partial d/\partial x_i = I$ and $\partial d/\partial x_j = -I$, which gives the signs in (3.8) and, for the gradient, $\partial E/\partial x_i = +\nabla_d \varphi$ and $\partial E/\partial x_j = -\nabla_d \varphi$. □

Three facts follow at once. (i) $\sum_i F_i = 0$ (Newton's third law), so the gradient is orthogonal to rigid translations. (ii) Each row of blocks of H sums to zero, so the two translation vectors $\tau_x = (1, 0, 1, 0, \dots)$ and $\tau_y = (0, 1, 0, 1, \dots)$ lie in $\ker H$. (iii) The radial part $V'' uu^{\mathsf{T}}$ is positive (since $V'' > 0$), while the transverse part V'_{sf}/r is negative (repulsion), which is what allows saddles.

The force in the code, with the external (disk) terms trimmed:

[3] Evaluated with `python -I -c "import math; a=math.sqrt(2/math.sqrt(3)); print(a, 16*a, 9*math.sqrt(3)*a)"`, output `1.074569931823542 17.193118909176672 16.750887463837792`.

lib.rs, lines 105–121 (external-potential part trimmed)

```
    /// Force `F = -grad E` written into `f`.
    pub fn force(&self, x: &[f64], f: &mut [f64]) {
        let n = x.len() / 2;
        f.iter_mut().for_each(|v| *v = 0.0);
        for i in 0..n {
            for j in (i + 1)..n {
                let (dx, dy) = self.delta(x[2 * i], x[2 * i + 1], x[2 * j], x[2 * j + 1]);
                let r = (dx * dx + dy * dy).sqrt();
                let (_, dv, _) = self.pair(r);
                if dv != 0.0 {
                    let (fx, fy) = (-dv * dx / r, -dv * dy / r);
                    f[2 * i] += fx;
                    f[2 * i + 1] += fy;
                    f[2 * j] -= fx;
                    f[2 * j + 1] -= fy;
                }
            }
            // [...] external potential U(r) for the disk geometries (lines 122-131)
```

and the Hessian, exactly (3.8):

lib.rs, lines 135–160 (external-potential part trimmed)

```
    /// Hessian of `E` (dense, row-major, `2n x 2n`).
    pub fn hessian(&self, x: &[f64]) -> Vec<f64> {
        let n = x.len() / 2;
        let m = 2 * n;
        let mut h = vec![0.0; m * m];
        let mut add = |a: usize, b: usize, v: f64| h[a * m + b] += v;
        for i in 0..n {
            for j in (i + 1)..n {
                let (dx, dy) = self.delta(x[2 * i], x[2 * i + 1], x[2 * j], x[2 * j + 1]);
                let r = (dx * dx + dy * dy).sqrt();
                let (_, dv, d2v) = self.pair(r);
                if r >= self.rc {
                    continue;
                }
                let u = [dx / r, dy / r];
                for a in 0..2 {
                    for b in 0..2 {
                        let id = if a == b { 1.0 } else { 0.0 };
                        let k = d2v * u[a] * u[b] + dv / r * (id - u[a] * u[b]);
                        add(2 * i + a, 2 * i + b, k);
                        add(2 * j + a, 2 * j + b, k);
                        add(2 * i + a, 2 * j + b, -k);
                        add(2 * j + a, 2 * i + b, -k);
                    }
                }
            }
            // [...] bowl / wall terms for the disk geometries (lines 161-183)
```

Formulas are checked against finite differences in the crate's unit tests: `force_matches_finite_difference_of_energy` (central difference $h = 10^{-6}$, tolerance 10^{-6}) and `hessian_matches_finite_difference_of_force` (tolerance 10^{-5}), on a lattice perturbed by Gaussian noise of standard deviation 0.05. We ran the whole test suite once for this book (2 build jobs, 2 test threads):

cargo test -q –release -j 2 –offline -p qf-yukawa-crystal –manifest-path .../Cargo.toml – –test-threads=2

```
running 10 tests
..........
```

```
test result: ok. 10 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.15s
[...] two further blocks "running 0 tests ... 0 passed; 0 failed"
```

The ten tests are the five of `lib.rs` (force and Hessian against finite differences, box energy against the direct lattice sum, a Jacobi eigenvalue check on a matrix with known spectrum, square above triangular) and the five of `jellium.rs` (Section 3.5). The line marked `[...]` is our summary of the omitted output (two further test targets with no tests, presumably the binary and the doc-tests; `-q` hides their names).

3.3 Gradient flow, stiffness and CVODE

3.3.1 Relaxation as an ODE

The relaxation is the *gradient flow*

$$\dot{x}(t) = F(x(t)) = -\nabla E(x(t)), \qquad x(0) = x_0 \in \mathbb{R}^{2N}. \tag{3.9}$$

Physically it is the overdamped motion of particles in a viscous fluid at zero temperature, which is how colloids move (Chapter 4). Mathematically it has a Lyapunov function:

$$\frac{d}{dt}E(x(t)) = \nabla E \cdot \dot{x} = -|\nabla E(x(t))|^2 \le 0. \tag{3.10}$$

The energy decreases strictly unless x is a critical point. Since E is bounded below on the box (it is a finite sum of terms bounded below by $V_{\mathrm{sf}} \ge 0$) and blows up when two particles collide, trajectories stay away from collisions and approach the set of critical points. Generic starting points converge to local minima; a nondegenerate saddle attracts only a measure-zero set of starting points (its stable manifold). A gradient flow ends where it ends: in whatever basin it started. This is why P3 expects most quenches to end in defected minima.

Near a critical point x^*, the linearisation $\dot{\delta} = -H(x^*)\delta$ decays along each eigenvector of H at rate equal to its eigenvalue λ_k. The Jacobian of the right-hand side is

$$J(x) = \frac{\partial F}{\partial x} = -H(x). \tag{3.11}$$

3.3.2 Why stiff, and why BDF

An ODE is *stiff* when its Jacobian has eigenvalues of very different sizes, so that an explicit method must take steps set by the fastest mode while the solution evolves on the slowest. For the perfect lattice at $\kappa = 2$ we computed the full spectrum of H with the formula of Proposition 3.3: two zero modes, then the smallest positive eigenvalue 2.355×10^{-2} (twice), and the largest 5.700, a ratio of about 242.[4] At defected minima the study recorded positive eigenvalues as small as 5.4×10^{-5} ($\kappa = 2$), so the slowest relaxation times are of order $1/(5.4 \times 10^{-5}) \approx 2 \times 10^4$ time units while the fastest are below one. And far from equilibrium, in a random start where two particles sit at distance $0.5a$, the local curvature $V''(0.5a)$ is larger still.

An explicit method such as forward Euler is stable for this linear problem only if the step satisfies $h < 2/\lambda_{\max}$; integrating to $t \sim 10^5$ would need on the order of 10^5 steps even where nothing happens. Backward differentiation formulas (BDF) are implicit. They are A-stable for orders 1 and 2 and A(α)-stable for orders 3 to 5, so in every case the whole negative real axis,

[4] A scratch check written for this chapter (not shipped with the book; it re-implements (3.8) in numpy and calls `numpy.linalg.eigvalsh`). Output: `kappa=2.0 N=288 lowest=[-2.77382284e-15 2.77729229e-15 2.35534976e-02 2.35534976e-02] largest=5.700435e+00 ratio_max_over_third=2.4202e+02`. The lowest positive value agrees with the study's control value 0.0236.

where the eigenvalues $-\lambda_k$ of $J = -H$ lie, is inside the stability region, and the step size is chosen by accuracy alone. The price is a nonlinear solve per step,

$$x_{n+1} - h\beta_0 F(x_{n+1}) = (\text{known combination of past values}), \tag{3.12}$$

solved by Newton's method with the matrix $I - h\beta_0 J$. Here the Jacobian is available exactly: $J = -H$, from Proposition 3.3. An exact Jacobian costs one Hessian evaluation ($O(N^2)$ pair work plus a dense $2N \times 2N$ matrix) but makes Newton converge in few iterations.

3.3.3 CVODE in rusty-SUNDIALS

rusty-SUNDIALS [18] is a Rust implementation of the SUNDIALS solver family. Its `cvode` crate provides CVODE's variable-step BDF method of orders 1 to 5 in Nordsieck form (`constants.rs`, `nordsieck.rs`), and forms the Newton matrix $M = I - \gamma J$ from a user-supplied dense Jacobian or, if none is given, from finite differences (`solver.rs`). The relaxation driver, verbatim:

lib.rs, lines 252–300

```
/// Overdamped relaxation ‘dx/dt = F(x)‘ with CVODE (BDF, analytic Jacobian) until ‘max |F| <
    ftol‘ or ‘t > tmax‘.
pub fn relax(sys: &System, x0: &[f64], ftol: f64, tmax: f64) -> Result<Relaxed, String> {
    let s1 = *sys;
    let s2 = *sys;
    let rhs = move |_t: f64, y: &[f64], ydot: &mut [f64]| -> Result<(), String> {
        s1.force(y, ydot);
        Ok(())
    };
    let jac = move |_t: f64, y: &[f64], j: &mut DenseMat| -> Result<(), String> {
        let h = s2.hessian(y);
        let m = y.len();
        for c in 0..m {
            for r in 0..m {
                j.cols[c][r] = -h[r * m + c];
            }
        }
        Ok(())
    };
    let mut solver = Cvode::builder(Method::Bdf)
        .rtol(1e-10)
        .atol(1e-12)
        .max_steps(2_000_000)
        .jacobian(jac)
        .build(rhs, 0.0, SerialVector::from_slice(x0))
        .map_err(|e| format!("CVODE setup failed: {e}"))?;
    let mut f = vec![0.0; x0.len()];
    let mut tout = 1.0;
    loop {
        let (t, y) = solver
            .solve(tout, Task::Normal)
            .map_err(|e| format!("CVODE failed at t = {tout}: {e}"))?;
        let y = y.to_vec();
        sys.force(&y, &mut f);
        let fmax = f.iter().fold(0.0_f64, |a, &b| a.max(b.abs()));
        if fmax < ftol || t >= tmax {
            let mut x = y;
            sys.wrap(&mut x);
            return Ok(Relaxed {
                x,
                t,
                max_force: fmax,
                steps: solver.num_steps(),
                rhs_evals: solver.num_rhs_evals(),
```

```
                converged: fmax < ftol,
            });
        }
        tout *= 2.0;
    }
}
```

Points to notice:

- The right-hand side is the force; the Jacobian callback writes $-H$ into CVODE's column-major dense matrix (`j.cols[c][r]` is row r, column c).
- Tolerances are tight (10^{-10} relative, 10^{-12} absolute): we want the end point, not a pretty trajectory, but a sloppy trajectory can cross into a different basin.
- The solver is asked for output at $t = 1, 2, 4, 8, \ldots$ (`tout *= 2.0`). The doubling matches the multiscale spectrum: each output is a checkpoint, and the time horizon grows geometrically up to $t_{\max} = 10^7$ (`TMAX` in `main.rs`).
- The run stops when $\max_k |F_k| < 10^{-9}$ (`FTOL` in `main.rs`). A run that hits $t_{\max}$ first is returned with `converged: false`, and the summary counts it; it is not silently dropped.

Two absences, for the record: the rusty-SUNDIALS workspace has no KINSOL (a Newton solver for $F(x) = 0$ directly), and its `crates/ida` is a documented fixed-step prototype that the study does not use.

3.3.4 What the stopping rule guarantees

Proposition 3.4. *[Tier C] Suppose E is exactly quadratic near a local minimum x^*, $E(x) = E(x^*) + \frac{1}{2}\delta^\mathsf{T} H \delta$ with $\delta = x - x^*$, $H \succeq 0$, $\ker H$ spanned by translations, and smallest positive eigenvalue λ_*. If the gradient $g = \nabla E(x) = H\delta$ satisfies $\max_k |g_k| < \varepsilon$, then*

$$0 \le E(x) - E(x^*) \le \frac{|g|^2}{2\lambda_*} < \frac{2N\varepsilon^2}{2\lambda_*}. \tag{3.13}$$

Human proof. Translations do not change E, so we may take $\delta \perp \ker H$. Then $\delta = H^+ g$ and $E(x) - E(x^*) = \frac{1}{2} g^\mathsf{T} H^+ g \le |g|^2/(2\lambda_*)$, and $|g|^2 \le 2N \max_k |g_k|^2$. □

With $N = 288$, $\varepsilon = 10^{-9}$ and $\lambda_* \ge 5.4 \times 10^{-5}$ this gives $E(x) - E(x^*) < 5.3 \times 10^{-12}$, i.e. less than 2×10^{-14} per particle, far below the P1 tolerance $10^{-9}|e_{\text{tri}}| \approx 4 \times 10^{-10}$. The argument is **[Tier C]**: real energies are not exactly quadratic and the bound uses the measured λ_*, but it shows that the stopping rule is not the weak point.

3.3.5 The Hessian spectrum as a minimum check

A converged run is a critical point; it could still be a saddle. The second-order test settles it.

Proposition 3.5. *Let E be C^2 near a critical point x^*, invariant under translations, with $H(x^*) \succeq 0$ and $\ker H(x^*)$ equal to the translation plane $\operatorname{span}\{\tau_x, \tau_y\}$. Then x^* is a strict local minimum modulo translation: there is $\rho > 0$ with $E(x) > E(x^*)$ for every x with $0 < \operatorname{dist}(x, x^* + \operatorname{span}\{\tau_x, \tau_y\}) < \rho$.*

Human proof. Restrict E to the affine slice $x^* + W$, $W = \operatorname{span}\{\tau_x, \tau_y\}^\perp$. There the Hessian is positive definite, so x^* is a strict local minimum on the slice (the usual second-order sufficient condition). Every point near the orbit $x^* + \operatorname{span}\{\tau_x, \tau_y\}$ is a translate of a point of the slice near x^*, and E is translation invariant. □

On a torus, rotations are *not* symmetries (they do not preserve the box), so for a periodic configuration we expect exactly two zero modes. (For the disk with a harmonic bowl, translations are broken and a rotation is a symmetry, so one expects one zero mode there; the disk runs were not Hessian-checked.)

The eigenvalues come from cyclic Jacobi rotations (`symmetric_eigenvalues`, `lib.rs` lines 302–343), which are slow ($O(m^3)$ per sweep, $m = 576$) but simple, and accurate for small eigenvalues of symmetric matrices. The crate checks it on the tridiagonal matrix with known eigenvalues $2 - 2\cos(k\pi/(m+1))$. The verdict on each run:

main.rs, lines 65–72

```
/// Lowest Hessian eigenvalues (after the two translation zero modes are included) for a
    periodic configuration.
fn hessian_summary(sys: &System, x: &[f64]) -> (f64, f64, f64, usize) {
    let m = x.len();
    let ev = symmetric_eigenvalues(&sys.hessian(x), m);
    let scale = ev[m - 1].abs();
    let neg = ev.iter().filter(|&&v| v < -1e-9 * scale).count();
    (ev[0], ev[1], ev[2], neg)
}
```

An eigenvalue counts as negative if it is below $-10^{-9}\lambda_{\max}$; the three lowest eigenvalues are recorded so that one can see the two translation modes and the gap above them.

3.3.6 Annealing, outside SUNDIALS

A $T = 0$ quench gets stuck. To give the lattice a chance, the study first *anneals*: overdamped Langevin dynamics

$$dX_t = F(X_t)\,dt + \sqrt{2T(t)}\,dW_t, \tag{3.14}$$

with W a standard Brownian motion in $\mathbb{R}^{2N}$ and a temperature lowered from $T_0 = 0.05$ to $T_1 = 10^{-5}$. At fixed T the stationary law is the Boltzmann distribution $\propto e^{-E/T}$, so slow cooling concentrates near low minima; how slow is "slow enough" is exactly what the study tests (P2).

Equation (3.14) is a stochastic differential equation, not an ODE. Brownian paths are nowhere differentiable, so there is no smooth right-hand side for CVODE to interpolate, and CVODE's local error estimates, which compare polynomial predictors with correctors, have no meaning. A stochastic scheme is needed. The code uses Euler–Maruyama, $X_{k+1} = X_k + F(X_k)\,\Delta t + \sqrt{2T_k\Delta t}\,\xi_k$ with ξ_k standard Gaussian, and a geometric schedule $T_{k+1} = T_k(T_1/T_0)^{1/K}$:

lib.rs, lines 488–503

```
    let mut rng = Rng::new(seed);
    let mut x = x0.to_vec();
    let mut f = vec![0.0; x.len()];
    let ratio = (t1 / t0).powf(1.0 / steps as f64);
    let mut temp = t0;
    for _ in 0..steps {
        sys.force(&x, &mut f);
        let amp = (2.0 * temp * dt).sqrt();
        for k in 0..x.len() {
            let d = (f[k] * dt + amp * rng.gauss()).clamp(-cap, cap);
            x[k] += d;
        }
        sys.wrap(&mut x);
        temp *= ratio;
    }
    x
```

The study calls it with $K = 300\,000$ steps, $\Delta t = 0.01$ and `cap` $= 0.05$ (`anneal(sys, x0, 0.05, 1e-5, 300_000, 0.01, 0.05, seed)` in `main.rs`). Note the `clamp`: each coordinate increment

is capped at 0.05 so that a close encounter cannot throw a particle across the box. With the cap the scheme is a *capped* Euler–Maruyama scheme, not a consistent discretisation of (3.14); it is a heuristic search, which is all P2 asks of it. Every annealed state is then quenched with CVODE as above, so the reported end states are critical points of E, whatever the annealing did.

3.4 Results of the preregistered study

Every number in this section is **[Tier X]** and comes from the study's recorded results (rusty-SUNDIALS commit `1f46e70`). Initial conditions are of three kinds: the perfect and noisy lattices (controls), uniform random points with minimum separation $0.5a$, and points derived from the Hugging Face dataset `colabfit/OC20_IS2RES_val_id` [16] (CC-BY-4.0, shard `co/co_10.parquet`): atomic structures rotated at random in 3D, projected onto the plane and accepted with minimum separation $0.5a$. The dataset supplies disorder only; nothing depends on its chemistry.

3.4.1 References

	e_{tri}	e_{sq}	$e_{\mathrm{sq}} - e_{\mathrm{tri}}$
$\kappa = 2$	0.3935333668	0.4009647699	7.43×10^{-3}
$\kappa = 1$	1.6076092172	1.6171455730	9.54×10^{-3}

Table 3.1: Lattice energies per particle at density one, shifted-force Yukawa with $r_c = 8$ (**[Tier X]**). The periodic-box energy of the perfect lattice and the direct lattice sum agree to about 10^{-14}. The Python companion of Section 3.6 reproduces $e_{\mathrm{tri}}(\kappa = 2)$ independently.

The triangular lattice is below the square one at both screening lengths, as the theorem (and, for lattices alone, the older results of Rankin, Cassels and Montgomery [5, 6, 7] on Epstein zeta and theta functions) predicts. Figure 3.1 shows the references.

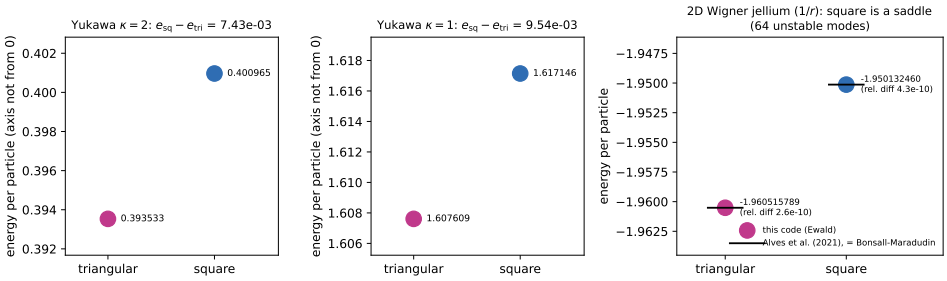


Figure 3.1: Reference energies per particle at density one (the vertical axes do not start at zero). Left and middle: shifted-force Yukawa at the two values of κ, triangular below square. Right: the jellium extension of Section 3.5. **[Tier X]**.

3.4.2 Controls

The negative controls matter most. The square lattice is a critical point of E (by symmetry every particle feels zero net force), so a code that only checked $\max |F|$ would accept it as an equilibrium. The Hessian test rejects it with dozens of negative eigenvalues. The incommensurate box cannot hold the triangular lattice, and indeed no relaxed state there reaches e_{tri} or is defect-free: the defect counter and the energy gap both fire.

control	expected	observed
perfect lattice (positive)	stays at e_{tri}	gap 5.2×10^{-14}
lattice + noise $0.1a$ (positive)	relaxes back to e_{tri}	gap 2.5×10^{-14}
Hessian at the lattice (positive)	exactly 2 zero modes, then a gap	2 zero modes, then 0.0236
square lattice in 17×17 box (negative)	a saddle: negative eigenvalues	72 negative eigenvalues ($\kappa = 2$), 64 ($\kappa = 1$)
incommensurate square box, $N = 289$ (negative)	every relaxed state defected, above e_{tri}	all defected, minimum gap 9.6×10^{-4} ($\kappa = 2$)

Table 3.2: Positive and negative controls ([Tier X]). Each check is shown to pass where it must and to fire where it must before any production number is read.

3.4.3 The three predictions

P1 (no undercut): PASS. At $\kappa = 2$, 0 undercuts in 100 relaxations, 20 of them in the incommensurate control box. At $\kappa = 1$, 0 undercuts in 50 relaxations. (Each run of the main study is counted twice, as a $T = 0$ quench and as an annealed-then-quenched state.) Figure 3.2 shows every relaxation's gap.

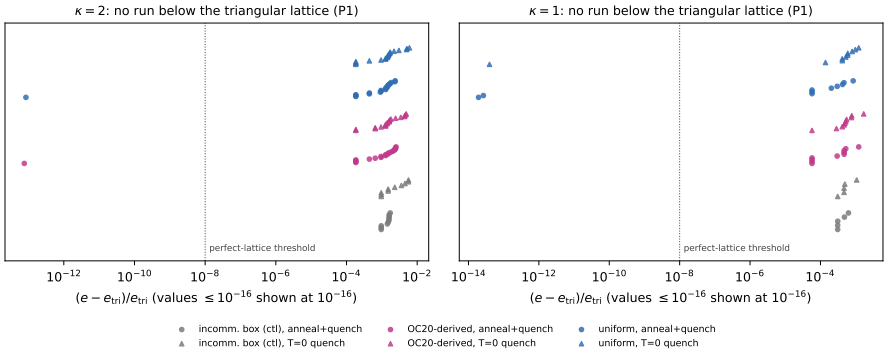


Figure 3.2: Each relaxation's relative energy gap $(e - e_{\text{tri}})/e_{\text{tri}}$, one marker per run, on a logarithmic axis. P1 concerns the sign of the gap: no gap is negative. Markers at the far left are runs that reached the lattice to round-off. [Tier X].

P2 (annealing reaches the lattice at least once per family). At $\kappa = 2$: PASS, narrowly. One of 20 OC20-derived and one of 20 uniform annealed runs reached the perfect lattice (zero defects and $|e - e_{\text{tri}}| < 10^{-8}|e_{\text{tri}}|$). At $\kappa = 1$: **FAIL as preregistered**: 0 of 10 (OC20-derived) and 2 of 10 (uniform). The failure is a statement about the annealing schedule (3×10^5 steps of capped Euler–Maruyama), which is a weak instrument at $N = 288$. It says nothing about the theorem in either direction, and it is reported, not hidden, as the preregistration requires.

P3 (quenches are glassy): as expected. At $\kappa = 2$, 15 of 20 (OC20-derived) and 14 of 20 (uniform) $T = 0$ quenches end with defects.

Hessian coverage. The final Hessian was computed for the 80 ($\kappa = 2$) and 40 ($\kappa = 1$) non-control runs. None has a negative eigenvalue. The two translation modes satisfy $|\lambda| \leq 1.5 \times 10^{-15}$ ($\kappa = 2$) and $\leq 1.8 \times 10^{-15}$ ($\kappa = 1$), and every other eigenvalue is at least 5.4×10^{-5} ($\kappa = 2$) and 5.0×10^{-5} ($\kappa = 1$). By Proposition 3.5, up to floating point, these end states are strict local minima modulo translation. Ten orders of magnitude separate the zero modes from the smallest

genuine eigenvalue, so the classification is not borderline. The incommensurate-box and disk runs were not Hessian-checked.

Read together: many distinct local minima exist, every one found lies above the lattice, and annealing finds the lattice only occasionally. The theorem provides none of this landscape information.

3.4.4 Open versus closed disk (confounded)

The Lean energy counts points in *closed* disks (Chapter 1); the D4 work asks whether open disks give the same answer. A tempting "experiment" is to confine $N = 300$ particles ($\kappa = 2$) in an open disk (harmonic bowl $U = kr^2/2$, $k = 0.02$) and in a closed disk (soft wall $U = \varepsilon(r - R)^4$ for $r > R$, $\varepsilon = 100$), anneal and quench, and compare. Results: interior defect fraction 0.086 (open) versus 0.033 (closed); edge defect fraction 0.965 versus 0.602; interior mean $|\psi_6|$ 0.913 versus 0.931.

Caution : a confounded comparison

These numbers do not measure an open-versus-closed effect. The bowl produces a non-uniform density; the wall compresses particles against the edge; and the two "edge" layers (within two spacings of the outermost particle) contain different numbers of particles. The study describes finite-size edge effects in two different set-ups. It is not a test of upstream's closed-disk counting, which is a question about measure-zero boundary terms in a limit $R \to \infty$ and is settled (or not) by proof, not by simulation.

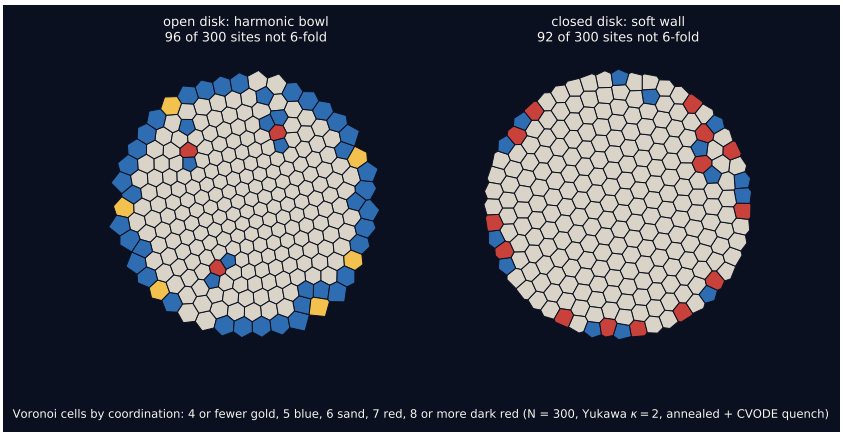


Figure 3.3: Open (harmonic bowl) versus closed (soft wall) disk, $N = 300$, $\kappa = 2$: Voronoi cells coloured by coordination, one run of each regenerated from its published seed. Defects concentrate at the edge in both cases. **[Tier X]**, and confounded as stated in the text.

The order parameters used throughout are those of `order` (`lib.rs`, lines 360–420): d_0 is the median nearest-neighbour distance, two particles are neighbours if they are closer than $1.35\,d_0$, a particle is a defect if it does not have exactly 6 neighbours, and $\psi_6(i) = \frac{1}{c_i}\sum_{j\sim i} e^{6\mathrm{i}\theta_{ij}}$ is the bond-orientational order of particle i, with θ_{ij} the angle of the bond. For the perfect triangular lattice $|\psi_6(i)| = 1$; for the square lattice $\psi_6(i) = 0$, since $\sum_{m=0}^{3} e^{6\mathrm{i}(m\pi/2)} = 0$.

3.5 The jellium extension: $s = 1$ with Ewald summation

3.5.1 Why a background

Here 'Coulomb' means the three-dimensional $1/r$ law restricted to the plane (Riesz $s = 1$); the planar Coulomb (logarithmic, $s = 0$) interaction is distinguished where it matters. For the

Coulomb potential $1/r$ (Riesz $s = 1$ in the plane) the lattice sum $\sum_{v\neq 0} |v|^{-1}$ diverges: there are about $2\pi r\,dr$ points at distance r, and $\int^{\infty} 2\pi r \cdot r^{-1}\,dr = \infty$. Every competitor of density one has infinite energy, and the theorem says nothing (`latticeEnergy_riesz_lt_top` needs $s > 2$, and the inequality $\infty \le \infty$ is empty). Physics removes the divergence with a uniform neutralising background of opposite charge: the *one-component plasma*, or *jellium*. Its energy per particle is finite, and for $s = 1$ the classical ground state is the two-dimensional Wigner crystal. Upstream's manuscript treats renormalized energies for $0 \le s < 2$ in its Section 8, following Petrache and Serfaty [4], and concludes that the triangular lattice minimizes them [1] **[Tier L]**. This is *not* formalised in Lean anywhere. The computation below is a numerical first step, **[Tier X]**.

3.5.2 The Ewald energy

Let N unit charges sit at $x_1, \dots, x_N$ in a periodic rectangle of area $\mathcal{A} = L_x L_y$, with background density $n = N/\mathcal{A}$. Choose a splitting parameter $\alpha > 0$ and write

$$\frac{1}{r} = \underbrace{\frac{\operatorname{erfc}(\alpha r)}{r}}_{\text{short range}} + \underbrace{\frac{\operatorname{erf}(\alpha r)}{r}}_{\text{smooth, long range}}. \tag{3.15}$$

Proposition 3.6. *With $S(k) = \sum_j e^{\mathrm{i}k\cdot x_j}$ and k ranging over the reciprocal lattice $\{(2\pi m/L_x, 2\pi n'/L_y)\}$, the energy of the neutral periodic system (particles plus background) is*

$$E = \frac{1}{2}{\sum_{i,j}}' \frac{\operatorname{erfc}(\alpha r_{ij})}{r_{ij}} + \frac{\pi}{\mathcal{A}} \sum_{k\neq 0} \frac{\operatorname{erfc}(k/2\alpha)}{k} |S(k)|^2 - \frac{N\alpha}{\sqrt{\pi}} - \frac{N^2\sqrt{\pi}}{\alpha\mathcal{A}}, \tag{3.16}$$

where $\sum'$ runs over all pairs and periodic images except $i = j$ in the same cell. E does not depend on α.

Human proof. **[Tier C]** (the classical derivation; we give its steps.) *Long-range part.* The function $\operatorname{erf}(\alpha r)/r$ is smooth and its two-dimensional Fourier transform is $\frac{2\pi}{k}\operatorname{erfc}\big(\frac{k}{2\alpha}\big)$. Summing it over all pairs and images (including $i = j$) is a sum over the periodic charge density, which by the Poisson summation formula equals $\frac{1}{2\mathcal{A}}\sum_k \frac{2\pi}{k}\operatorname{erfc}(\frac{k}{2\alpha})|S(k)|^2$. The $k \neq 0$ terms give the second term of (3.16). The $k = 0$ term diverges, like $2\pi N^2/(2\mathcal{A}k)$, and is cancelled exactly by the long-range interaction of the particles with the background and of the background with itself: a neutral system has no $k = 0$ Fourier component. *Self term.* The long-range sum included $i = j$ at distance 0, which is not a real interaction. Its value $\lim_{r\to 0}\operatorname{erf}(\alpha r)/r = 2\alpha/\sqrt{\pi}$, with the factor $\frac{1}{2}$ and N particles, gives $-N\alpha/\sqrt{\pi}$. *Background in the short-range part.* The background also interacts through the short-range kernel. With $\int_{\mathbb{R}^2} \frac{\operatorname{erfc}(\alpha r)}{r}\,d^2x = 2\pi\int_0^{\infty}\operatorname{erfc}(\alpha r)\,dr = \frac{2\sqrt{\pi}}{\alpha}$, the particle–background energy is $-Nn\frac{2\sqrt{\pi}}{\alpha}$ and the background–background energy is $+\frac{1}{2}Nn\frac{2\sqrt{\pi}}{\alpha}$; the sum is $-\frac{N^2\sqrt{\pi}}{\alpha\mathcal{A}}$. *Independence of α.* Only the splitting of $1/r$ changes with α, not the total. □

The code states the same formula in its header and implements it with a half-space k-sum (since $|S(-k)| = |S(k)|$, each k is taken once with weight 2):

jellium.rs, lines 4–9

```
//! Energy (with `A = Lx Ly`, `S(k) = sum_j exp(i k.x_j)`):
//! `E = 1/2 sum'_{i,j} erfc(a r_ij)/r_ij + (pi/A) sum_{k != 0} erfc(k/(2a))/k |S(k)|^2 - N a/
    sqrt(pi) - N^2 sqrt(pi)/(a A)`,
//! with the real-space sum over minimum images inside `rc` (requires `rc <= min(Lx, Ly)/2`).
//! The last term is what remains of the particle-background and background-background
    interactions after the
```

```
//! divergent ‘k = 0‘ parts cancel. This is the renormalized (jellium) energy of upstream’s
    Section 8 for ‘s = 1‘, up to
//! upstream’s normalisation constant.
```

The last sentence of that comment is a **[Tier C]** identification (the normalisation constant is not computed here). The constructor `Jellium::new` (lines 27–61) stores, for each half-space vector k with $|k| < k_{\mathrm{cut}}$, the coefficient `c = 2.0 * libm::erfc(k / (2.0 * alpha)) / k`; the factor 2 is the weight of the pair $\pm k$.

jellium.rs, lines 85–110

```
    /// Total energy.
    pub fn energy(&self, x: &[f64]) -> f64 {
        let n = x.len() / 2;
        let geo = self.geometry_system();
        let area = self.lx * self.ly;
        let mut real = 0.0;
        for i in 0..n {
            for j in (i + 1)..n {
                let (dx, dy) = geo.delta(x[2 * i], x[2 * i + 1], x[2 * j], x[2 * j + 1]);
                let r = (dx * dx + dy * dy).sqrt();
                if r < self.rc {
                    real += libm::erfc(self.alpha * r) / r;
                }
            }
        }
        let mut recip = 0.0;
        for &(kx, ky, c) in &self.kvecs {
            let (re, im) = self.structure(x, kx, ky);
            recip += c * (re * re + im * im);
        }
        recip *= std::f64::consts::PI / area;
        let nf = n as f64;
        real + recip
            - nf * self.alpha / std::f64::consts::PI.sqrt()
            - nf * nf * std::f64::consts::PI.sqrt() / (self.alpha * area)
    }
```

The real-space sum keeps only minimum images inside $r_c = 8$ and the reciprocal sum only $|k| < k_{\mathrm{cut}} = 7$, with $\alpha = 0.66$ (`tri_jellium` in `main.rs`). Both truncations are tiny: $\mathrm{erfc}(\alpha r_c) = \mathrm{erfc}(5.28) \approx 8.2 \times 10^{-14}$ and $\mathrm{erfc}(k_{\mathrm{cut}}/2\alpha) \approx 6.4 \times 10^{-14}$.[5]

The force follows by differentiating (3.16). The real-space pair force has magnitude $-\frac{d}{dr}\frac{\mathrm{erfc}(\alpha r)}{r} = \frac{\mathrm{erfc}(\alpha r)}{r^2} + \frac{2\alpha}{\sqrt{\pi}}\frac{e^{-\alpha^2 r^2}}{r}$. For the reciprocal part, with $\phi_i = k \cdot x_i$, $\partial_{x_i}|S(k)|^2 = 2\,\mathrm{Re}\big(\mathrm{i}k\, e^{\mathrm{i}\phi_i}\overline{S(k)}\big) = -2k\,\mathrm{Im}\big(e^{\mathrm{i}\phi_i}\overline{S(k)}\big)$ and $\mathrm{Im}(e^{\mathrm{i}\phi}\overline{S}) = \mathrm{Re}S\sin\phi - \mathrm{Im}S\cos\phi$. The code (lines 134–145) implements exactly this, with the comment `// Im(conj(S) e^{i ph}) = sre sin(ph) - sim cos(ph)` and the prefactor `pref = 2.0 * std::f64::consts::PI / area`, and the unit test `force_matches_finite_difference` checks it against the energy. Unlike the Yukawa case, the jellium relaxation does *not* pass an analytic Jacobian: `Jellium::relax` builds CVODE without `.jacobian(...)`, so CVODE forms its own difference-quotient Jacobian (the doc comment says “finite-difference Jacobian”). The Hessian used for the spectrum checks is `hessian_fd`: central finite differences of the analytic force with step 10^{-5}, symmetrised. Annealing uses the same capped Euler–Maruyama with 10^5 steps, because each force costs a k-sum.

[5] Printed by the scratch check of footnote 4: `erfc(0.66*8) = 8.201410123512745e-14 erfc(7/(2*0.66)) = 6.400438678401495e-14`.

3.5.3 Controls

The jellium extension was preregistered separately (rusty-SUNDIALS branch `physics/jellium-wigner-2026-10-11`; preregistration `e706f85`, code `3bc55fa`, credit to Alves et al. `b7e4380`). Its controls:

- *Madelung energies against the literature.* At density one ($r_s = 1/\sqrt{\pi}$) the energy per particle is $\eta\sqrt{\pi}$, with η the Madelung constant in Hartree$\times r_s$. The crate uses the ten-digit values of Alves, Bendazzoli, Evangelisti and Berger [9], which agree with Bonsall and Maradudin [8]:

  jellium.rs, lines 252–253

  ```
  pub const ETA_TRIANGULAR: f64 = -1.106102587;
  pub const ETA_SQUARE: f64 = -1.100244420;
  ```

  The Ewald code gives $e_{\mathrm{tri}} = -1.960515789320$ against the literature value -1.960515789825 (relative difference 2.6×10^{-10}), and $e_{\mathrm{sq}} = -1.950132460$ (relative difference 4.3×10^{-10}). This is the rounding level of nine-decimal published constants. **[Tier L]** for the literature values, **[Tier X]** for ours.

- *Square saddle.* The square lattice has 64 unstable modes (negative eigenvalues of `hessian_fd`), consistent with the imaginary phonon branch reported by Bonsall and Maradudin [8].

- *α-invariance.* Proposition 3.6 says E does not depend on α. The test perturbs the lattice by noise of standard deviation 0.2 and compares three values of α:

  jellium.rs, lines 304–311

  ```
      // A 4x range of the splitting parameter, each with its k-space cutoff scaled so
  both truncations stay ~1e-13.
      let e: Vec<f64> = [0.66, 1.32, 2.64]
          .iter()
          .map(|&al| Jellium::new(lx, ly, al, 8.0, 10.6 * al).energy(&x))
          .collect();
      for v in &e[1..] {
          assert!(((v - e[0]) / e[0]).abs() < 1e-10, "{:?}", e);
      }
  ```

  The energy changes by less than 10^{-10} relative over this $4\times$ range. A wrong self term or background term would break this test, because those terms depend on α explicitly: it is a control that fires on the most likely bugs.

3.5.4 The main jellium relaxation study

The preregistered prediction (J1) is the analogue of P1: no relaxed configuration below the triangular Madelung energy. Status of the main study: the main jellium relaxation study (prediction J1) was still running when this book was assembled.

3.6 A from-scratch Python companion

The Rust crate is the instrument of record; for learning, a script short enough to read in one sitting is better. The script below, `code/ch3_yukawa_small.py` (78 lines, numpy and scipy only), builds the same shifted-force Yukawa model in a small commensurate box, $n \times 2n$ triangular sites, $N = 2n^2$, area N, and minimises the energy with L-BFGS from a perturbed lattice and from five random starts.

Caution : L-BFGS is not CVODE

L-BFGS is a quasi-Newton *minimiser*: it takes line-search steps downhill using a low-rank curvature model. It does not follow the gradient-flow trajectory (3.9), so from the same start it can end in a different local minimum than CVODE. The script computes no Hessian, so it cannot distinguish a minimum from a saddle. Everything it prints is **[Tier X]**.

code/ch3_yukawa_small.py

```
"""Chapter 3 companion (Tier X): relax a small periodic Yukawa system with L-BFGS and compare
    with the lattice.

Model as in rusty-SUNDIALS qf-yukawa-crystal: V(r) = exp(-kappa r)/r, shifted-force cutoff rc,
    periodic box that
holds n x 2n triangular sites (N = 2 n^2) at density 1. L-BFGS is a minimiser, not the CVODE
    gradient flow.
Usage: python ch3_yukawa_small.py [n] [kappa] [rc]
"""
from __future__ import annotations
import sys
import numpy as np
from scipy.optimize import minimize
n = int(sys.argv[1]) if len(sys.argv) > 1 else 6
KAPPA = float(sys.argv[2]) if len(sys.argv) > 2 else 2.0
RC = float(sys.argv[3]) if len(sys.argv) > 3 else 6.0
A = np.sqrt(2.0 / np.sqrt(3.0))                          # triangular spacing at density 1
L = np.array([n * A, n * np.sqrt(3.0) * A])              # box n a x n sqrt(3) a, area N
N = 2 * n * n
v = lambda r: np.exp(-KAPPA * r) / r                      # noqa: E731
dv = lambda r: -np.exp(-KAPPA * r) * (KAPPA * r + 1.0) / r**2  # noqa: E731
VC, DVC = v(RC), dv(RC)
M = np.ceil(RC / L + 0.5).astype(int)                    # image shells needed beyond the minimum
    image
SHIFTS = np.array([(i * L[0], j * L[1]) for i in range(-M[0], M[0] + 1) for j in range(-M[1], M
    [1] + 1)])

def energy_grad(flat: np.ndarray) -> tuple[float, np.ndarray]:
    """E = (1/2) sum_{i,j,images} V_sf(r) and its gradient (sum over all images inside rc)."""
    x = flat.reshape(N, 2)
    d = x[:, None, :] - x[None, :, :]
    d = d - L * np.round(d / L)
    d = d[:, :, None, :] + SHIFTS[None, None, :, :]
    r = np.linalg.norm(d, axis=-1)
    inside = (r < RC) & (r > 1e-12)
    rs = np.where(inside, r, RC)
    e = np.where(inside, v(rs) - VC - (rs - RC) * DVC, 0.0)
    g = np.where(inside, dv(rs) - DVC, 0.0)
    return 0.5 * float(e.sum()), ((g / rs)[..., None] * d).sum(axis=(1, 2)).ravel()

def triangular() -> np.ndarray:
    return np.array([((i + 0.5 * (j % 2) + 0.25) * A, (j + 0.5) * A * np.sqrt(3.0) / 2.0)
                     for j in range(2 * n) for i in range(n)])

def lattice_sum() -> float:
    """Direct sum (1/2) sum_{v != 0} V_sf(|v|) over the infinite triangular lattice."""
    m = int(RC / (A * np.sqrt(3.0) / 2.0)) + 3
    tot = 0.0
    for i in range(-m, m + 1):
        for j in range(-m, m + 1):
            r = np.hypot(i * A + 0.5 * j * A, j * A * np.sqrt(3.0) / 2.0)
            if 0.0 < r < RC:
```

```
                tot += v(r) - VC - (r - RC) * DVC
    return 0.5 * tot

def defects(x: np.ndarray) -> int:
    d = x[:, None, :] - x[None, :, :]
    r = np.linalg.norm(d - L * np.round(d / L), axis=-1) + np.eye(N) * 1e9
    return int(((r < 1.35 * np.median(r.min(axis=1))).sum(axis=1) != 6).sum())

def random_start(rng: np.random.Generator, dmin: float) -> np.ndarray:
    pts: list[np.ndarray] = []
    while len(pts) < N:
        p = rng.uniform(0.0, 1.0, 2) * L
        if all(np.linalg.norm((p - q) - L * np.round((p - q) / L)) >= dmin for q in pts):
            pts.append(p)
    return np.array(pts)

def relax(x0: np.ndarray) -> tuple[float, float, int, np.ndarray]:
    res = minimize(energy_grad, x0.ravel(), jac=True, method="L-BFGS-B",
                   options={"maxiter": 50000, "gtol": 1e-10, "ftol": 1e-15})
    return res.fun / N, float(np.abs(res.jac).max()), res.nit, res.x.reshape(N, 2) % L

lat = triangular()
e_box = energy_grad(lat.ravel())[0] / N
print(f"N={N} kappa={KAPPA} rc={RC} box={L[0]:.6f}x{L[1]:.6f}")
print(f"e_tri box sum = {e_box:.12f}  direct lattice sum = {lattice_sum():.12f}")
rng = np.random.default_rng(0)
starts = [("perturbed lattice 0.1a", lat + 0.1 * A * rng.normal(size=lat.shape))]
starts += [(f"random start seed {s}", random_start(np.random.default_rng(s), 0.5 * A)) for s in
     range(1, 6)]
for name, x0 in starts:
    e, gmax, nit, x = relax(x0)
    print(f"{name:24s} e-e_tri = {e - e_box: .3e}  max|grad| = {gmax:.1e}  iters = {nit:5d}
    defects = {defects(x)}")
```

One design point differs from the Rust code. At $n = 6$ the box is 6.447×11.167, so $r_c = 6$ (or 8) exceeds $L_x/2$, and the minimum image alone would miss pairs. The script therefore adds image shells, $M = \lceil r_c/L + \frac{1}{2} \rceil$ copies in each direction beyond the minimum image, and sums over all images inside r_c, including a particle's own images. The Rust code can use the minimum image only because its box is large enough ($8 < 8.375$). The gradient is $\sum_{j,\text{images}} V'_{\text{sf}}(r)\, d/r$; the factor $\frac{1}{2}$ of the energy disappears because each pair appears twice in the double sum (Exercise 3.1).

We ran the script with its defaults ($n = 6$, $\kappa = 2$, $r_c = 6$) on two cores (`taskset -c 2,3`); it took 26.8 s of wall-clock time. The exact output:

python -I code/ch3_yukawa_small.py

```
N=72 kappa=2.0 rc=6.0 box=6.447420x11.167258
e_tri box sum = 0.393231076156  direct lattice sum = 0.393231076156
perturbed lattice 0.1a   e-e_tri =  1.332e-15  max|grad| = 1.2e-07  iters =    62  defects = 0
random start seed 1      e-e_tri =  7.772e-16  max|grad| = 4.0e-08  iters =    95  defects = 0
random start seed 2      e-e_tri =  1.903e-03  max|grad| = 4.8e-08  iters =   165  defects = 4
random start seed 3      e-e_tri =  2.220e-16  max|grad| = 7.5e-08  iters =   112  defects = 0
random start seed 4      e-e_tri =  6.661e-16  max|grad| = 3.6e-08  iters =   146  defects = 0
random start seed 5      e-e_tri =  1.833e-03  max|grad| = 2.1e-07  iters =   222  defects = 4
```

And with the study's cutoff, `python -I code/ch3_yukawa_small.py 6 2 8`:

python -I code/ch3_yukawa_small.py 6 2 8

```
N=72 kappa=2.0 rc=8.0 box=6.447420x11.167258
e_tri box sum = 0.393533366826  direct lattice sum = 0.393533366826
perturbed lattice 0.1a   e-e_tri =  1.388e-15  max|grad| = 1.2e-07  iters =    62  defects = 0
random start seed 1      e-e_tri =  7.772e-16  max|grad| = 7.5e-08  iters =    94  defects = 0
random start seed 2      e-e_tri =  1.903e-03  max|grad| = 1.3e-07  iters =   160  defects = 4
```

```
random start seed 3      e-e_tri =  8.882e-16  max|grad| = 5.6e-08  iters =   114  defects = 0
random start seed 4      e-e_tri =  9.437e-16  max|grad| = 4.4e-08  iters =   145  defects = 0
random start seed 5      e-e_tri =  1.833e-03  max|grad| = 4.7e-08  iters =   221  defects = 4
```

How to read this output honestly:

- *An independent reference check.* With $r_c = 8$ the script's box energy and direct lattice sum, 0.393533366826, agree with the Rust reference $e_{\mathrm{tri}}(\kappa = 2) = 0.3935333668$ of Table 3.1, from an independent implementation in another language with a different image scheme and $N = 72$ instead of 288. (The lattice energy per particle does not depend on N for a commensurate box.) With $r_c = 6$ the value is 0.393231076156: the cutoff changes the fourth significant digit at $\kappa = 2$, which is why both codes compare equally truncated energies.

- *The stopping rule was not met as asked.* The script requests `gtol=1e-10`, but the final $\max|\nabla E|$ is between 4×10^{-8} and 2×10^{-7}: L-BFGS stopped on its relative energy-decrease criterion `ftol`. (A scratch rerun of two starts, the perturbed lattice and seed 2, printed scipy's message `CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH` for both.) This is looser than the Rust rule $\max|F| < 10^{-9}$. For the lattice runs the energy agrees with e_{tri} to about 10^{-15} anyway, as Proposition 3.4 leads one to expect (the error is quadratic in the gradient).

- *No undercut*, in twelve relaxations. Three of the five random starts (seeds 1, 3, 4) reached the lattice (within 10^{-15}, zero defects); two (seeds 2, 5) ended 1.8–1.9 $\times 10^{-3}$ above it with 4 defects. Do *not* compare "2 of 5 defected" with P3 (15 of 20 quenches defected): system size (72 versus 288), algorithm (L-BFGS versus gradient flow) and cutoff all differ.

- *No designed negative control.* The two defected end states show that the defect counter and the energy gap *can* fire, but that is incidental, not designed. By the standard of Section 3.1 the script is incomplete until you add one; Exercises 3.5 and 3.6 do that.

3.7 How to reproduce the Rust study

All commands are run from the rusty-SUNDIALS checkout (branch `physics/jellium-wigner-2026-10-11` for the jellium commands). The binary is `yukawa_crystal` (`[[bin]]` in the crate's `Cargo.toml`); every subcommand takes the crate directory and writes JSON lines under `<crate_dir>/results/` (`main.rs`, lines 1–3: the "Experiment driver for the preregistered study `results/preregistration.json` (YUK-TRI-1)"). The number of worker threads is read from the environment:

main.rs, lines 260–265

```
    // Worker threads: YUKAWA_THREADS if set, else 2 (the machine is shared; the owner asked
    for 2 cores overnight).
    let threads = std::env::var("YUKAWA_THREADS")
        .ok()
        .and_then(|v| v.parse::<usize>().ok())
        .filter(|&t| t > 0)
        .unwrap_or(2);
```

The workflow:

Reproduction (one command per line; run from the repository root)

```
cargo test --release -p qf-yukawa-crystal
cargo build --release -p qf-yukawa-crystal

# initial conditions: OC20 shard co/co_10.parquet from colabfit/OC20_IS2RES_val_id
```

```
python crates/qf-yukawa-crystal/scripts/hf_initial_conditions.py co_10.parquet crates/qf-yukawa
    -crystal/data/initial

YUKAWA_THREADS=2 target/release/yukawa_crystal refs     crates/qf-yukawa-crystal
YUKAWA_THREADS=2 target/release/yukawa_crystal controls crates/qf-yukawa-crystal
YUKAWA_THREADS=2 target/release/yukawa_crystal main     crates/qf-yukawa-crystal 2 20
YUKAWA_THREADS=2 target/release/yukawa_crystal main     crates/qf-yukawa-crystal 1 10
YUKAWA_THREADS=2 target/release/yukawa_crystal disk     crates/qf-yukawa-crystal
YUKAWA_THREADS=2 target/release/yukawa_crystal snapshot crates/qf-yukawa-crystal
python crates/qf-yukawa-crystal/scripts/summarize.py crates/qf-yukawa-crystal

YUKAWA_THREADS=2 target/release/yukawa_crystal jel-controls crates/qf-yukawa-crystal
YUKAWA_THREADS=2 target/release/yukawa_crystal jel-main     crates/qf-yukawa-crystal 20
```

The initial-conditions script writes 20 periodic ($N = 288$), 10 square-box ($N = 289$) and 10 disk ($N = 300$) CSV files plus a `provenance.json`; every command except `refs` and `controls` reads them. `main <dir> <kappa> <nrun>` runs `nrun` OC20-derived and `nrun` uniform starts, each quenched and annealed-then-quenched with Hessian, plus `nrun/2` incommensurate-box controls; the counts of Section 3.4 correspond to `nrun` $= 20$ at $\kappa = 2$ ($2 \cdot (20 + 20) + 2 \cdot 10 = 100$) and `nrun` $= 10$ at $\kappa = 1$ ($2 \cdot (10 + 10) + 2 \cdot 5 = 50$). `snapshot` re-runs selected published runs with their seeds for the figures; `summarize.py` checks P1–P3 with the preregistered tolerances (`REL_TOL_UNDERCUT = 1e-9`, `REL_TOL_LATTICE = 1e-8`); the jellium commands write under `results/jellium/`. Run times: the only times measured for this book are those of the unit tests (0.15 s for the ten tests, after compilation) and of the Python companion (26.8 s). The production commands perform hundreds of CVODE relaxations of a 576-dimensional system with dense Jacobians and 3×10^5-step anneals; we did not time them here and quote no figure. Start with `refs` and `controls`.

3.8 The laboratory analysis script

The tiered paper [20] proposes laboratory tests with colloids and dusty plasmas (protocols L1–L4, **[Tier C]** proposals; none has been run; Chapter 4 discusses the physics). They share one analysis script, `papers/universal_optimality_tiered/lab/lab_analysis.py`, which takes tracked 2D positions and computes the same observables as the simulations: density from the mean Voronoi cell area of the particles over which the energy is averaged; coordination with the $1.35\,d_0$ rule; local and global $|\psi_6|$; and the interaction energy per interior particle for a stated law (`dipolar` r^{-3}, or `yukawa:<kappa>` with κ in the experiment's units), truncated at r_c with a uniform tail correction $\frac{1}{2}\int_{r_c}^{\infty} 2\pi r\, V(r)\, dr$ and compared with a triangular patch of the same density and truncation. Its self-test feeds it four synthetic frames: a triangular patch in arbitrary units (scaled by 3.7), the same with Gaussian noise 0.08, a square patch, and a Yukawa case that checks the unit conversion of κ. We ran `taskset -c 2,3 python -I lab_analysis.py -selftest`; exit code 0. The output is below; we left out fields, marked `[...]` (our insertion), that repeat across blocks or are not discussed.

lab_analysis.py –selftest

```
{
 "triangular": {
  "n": 1600,
  "measured_density_input_units": 0.07304601899196492,
  "law": "dipolar",
  "law_density1": "dipolar",
  "rcut": 6.0,
  "counted": 728,
  "defect_fraction": 0.0,
  "psi6_local_mean": 1.0,
  "psi6_global": 1.0,
  "d0": 1.0745699318235395,
```

```
  [...]
  "e_per_particle": 4.464518999036898,
  "e_tri_same_density": 4.464518999036646,
  "rel_gap": 5.6399935141796596e-14,
  "prediction_e_ge_e_tri": true
 },
 "noisy_triangular": {
  "n": 1600,
  "measured_density_input_units": 1.0006342525142475,
  [...]
  "defect_fraction": 0.2625994694960212,
  "psi6_local_mean": 0.8244091272609662,
  "psi6_global": 0.8076197599391082,
  [...]
  "rel_gap": 0.029497434514365446,
  "prediction_e_ge_e_tri": true
 },
 "square": {
  "n": 1600,
  "measured_density_input_units": 1.0,
  [...]
  "defect_fraction": 1.0,
  "psi6_local_mean": 1.8369701987210297e-16,
  "psi6_global": 1.8369701987210538e-16,
  [...]
  "rel_gap": 0.011590527342694872,
  "prediction_e_ge_e_tri": true
 },
 "yukawa_units": {
  "n": 1600,
  "measured_density_input_units": 0.07304601899196492,
  "law": "yukawa:0.5405405405405405",
  "law_density1": "yukawa:2.0",
  [...]
  "defect_fraction": 0.0,
  [...]
  "e_per_particle_truncated": 0.39353696637141355,
  "e_tri_truncated": 0.3935369663713675,
  "tail_correction": 9.65130619565578e-06,
  [...]
  "rel_gap": 1.1707445433997635e-13,
  "prediction_e_ge_e_tri": true
 }
}
SELFTEST PASS
```


What the self-test shows, and what it does not:

- *Unit invariance.* The triangular patch was given in units 3.7 times too large (density $0.0730\ldots$ in input units, about $1/3.7^2$); after rescaling, the defect fraction is 0, $|\psi_6| = 1$ and the relative gap is 5.6×10^{-14}. The Yukawa case recovers $\kappa_1 = 2$ from $\kappa_{\text{input}} = 2/3.7$.

- *Thermal noise raises the energy.* The noisy patch has relative gap $+0.029$ and local $|\psi_6| = 0.82$.

- *The defect detector fires on the square patch*: defect fraction 1.0, $\psi_6 \approx 10^{-16}$.

- *What it does not show*: no self-test case returns `prediction_e_ge_e_tri = false`. Every input is a configuration at or above the lattice energy, so the energy comparison is never shown to fire. A too-short cutoff does make it fire (Exercise 3.9).

The Yukawa value 0.393537 (plain truncation at $r_c = 6$) is not the same quantity as the shifted-force values 0.393231 and 0.393533 of Section 3.6: only equal truncation conventions may be compared.

3.9 Exercises

All exercises are numerical experiments; their results are **[Tier X]**. Use at most two cores (`taskset -c 2,3`). Before you report any number, run a positive and a negative control.

Exercise 3.1 (Derivatives)**.** Derive ∇E for the image-sum energy $E = \frac{1}{2}\sum_{i,j}\sum_s{}' V_{\mathrm{sf}}(|x_i - x_j + s|)$ of the Python script and show that the factor $\frac{1}{2}$ disappears. Then check `energy_grad` against central finite differences at a random configuration (step 10^{-6}), and verify Proposition 3.3 by finite differences of the gradient for one 2×2 block.

Exercise 3.2 (Varying κ)**.** Run the Python script with $\kappa \in \{0.5, 1, 2, 4\}$ at $n = 6$, $r_c = 8$. Record e_{tri} and how many of the five random starts reach it. Is there a trend? What happens to the image-shell count M and to the run time as $\kappa \to 0$ with r_c fixed, and why must r_c grow as κ decreases for the truncation to stay at the same level?

Exercise 3.3 (Varying N)**.** Run $n = 4, 5, 6, 7, 8$ at $\kappa = 2$. Plot the fraction of random starts that reach the lattice against $N = 2n^2$, with more seeds (say 20) for small n. Explain, in terms of the number of local minima, why the fraction should fall with N. Is your result comparable with P3? List every difference between your experiment and the Rust study.

Exercise 3.4 (Varying the cutoff)**.** At $n = 6$, $\kappa = 1$, compute e_{tri} with $r_c \in \{3, 4, 6, 8, 10\}$ and compare $|e_{\mathrm{tri}}(r_c) - e_{\mathrm{tri}}(10)|$ with $e^{-\kappa r_c}/r_c$ and with the unshifted sum of footnote 2; explain why the shift dominates. Then replace the shifted-force potential by plain truncation ($V(r)$ for $r < r_c$, 0 beyond) and observe what happens to L-BFGS: the energy is discontinuous at r_c. Why does that break a gradient-based minimiser, and why does it not matter for the shifted-force form?

Exercise 3.5 (A designed negative control)**.** Reproduce the incommensurate-box negative control in Python. Put $N = n^2$ particles in a square box of side n (density one) and relax from random starts. Check whether any end state is defect-free (at small n a strained, 6-coordinated state is possible) and whether every one lies above the triangular value e_{tri} computed in a commensurate box. Then plant a bug that affects only one side of a comparison, e.g. drop the factor $\frac{1}{2}$ in `energy_grad` but not in `lattice_sum`, and check that the box-sum versus direct-sum line of the output now disagrees. (Dropping it everywhere would double e and e_{tri} alike, and the comparison would not fire.) A control that does not fire on a planted bug is not a control.

Exercise 3.6 (Find the square-lattice saddle)**.** In the square box of side n with $N = n^2$, place the perfect square lattice. (a) Show that $\max|\nabla E|$ is at round-off level, so a gradient-based stopping rule accepts it. (b) Build the Hessian from (3.8) and count its negative eigenvalues with `numpy.linalg.eigvalsh` using the criterion $\lambda < -10^{-9}\lambda_{\max}$. The Rust study found 72 at $\kappa = 2$ and 64 at $\kappa = 1$ for $N = 289$, $r_c = 8$; your count at smaller n will differ. (c) Perturb the lattice by 10^{-3} along the eigenvector of the most negative eigenvalue and relax. Where does it go, and how far below e_{sq}? (d) At the end state, check that exactly two eigenvalues are at round-off level, with eigenvectors close to τ_x, τ_y; then put the particles in a harmonic bowl instead and explain why one zero mode remains.

Exercise 3.7 (L-BFGS versus gradient flow)**.** Integrate the gradient flow (3.9) for the $n = 4$ system with `scipy.integrate.solve_ivp`, once with `method="RK45"` and once with `method="BDF"` and the analytic Jacobian $-H$. Compare the number of right-hand-side evaluations needed to reach $\max|F| < 10^{-9}$. From the same random start, do L-BFGS and the gradient flow end in the same state? Relate the RK45 cost to the stability bound $h < 2/\lambda_{\max}$.

Exercise 3.8 (Ewald by hand)**.** Write a Python version of (3.16) for $N = 72$ in the box of the script. With the minimum image you need $r_c \le L_x/2 \approx 3.2$, so choose α large enough

that $\operatorname{erfc}(\alpha r_c)$ is negligible, and raise k_{cut} accordingly. (a) Check the Madelung energy of the triangular lattice against $\eta\sqrt{\pi}$ with $\eta = -1.106102587$. (b) Check α-invariance over a $4\times$ range, scaling k_{cut} with α. (c) Plant a bug: drop the background term $-N^2\sqrt{\pi}/(\alpha\mathcal{A})$, and show that the α-invariance test fires.

Exercise 3.9 (Make the lab self-test fire)**.** Find an input for which `lab_analysis.py` reports `prediction_e_ge_e_tri` as `false`, and explain why this is a failure of the analysis assumptions and not of the theorem. Hint: a wrong κ alone will not do it, because the reference uses the same law. A cutoff shorter than the square lattice's second-neighbour distance will: in a scratch call of `analyse` on the self-test's 40×40 square patch with the dipolar law and $r_c = 1.1$, we obtained a truncated energy 2.0 against 2.4178 for the triangular patch, and the flag `False`. Why does the comparison invert?

Chapter 4

Physics: from fluid dynamics to plasmas, and what rusty-SUNDIALS brings

The theorem of the previous chapters is a statement about exact potentials and infinite point configurations. Physics never offers either. It offers finite systems at finite temperature, with interaction laws that are only approximately of the ideal form, confined by walls, bowls or electrodes, and evolving by a dynamics that may or may not seek the minimum of anything. This chapter asks where, in real systems, the triangular lattice appears, which of those systems are actually inside the class of the theorem, and which only resemble it.

The primary domain is *fluid dynamics*: two-dimensional incompressible flow and its point vortices (Section 4.1), and colloidal monolayers at fluid interfaces, whose overdamped motion through a viscous liquid is exactly the gradient-flow model used in Chapter 3 (Section 4.2). We then visit plasmas (Section 4.3), summarise four laboratory protocols (Section 4.4), and ask what the solver library rusty-SUNDIALS [18] does well, does badly, and does not yet do for this physics (Section 4.5). A short survey of other domains (Section 4.6) and exercises close the chapter.

Caution : how to read the tiers in this chapter

Almost nothing in this chapter is Tier A. The kernel-checked statements of Chapter 2 are about the functions $g(t) = t^{-s/2}$ (Riesz, $s > 0$) and about *admissible* potentials in general. Whether a given laboratory system *is* described by such a potential is a modelling statement: Tier C when we argue it, Tier L when a cited paper reports it. An observation in a cited experiment is Tier L; an analogy between two systems is Tier C; a number we computed in floating point is Tier X. A physical system that "looks triangular" is never evidence for the theorem, and a physical system that does not is never evidence against it.

4.1 Two-dimensional Euler flow and point vortices

4.1.1 The vorticity equation

An ideal (inviscid), incompressible fluid in the plane has velocity $u = (u_1, u_2)$ obeying the Euler equations

$$\partial_t u + (u \cdot \nabla) u = -\nabla p, \qquad \nabla \cdot u = 0, \tag{4.1}$$

with the density set to one. Incompressibility lets us write u through a *stream function* ψ[1],

$$u = \nabla^{\perp}\psi := (\partial_y\psi,\, -\partial_x\psi), \tag{4.2}$$

and the scalar *vorticity* $\omega = \partial_x u_2 - \partial_y u_1$ then satisfies

$$-\Delta\psi = \omega. \tag{4.3}$$

Taking the curl of (4.1) eliminates the pressure. In two dimensions the vortex-stretching term vanishes and one is left with pure transport,

$$\partial_t\omega + u\cdot\nabla\omega = 0. \tag{4.4}$$

Vorticity is carried by the flow; it is neither created nor destroyed. The whole dynamics is: given ω, solve the Poisson equation (4.3) for ψ, form u by (4.2), and move ω along u. This is textbook fluid mechanics **[Tier C]** (we state it without a citation; any fluid dynamics text derives it).

4.1.2 The Green's function of $-\Delta$ in the plane

To solve (4.3) on the whole plane we need a function G with $-\Delta G = \delta_0$.

Proposition 4.1. *The function* $G(x) = -\frac{1}{2\pi}\log|x|$ *satisfies* $-\Delta G = \delta_0$ *in the sense of distributions on* $\mathbb{R}^2$.

Human proof. For $x \neq 0$ write $r = |x|$. A radial function $f(r)$ has $\Delta f = f'' + f'/r = r^{-1}(rf')'$. With $f = \log r$, $rf' = 1$, so $\Delta\log r = 0$ away from the origin. It remains to find the weight of the delta. Let φ be a smooth test function with compact support. By Green's second identity on the annulus $\epsilon < |x| < R$ (with R beyond the support of φ),

$$\int_{|x|>\epsilon} G\,\Delta\varphi\,dx = \oint_{|x|=\epsilon} \big(G\,\partial_n\varphi - \varphi\,\partial_n G\big)\,d\ell,$$

where n is the normal pointing *into* the small disk, i.e. $\partial_n = -\partial_r$. Now $\partial_r G = -1/(2\pi r)$, so $-\varphi\,\partial_n G = \varphi\,\partial_r G = -\varphi/(2\pi\epsilon)$ on the circle, whose length is $2\pi\epsilon$; this term tends to $-\varphi(0)$. The first term is bounded by $\frac{1}{2\pi}|\log\epsilon|\cdot\|\nabla\varphi\|_\infty\cdot 2\pi\epsilon \to 0$. Hence $\int G\,\Delta\varphi = -\varphi(0)$, which is $\langle -\Delta G, \varphi\rangle = \varphi(0)$. □

The physical reading is Gauss's law: the flux of $-\nabla G$ through any circle around the origin is $2\pi r\cdot\frac{1}{2\pi r} = 1$, independent of r. In three dimensions the same computation gives $1/(4\pi|x|)$; in two it gives a *logarithm*. This single fact is why the planar point-vortex system is the logarithmic case of our energy problem.

4.1.3 Kirchhoff's point vortices

Concentrate the vorticity at N points: $\omega = \sum_{j=1}^{N}\Gamma_j\,\delta_{x_j}$, where Γ_j is the *circulation* of vortex j (positive for counter-clockwise rotation). By Proposition 4.1,

$$\psi(x) = -\frac{1}{2\pi}\sum_{j}\Gamma_j\log|x - x_j|.$$

Each vortex is moved by the velocity due to all the *others*; its own field is circular around it and does not move it (this is the standard regularisation, Tier C). Writing $x_i = (x_i, y_i)$ and $r_{ij} = |x_i - x_j|$, (4.2) gives the Kirchhoff–Helmholtz equations

$$\dot{x}_i = -\frac{1}{2\pi}\sum_{j\neq i}\Gamma_j\,\frac{y_i - y_j}{r_{ij}^2}, \qquad \dot{y}_i = \frac{1}{2\pi}\sum_{j\neq i}\Gamma_j\,\frac{x_i - x_j}{r_{ij}^2}. \tag{4.5}$$

[1]The stream function ψ of this section is unrelated to the bond-orientational order parameter ψ_6 used later in this chapter and in Chapter 3.

Proposition 4.2 (Hamiltonian structure)**.** *Let*

$$H = -\frac{1}{2\pi}\sum_{i<j}\Gamma_i\Gamma_j \log r_{ij} = \frac{1}{2}\sum_{i\neq j}\Gamma_i\Gamma_j\, G(x_i - x_j). \tag{4.6}$$

Then (4.5) *is equivalent to* $\Gamma_i\dot{x}_i = \partial H/\partial y_i$ *and* $\Gamma_i\dot{y}_i = -\partial H/\partial x_i$ *(for* $\Gamma_i \neq 0$*).*

Human proof. $\partial_{y_i}\log r_{ij} = (y_i - y_j)/r_{ij}^2$, and y_i appears in the terms with pair $\{i,j\}$ for every $j \neq i$. So $\partial H/\partial y_i = -\frac{1}{2\pi}\sum_{j\neq i}\Gamma_i\Gamma_j(y_i - y_j)/r_{ij}^2$; divide by Γ_i and compare with (4.5). The x-equation is the same computation with a sign. □

So $(x_i, \Gamma_i y_i)$ are canonical coordinates: the plane of each vortex is its own phase space. The Hamiltonian is conserved. So are the *linear impulse* $P = \sum_i \Gamma_i x_i$, $Q = \sum_i \Gamma_i y_i$, and the *angular impulse* $I = \sum_i \Gamma_i |x_i|^2$. These three come from translation and rotation invariance; in fact they are conserved for *any* velocity law of the form $\dot{x}_i = \sum_{j\neq i}\Gamma_j\,(x_i - x_j)^{\perp} f(r_{ij})$, a fact the code below uses as a negative control (Exercise 4.3).

Remark 4.3 (Two different divergences). The kinetic energy of the flow is $\frac{1}{2}\int |u|^2 = \frac{1}{2}\int \omega\psi$ (integrate by parts). For point vortices this is $\frac{1}{2}\sum_{i,j}\Gamma_i\Gamma_j G(x_i - x_j)$, *including* $i = j$, where $G(0) = +\infty$. Dropping the diagonal (the *self-energy* of each vortex) gives (4.6); it is a renormalisation by an infinite constant that does not depend on positions. A second, independent divergence appears for infinitely many vortices of one sign at positive density: the pair sum $\sum_j \log r_{ij}$ diverges at large distances. The first divergence is local and harmless for dynamics; the second is the one that puts the problem outside the theorem.

4.1.4 Why this is the case $s = 0$, and why it is outside the theorem

For like-signed vortices, $\Gamma_i\Gamma_j > 0$, the pair energy $-\frac{\Gamma_i\Gamma_j}{2\pi}\log r$ *decreases* with distance: like vortices repel in the energy sense, exactly as like charges do. In the Riesz family r^{-s}, the logarithm is the $s \to 0$ limit after subtracting a constant:

$$\frac{r^{-s} - 1}{s} \longrightarrow -\log r \qquad (s \to 0^+).$$

In the language of Chapter 1, with $t = r^2$, the logarithmic potential is $g(t) = -\frac{1}{2}\log t$. It fails `AdmissiblePotential` **[Tier C]** (an elementary check, Exercise 4.4): it is negative for $t > 1$. Moreover, every nonzero lattice point has $|a|^2 \geq 2/\sqrt{3} > 1$, so the real lattice sum $\sum_{a\neq 0} -\frac{1}{2}\log|a|^2$ diverges to $-\infty$. (In Lean, `ENNReal.ofReal` truncates each negative term to 0, so `latticeEnergy` of this g is 0.) Either way, the theorem does not apply (the logarithm is not a nonnegative completely monotone function). The theorem of Chapter 2 says nothing about it.

What replaces it is a *renormalised* energy, in which the infinite interaction with distant particles is cancelled by a neutralising background. Upstream's manuscript states, in its Section 8 and after Petrache and Serfaty [4], that for $0 \leq s < 2$ (the logarithm included) the triangular lattice minimises the renormalised energy [1] **[Tier L]**. That argument is in the AI-generated, not peer-reviewed manuscript only; it is *not* in Lean anywhere. Chapter 5 explains the construction and the open endpoint $s = 2$.

4.1.5 Rotating frames: the vortex lattice as a logarithmic jellium

A family of N vortices that rotates rigidly at angular velocity Ω about the origin satisfies $\dot{x}_i = -\Omega y_i$, $\dot{y}_i = \Omega x_i$. Insert this into Proposition 4.2: $\Gamma_i(-\Omega y_i) = \partial H/\partial y_i$ and $\Gamma_i\Omega x_i = -\partial H/\partial x_i$, that is,

$$\nabla_{x_i}\Big(H + \frac{\Omega}{2}\,I\Big) = 0 \quad \text{for all } i, \qquad I = \sum_i \Gamma_i |x_i|^2. \tag{4.7}$$

Rigidly rotating vortex configurations are exactly the critical points of $H_\Omega = H + \frac{\Omega}{2} I$. For N equal vortices $\Gamma > 0$ and $\Omega > 0$,

$$H_\Omega = -\frac{\Gamma^2}{2\pi} \sum_{i<j} \log r_{ij} + \frac{\Omega\Gamma}{2} \sum_i |x_i|^2,$$

a logarithmic repulsion in a harmonic bowl. The bowl is precisely the potential of a uniform neutralising background. Indeed, let a uniform disk D carry vorticity density $-\Gamma n_b$. A vortex Γ at $x \in D$ has interaction energy $V(x) = -\Gamma^2 n_b \int_D G(x-y)\,dy$ with it, and by Proposition 4.1, $-\Delta_x \int_D G(x-y)\,dy = 1$ inside D, so $\Delta V = \Gamma^2 n_b$. On the other hand $\Delta\big(\frac{\Omega\Gamma}{2}|x|^2\big) = 2\Omega\Gamma$. Both are radial, so they agree up to a constant exactly when

$$n_b = \frac{2\Omega}{\Gamma}. \tag{4.8}$$

The rotating frame therefore turns the like-sign vortex gas into a *two-dimensional one-component plasma*: a logarithmic jellium with background density $2\Omega/\Gamma$ **[Tier C]**. If the vortex density equals the background density, the system is neutral on average, and the large-scale divergence of Remark 4.3 cancels. In a quantum superfluid, every vortex carries the same quantised circulation $\Gamma = h/m$, and (4.8) becomes the familiar statement that a rotating superfluid mimics solid-body rotation with a vortex density $2\Omega m/h$ (this is often called Feynman's rule; we give no citation for it here) **[Tier C]**.

Caution : what the analogy does and does not say

The identification "rotating vortices = log-jellium" is an exact rewriting for *point* vortices in an *unbounded* ideal fluid. Real vortices have cores, real containers have walls (which add image vortices), and real superfluids are described by a Gross–Pitaevskii or similar field theory, not by point particles. Moreover, the renormalised-energy statement for the logarithm is Tier L through [1] only. So "the triangular vortex lattice of a rotating superfluid is explained by universal optimality" is *not* a claim of this book. The claim is weaker: the energy that a rotating vortex array makes stationary is, after an exact rewriting, the $s = 0$ jellium energy that Chapter 5 discusses **[Tier C]**.

4.1.6 Experiments: vortex crystals and vortex lattices

Physics : vortex crystals in a magnetised electron column

Fine, Cass, Flynn and Driscoll observed the relaxation of two-dimensional turbulence to *vortex crystals* in a magnetised pure-electron column [14] **[Tier L]**. Why is an electron column a two-dimensional fluid? In a strong axial field $B\hat{z}$, the electrons' guiding centres drift with $v = E \times B/B^2$. With $E = -\nabla\phi$ and $B = B\hat{z}$, $v = \big(-\partial_y\phi,\ \partial_x\phi\big)/B$. Compare (4.2): $\psi = -\phi/B$. Poisson's law for the electron density n, $\Delta\phi = en/\varepsilon_0$, then gives $\omega = -\Delta\psi = en/(\varepsilon_0 B)$: *the electron density is the vorticity* **[Tier C]** (the standard guiding-centre drift argument, not taken from [14]). The drift dynamics is the 2D Euler equation (4.4), with the conducting wall of the trap fixing the boundary condition on ϕ. What [14] reports is stated in its title: two-dimensional turbulence relaxes to "vortex crystals". We quote nothing more from that paper here: no number of vortices, no arrangement, no lifetime. The physics we add is our own reasoning **[Tier C]**. Ideal drift dynamics is *conservative*: H is not minimised by dissipation. An ordered vortex pattern that persists in such a flow is naturally read as a rigidly rotating state, i.e. a critical point of H_Ω in (4.7), here in a bounded domain with a wall. The connection with jellium-type

energy minimisation is an analogy **[Tier C]**.

Physics : vortex lattices in a rotating Bose–Einstein condensate

Abo-Shaeer, Raman, Vogels and Ketterle observed vortex lattices in rotating Bose–Einstein condensates, the vortices arranging themselves in a triangular (Abrikosov) pattern [15] **[Tier L]**. Unlike the ideal point-vortex flow, a real condensate has dissipation, so its vortex array can settle into a state of low energy in the rotating frame **[Tier C]** (our gloss, not a statement taken from [15]). In the rotating frame, the energy that the lattice minimises contains the term $-\Omega L_z$, whose effect on the vortices is the harmonic, neutralising bowl of (4.7). The resulting "logarithmic jellium" picture is an analogy, with all the caveats of the box above **[Tier C]**: cores, the finite and inhomogeneous condensate, and the edge.

4.1.7 A computational check: conservation in a point-vortex code

Before any physics is extracted from a simulation, one should check that it conserves what the equations conserve, and that the check *can fail*. The script below (57 lines, `code/ch4_point_vortices.py` in the book directory) integrates (4.5) with `scipy.integrate.solve_ivp`. It runs three experiments.

1. *Known answer (positive control).* A regular ring of n equal vortices ($\Gamma = 1$) of radius R rotates rigidly at $\Omega = (n-1)\Gamma/(4\pi R^2)$ (Exercise 4.2). For $n = 5$, $R = 1$: $\Omega = 1/\pi$.

2. *Conservation.* Eight vortices of random signs and strengths, integrated to $t = 10$ with an explicit Runge–Kutta method (DOP853) and with an implicit BDF method; the drifts of H, P, Q, I over 201 output times are printed.

3. *Negative control.* The same system with the kernel $1/r$ in place of $1/r^2$ in (4.5) (`power=1`). This is no longer the flow generated by the logarithmic H, so H must drift, while P, Q, I should still be conserved by symmetry.

Listing 4.1: `code/ch4_point_vortices.py` (verbatim).

```
"""N point vortices in the plane (Kirchhoff): conservation checks with a known answer and a
    negative control."""
import numpy as np
from scipy.integrate import solve_ivp
from scipy.optimize import OptimizeResult as OdeResult


def rhs(t: float, z: np.ndarray, gam: np.ndarray, power: int = 2) -> np.ndarray:
    """Velocity of each vortex: u_i = -(1/2pi) sum_j G_j (y_i-y_j)/r^p, v_i = (1/2pi) sum_j G_j
     (x_i-x_j)/r^p."""
    x, y = z[0::2], z[1::2]
    dx, dy = x[:, None] - x[None, :], y[:, None] - y[None, :]
    r2 = dx**2 + dy**2
    np.fill_diagonal(r2, 1.0)
    k = gam[None, :] / (2 * np.pi * r2 ** (power / 2))
    np.fill_diagonal(k, 0.0)
    out = np.empty_like(z)
    out[0::2], out[1::2] = -(k * dy).sum(1), (k * dx).sum(1)
    return out


def invariants(z: np.ndarray, gam: np.ndarray) -> tuple[float, float, float, float]:
```

```
    """Hamiltonian H = -(1/2pi) sum_{i<j} G_i G_j log r_ij, impulses P, Q and angular impulse I
    ."""
    x, y = z[0::2], z[1::2]
    r = np.hypot(x[:, None] - x[None, :], y[:, None] - y[None, :])
    iu = np.triu_indices(len(gam), 1)
    h = -np.sum(np.outer(gam, gam)[iu] * np.log(r[iu])) / (2 * np.pi)
    return h, float(gam @ x), float(gam @ y), float(gam @ (x**2 + y**2))


def drift(sol: OdeResult, gam: np.ndarray) -> list[float]:
    """Max over output times of |invariant(t) - invariant(0)| (relative for H and I)."""
    inv = np.array([invariants(sol.y[:, k], gam) for k in range(sol.y.shape[1])])
    d = np.abs(inv - inv[0]).max(0)
    return [d[0] / abs(inv[0, 0]), d[1], d[2], d[3] / abs(inv[0, 3])]


# Known answer: a ring of n equal vortices (G = 1, radius R) rotates rigidly at omega = (n-1)
    /(4 pi R^2).
n, R, T = 5, 1.0, 20.0
th = 2 * np.pi * np.arange(n) / n
z0 = np.column_stack([R * np.cos(th), R * np.sin(th)]).ravel()
ring = solve_ivp(rhs, (0, T), z0, args=(np.ones(n),), method="DOP853", rtol=1e-12, atol=1e-12)
angle = np.unwrap(np.arctan2(ring.y[1], ring.y[0]))[-1]
print(f"ring n=5: omega numeric {angle / T:.12f}  exact {(n - 1) / (4 * np.pi * R**2):.12f}")

# Random mixed-sign system, two integrators.
rng = np.random.default_rng(2026)
N = 8
gam = rng.choice([-1.0, 1.0], N) * rng.uniform(0.5, 1.5, N)
z0 = rng.uniform(-1, 1, 2 * N)
t_eval = np.linspace(0, 10, 201)
for method in ("DOP853", "BDF"):
    sol = solve_ivp(rhs, (0, 10), z0, args=(gam,), method=method, rtol=1e-10, atol=1e-12,
    t_eval=t_eval)
    dH, dP, dQ, dI = drift(sol, gam)
    print(f"{method:6s} nfev={sol.nfev:6d}  rel dH={dH:.2e}  dP={dP:.2e}  dQ={dQ:.2e}  rel dI={
    dI:.2e}")

# Negative control: a 1/r kernel instead of 1/r^2 is not the gradient of the log Hamiltonian.
bad = solve_ivp(rhs, (0, 10), z0, args=(gam, 1), method="DOP853", rtol=1e-10, atol=1e-12,
    t_eval=t_eval)
print("control (1/r kernel): rel dH={:.2e}  dP={:.2e}  dQ={:.2e}  rel dI={:.2e}".format(*drift(
    bad, gam)))
```

Numerics : output of the point-vortex script (Tier X)

Command, exactly as run (two cores):

```
taskset -c 2,3 /home/callensxavier_gmail_com/AutoevolveAI/.venv/bin/python -I /home/
    callensxavier_gmail_com/AutoevolveAI/.claude/worktrees/openai-math-discovery/papers/
    textbook_universal_optimality/code/ch4_point_vortices.py
```

Output, verbatim:

```
ring n=5: omega numeric 0.318309886184  exact 0.318309886184
DOP853 nfev= 57329  rel dH=1.92e-08  dP=7.55e-15  dQ=4.44e-15  rel dI=9.68e-11
BDF    nfev=166250  rel dH=5.78e-07  dP=2.60e-14  dQ=2.02e-14  rel dI=4.05e-09
control (1/r kernel): rel dH=3.61e-02  dP=2.11e-15  dQ=8.88e-16  rel dI=8.21e-11
```

How to read it **[Tier X]**:

- The ring's rotation rate agrees with $1/\pi$ to the twelve printed digits. The code's velocity

law and its normalisation $1/2\pi$ are therefore right.

- With DOP853 the linear impulses are conserved to round-off ($\sim 10^{-14}$), the angular impulse to about 10^{-10} relative and the Hamiltonian to about 2×10^{-8} relative. H is the most sensitive invariant here because it contains $\log r_{ij}$, which is large when two vortices of opposite sign come close.

- BDF conserves everything less well (about $30\times$ worse for H) and uses about three times as many right-hand-side evaluations. This is *not* a verdict on BDF in general. Without a supplied Jacobian, SciPy's BDF builds one by finite differences, and those evaluations are counted in `nfev`. More importantly, the point-vortex flow is Hamiltonian, not dissipative: the eigenvalues of its Jacobian lie on or near the imaginary axis, the problem is not stiff, and an implicit method built for stiff decay brings cost without benefit. The gradient flows of Chapter 3 are the opposite case (Section 4.5).

- The negative control does what it should: with the wrong kernel, H drifts by 3.6×10^{-2}, more than six orders of magnitude above the correct run, while P, Q and I remain conserved. A conservation test that checked only P, Q, I would have passed the wrong code. Only H tests that the kernel is the gradient of the logarithm.

4.2 Colloids at fluid interfaces

Colloids are particles of roughly micrometre size suspended in a liquid. Confined to a plane—by gravity onto a flat glass surface, or by surface tension at a liquid–air interface—and made to repel each other, they form two-dimensional crystals that can be filmed particle by particle. They are the cleanest laboratory realisation of the theorem's setting that we know of, for two reasons. The interaction law can be tuned and is close to a pure power law in one important case; and the dynamics is overdamped motion in a viscous fluid, which is the gradient flow that our numerics integrate.

4.2.1 Superparamagnetic colloids: a Riesz $s = 3$ system

Zahn, Wille, Maret, Sengupta and Nielaba studied superparamagnetic colloids at a flat air–water interface and measured the elastic properties of the resulting two-dimensional crystals by video microscopy [10] **[Tier L]**. A magnetic field B perpendicular to the interface induces in each particle a magnetic moment $m \propto \chi B$, where χ is the particle's susceptibility. All moments are parallel and perpendicular to the plane, so for two particles at in-plane distance r the dipole–dipole energy is

$$U(r) = \frac{\mu_0}{4\pi} \frac{m^2}{r^3} \tag{4.9}$$

(the angular factor $1-3\cos^2\theta$ of the dipole law equals one because the moments are perpendicular to their separation) **[Tier C]**. This is a Riesz potential with $s = 3$. In the language of Chapter 1, $g(t) = t^{-3/2}$ up to a positive constant. The Lean results of Chapter 2 are stated for $t^{-s/2}$ itself; multiplying by a positive constant multiplies every energy by that constant and changes no inequality (our step, **[Tier C]**, and trivial). For $t^{-3/2}$ they say **[Tier A]** (kernel-checked, conditional on the fidelity of the definitions):

- g is admissible (`riesz_admissible`, which needs $s > 0$), so the triangular lattice minimises the lower energy among all locally finite configurations of density one (`triangular_riesz_optimal`, D1);

- its lattice energy is finite (`latticeEnergy_riesz_lt_top`, which needs $s > 2$; D1b);

- the same holds at any density ρ, with the energy scaled accordingly (D2, theorem `triangular_optimal_any_density`). For $s = 3$ the lattice energy at density ρ is $\rho^{3/2}$ times its value at density one (`riesz_lattice_homogeneous`), and that scaled value is a lower bound for every competitor of density ρ (`riesz_any_density_homogeneous`).

The Tier A content is the statement about $t^{-3/2}$. That the colloids interact by (4.9) is a modelling assumption **[Tier C]**, with known corrections: particles have finite size, their moments induce each other (mutual induction), and the interface may not be perfectly flat.

It is useful to have a single dimensionless number for such a system. At area density n the typical spacing is $\propto n^{-1/2}$, so the typical dipolar energy is $\propto \mu_0 m^2 n^{3/2}$, and the ratio to the thermal energy is a coupling of the form

$$\Gamma_{\mathrm{mag}} \;\propto\; \frac{\mu_0\,(\chi B)^2\,n^{3/2}}{k_B T}.$$

Raising B at fixed n cools the system in the only sense that matters: it increases the coupling. A slow ramp of B is the laboratory analogue of the simulated annealing of Chapter 3 **[Tier C]**.

4.2.2 Charged colloids in cavities

Bubeck, Neser, Bechinger and Leiderer studied small two-dimensional colloidal crystals confined in circular, hexagonal and square cavities [12] **[Tier L]**. The tiered paper [20], summarising the publisher's abstract (its full text was read neither there nor here), reports that these clusters contained about 10 to 100 particles and that their structure was dominated by the shape of the confining region. For us, such cavities are the experimental counterpart of the "closed disk" of Chapter 3, in which a soft wall holds the particles. Edge effects are not a nuisance here; they are the phenomenon. A cavity of N particles has a boundary layer whose share of particles decreases only like $N^{-1/2}$.

4.2.3 Hydrodynamics: from Stokes drag to the gradient flow

Why does a colloidal monolayer relax by the gradient flow $\dot{x} = -\nabla E$ used in Chapter 3? Because it moves through a viscous liquid at very small Reynolds number.

Stokes drag. A sphere of radius a moving at velocity v through an unbounded liquid of viscosity η, slowly enough that inertia of the liquid is negligible, feels the drag $F_{\mathrm{drag}} = -\gamma v$ with $\gamma = 6\pi\eta a$ (Stokes' law, textbook result, no-slip boundary condition) **[Tier C]**. A particle sitting *at* an air–water interface is only partly immersed, and its drag coefficient differs from the bulk value; we leave γ symbolic and quote no number.

The Langevin equation. Newton's law for particle i, of mass m, at temperature T reads

$$m\,\ddot{x}_i = -\gamma\,\dot{x}_i - \nabla_{x_i} E(x_1,\dots,x_N) + \xi_i(t), \qquad \langle \xi_i(t) \otimes \xi_j(t') \rangle = 2\gamma k_B T\,\delta_{ij}\,\mathrm{Id}\;\delta(t-t'), \quad (4.10)$$

where ξ is the random force of the solvent molecules. Its strength is fixed by the fluctuation–dissipation theorem: with this choice, the stationary distribution is the Boltzmann distribution $\propto e^{-E/k_B T}$ **[Tier C]**.

The overdamped limit. Without the forces $-\nabla E$ and ξ, the velocity decays as e^{-t/τ_m} with $\tau_m = m/\gamma$. For a micrometre colloid in water, τ_m is many orders of magnitude shorter than the time the particle needs to move by a fraction of its spacing (Exercise 4.7, where you

estimate it yourself). On time scales $\gg \tau_m$ the inertial term is negligible, and (4.10) becomes the *overdamped Langevin equation*

$$\gamma\,\dot{x}_i = -\nabla_{x_i} E + \xi_i, \qquad D = \frac{k_B T}{\gamma} \quad \text{(Einstein relation)}. \tag{4.11}$$

At $T = 0$ it is the gradient flow $\dot{x} = -\gamma^{-1}\nabla E$. Rescaling time by γ gives exactly the equation $\dot{x} = -\nabla E$ that rusty-SUNDIALS integrates with CVODE in Chapter 3. The annealing of Chapter 3 is modelled on (4.11) at slowly decreasing T, but the code uses a capped Euler–Maruyama scheme (heuristic annealing, see Chapter 3), which is not a consistent discretisation of it **[Tier C]**.

Hydrodynamic interactions. The model above treats each particle as feeling only its own drag. In reality, a moving colloid sets the liquid around it in motion, and that flow drags its neighbours. To leading order this is described by a *mobility tensor* M_{ij} (Oseen or Rotne–Prager forms; see the literature), and the zero-temperature dynamics becomes $\dot{x}_i = -\sum_j M_{ij}\nabla_{x_j} E$. Two facts survive **[Tier C]**:

1. If M is symmetric positive definite, $\frac{d}{dt}E = -\sum_{ij} \nabla_i E \cdot M_{ij}\nabla_j E \le 0$: the energy still decreases.
2. The stationary points ($\nabla E = 0$) and their stability (the Hessian) do not depend on M.

Hydrodynamics therefore changes the *paths* to the minimum and the efficiency of annealing, not the minimum. This is why the question "what is the ground state?" belongs to the energy alone. It is also why the theorem, which is about the energy alone, is relevant to colloids at all.

4.3 Plasmas

4.3.1 Dusty-plasma crystals: the Yukawa potential

Thomas, Morfill, Demmel, Goree, Feuerbacher and Möhlmann observed Coulomb crystallisation of charged microparticles levitated in a plasma [11] **[Tier L]**. The grains acquire a large negative charge from the plasma, and the surrounding ions and electrons screen it. In the simplest (Debye–Hückel) model the screened pair potential is of Yukawa form,

$$V(r) = \frac{e^{-\kappa r}}{r} \quad \text{(up to a positive constant; } \kappa \text{ the inverse screening length)}$$

[Tier C]. As a function of $t = r^2$, $g(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$. Is it admissible? The function $1/\sqrt{t}$ is completely monotone; $e^{-\kappa\sqrt{t}}$ is completely monotone because $e^{-\kappa u}$ is completely monotone in u and $\sqrt{t}$ has a completely monotone derivative; products of completely monotone functions are completely monotone. So g satisfies the hypotheses of the upstream theorem **[Tier C]** (Exercise 4.9). This step is *not* in our Lean files, which is why every statement about Yukawa crystals in this book is Tier C at best. The numerical study of Chapter 3 is for exactly this potential (with a shifted-force cutoff at $r_c = 8$): at $\kappa = 2$, with $N = 288$ in a periodic box at density one, the reference energies per particle are $e_{\text{tri}} = 0.3935333668$ and $e_{\text{sq}} = 0.4009647699$ (each pair counted once, as in Chapter 3: one half of Chapter 1's E_g), and none of 100 relaxations (20 of them incommensurate-box controls) fell below e_{tri} **[Tier X]**.

Real dusty plasmas leave the theorem's class in several ways **[Tier C]** (general knowledge, not cited here). The ion flow past a grain builds a wake that attracts other grains, chiefly those below it; the wake force is non-reciprocal, so there is no pair energy at all. Grains move out of the plane; and far from a grain, the potential is not Yukawa. These confounds are listed again in protocol L2 below.

4.3.2 Electrons on liquid helium

Grimes and Adams observed a liquid-to-crystal phase transition of a classical two-dimensional sheet of electrons held above the surface of liquid helium [13] **[Tier L]**. The pair interaction is Coulomb, $e^2/(4\pi\varepsilon_0 r)$. Here 'Coulomb' means the three-dimensional $1/r$ law restricted to the plane (Riesz $s = 1$); the planar Coulomb (logarithmic, $s = 0$) interaction is distinguished where it matters. It is Riesz with $s = 1$, so the infinite lattice sum diverges and the system is neutralised by positive charge on electrodes *below* the sheet. This is jellium in the sense of Chapter 5—but with a caveat that the clean mathematical jellium does not have.

Image charges change the exponent. Model the electrode as a grounded conducting plane a distance d below the sheet. Each electron then has an image of opposite charge ($+e$ for an electron) at distance $2d$ from it, vertically below. Two electrons at in-plane distance r interact directly and through each other's images:

$$U(r) = \frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{\sqrt{r^2 + 4d^2}}\right) \approx \frac{e^2}{4\pi\varepsilon_0}\frac{2d^2}{r^3} \qquad (r \gg d)$$

[Tier C] (Exercise 4.10). At distances large compared with d, the interaction becomes *dipolar*, $s = 3$, and the system leaves the jellium regime and enters the finite-energy class of the theorem. Electrons on helium are therefore jellium only when d is much larger than the interparticle spacing.

4.3.3 Wigner crystal energetics

For the clean 2D Wigner crystal (point charges with $1/r$ interaction in a plane, uniform neutralising background in the same plane) the energies of the triangular and square lattices were computed by Bonsall and Maradudin [8], who found the triangular lattice lower and the square lattice dynamically unstable (an imaginary phonon branch) **[Tier L]**. Alves, Bendazzoli, Evangelisti and Berger gave ten-digit Madelung energies by a real-space method [9] **[Tier L]**.

Numerics : jellium controls from Chapter 3 (Tier X)

Our Ewald implementation (rusty-SUNDIALS branch `physics/jellium-wigner-2026-10-11`) reproduces the published values: triangular energy per particle -1.960515789320 against the literature value -1.960515789825 (relative difference 2.6×10^{-10}); square -1.950132460, relative difference 4.3×10^{-10} from the literature; the energy is invariant to better than 10^{-10} over a $4\times$ range of the Ewald splitting parameter; and the square lattice is a saddle point with 64 unstable modes, consistent with [8]. The main relaxation study: the main jellium relaxation study (prediction J1) was still running when this book was assembled.

These are controls, not results: they show that the code computes the energy that the literature computes. They say nothing about whether some non-lattice configuration could do better; that is the question the theorem answers (for $s > 0$ in the finite-energy class) and that Chapter 5 discusses for $s = 1$.

4.4 Laboratory protocols L1–L4

The tiered paper [20] proposes four experiments. We summarise them here with their caveats. They are *proposals* **[Tier C]**; none has been run. An experiment cannot test the Lean statement, which is about exact potentials and infinite configurations. What it can test is whether real two-dimensional repulsive systems behave as the theorem and the numerics suggest.

All four share one analysis script, `lab/lab_analysis.py` (in the tiered paper's directory), which takes tracked particle coordinates and computes the same observables as the simulations:

- the *density*, from the mean Voronoi cell area of the same interior particles over which the energy is averaged;
- the *coordination*, by the $1.35\,d_0$ neighbour rule, and the local and global bond-orientational order $|\psi_6|$;
- the *energy* per interior particle for a stated interaction law, truncated at r_c with a uniform tail correction, compared with a triangular lattice at the same density and the same truncation.

The script's self-test passes: a perfect lattice gives a gap at round-off level, thermal noise raises the energy above the lattice, and a square lattice is detected and placed above it. Two caveats bear on every kill criterion. The truncated potential is not exactly completely monotone, so the comparison is between equally truncated energies. And because at $s = 3$ the energy scales like $\rho^{3/2}$, a relative density error $\delta\rho/\rho$ becomes a relative energy error of about $1.5\,\delta\rho/\rho$ (Exercise 4.11), which must enter the error bar. Each protocol should be preregistered with the script, a predicted bracket and its kill criterion.

L1. Superparamagnetic colloids at a flat air–water interface (Riesz $s = 3$). Following [10]. *Procedure:* ramp B slowly from the fluid to the crystal at fixed area fraction; track positions by video microscopy; record χ and B so that the prefactor of (4.9) is known. *Predictions:* (a) at large coupling, $|\psi_6| \to 1$ and the interior defect fraction $\to 0$; (b) the measured dipolar energy per interior particle exceeds the triangular value at the measured density, by an amount that shrinks as B grows (the thermal excess); (c) a particle removed with optical tweezers leaves a vacancy that relaxes locally, without global restructuring; (d) a square arrangement imposed with holographic tweezers and released relaxes away from square—the analogue of the square-saddle control computed for Yukawa and jellium (the $s = 3$ square lattice itself was not computed). *Kill criterion:* a slowly ramped, defect-free interior whose dipolar energy lies below the triangular value beyond the measurement error, with the law's prefactor independently confirmed.

L2. Dusty-plasma monolayer (Yukawa). Following [11]. The radial confinement is close to parabolic, the analogue of the "open bowl" of Chapter 3. *Procedure:* measure κ and the coupling from the phonon spectrum or the pair correlation; image the monolayer; run the shared script; simulate the bowl model at the measured κ and N. *Prediction:* a near-triangular interior with defects concentrated at the rim; the interior defect fraction is compared with simulation at matched κ and N. *Confounds* **[Tier C]** (general knowledge, not cited here): ion-wake attraction (non-reciprocal, chiefly between vertically adjacent grains, able to drive a mode-coupling instability in a monolayer), out-of-plane motion, and departures from a pure Yukawa law at large distance all move the system outside the theorem's class. Neutral-gas drag damps the dynamics but does not change the interaction law.

L3. Charged colloids in circular, hexagonal and square cavities (closed disk). Following [12]. *Procedure:* repeat the circular-cavity case at $N \approx 300$, image the interior, compare with closed-wall simulations at matched N and wall stiffness. *Prediction:* the edge defect fraction exceeds the interior one, as in the closed-disk simulations of Chapter 3. Compared with L2 at matched N and density, this is the experimental open-versus-closed comparison—with the confounds that the simulated comparison already has (non-uniform density in the bowl, compression against the wall, different edge-layer sizes).

L4. Electrons on liquid helium (jellium, $s = 1$). Following [13]. The neutralising charge sits on electrodes a distance d below the sheet, so this is the jellium case only when d is much larger than the spacing; otherwise, as computed above, image charges make the interaction dipolar at distances $\gg d$. *Prediction:* the crystal is triangular. The evidence in [13] is spectroscopic, not a direct image **[Tier C]** (our reading of [13]), so a test compares spectroscopic signatures with the triangular Madelung and phonon structure [8]. Direct imaging would need a detection method that the protocol does not specify.

Caution : what a positive outcome would mean

Suppose L1 is carried out and every prediction holds. That would show that a real dipolar monolayer behaves as the ideal model predicts. It would *not* confirm the theorem (which needs no confirmation of that kind and admits none), and it would *not* confirm the model (a defect-free triangular interior is also what many non-ideal interactions produce). The kill criterion is asymmetric on purpose: only an undercut of the triangular energy, with the law independently confirmed, would be informative, and it would point at the model, not at the proof.

4.5 rusty-SUNDIALS: implications and opportunities

rusty-SUNDIALS [18] is a Rust re-implementation of parts of the SUNDIALS suite. The numerical study of Chapter 3 (crate `qf-yukawa-crystal`) uses its CVODE. This section asks what the physics of this chapter demands from such a library, and what the library currently provides. Every statement about the code below was checked against the files named; the proposals are **[Tier C]**.

4.5.1 Stiffness of particle relaxation

Linearise the gradient flow $\dot{x} = F(x) = -\nabla E(x)$ around a minimum x^*: $\dot{\delta x} = -\mathcal{H}\,\delta x$, with $\mathcal{H}$ the Hessian. If $\lambda_1 \le \cdots \le \lambda_{2N}$ are its eigenvalues, the Jacobian of the ODE has eigenvalues $-\lambda_k$. In a crystal:

- the two translation modes have $\lambda = 0$ (in our runs, $|\lambda| \le 1.5 \times 10^{-15}$ at $\kappa = 2$) **[Tier X]**;
- the long-wavelength phonons have small λ; for a crystal their smallest value decreases as the box grows, because the softest wavenumber scales like $1/L$ **[Tier C]**. At the single size studied ($N = 288$), the perfect lattice's smallest non-zero eigenvalue was 0.0236 **[Tier X]**. Separately, over the 80 non-control relaxed runs at $\kappa = 2$, most of them defected, every non-translation eigenvalue was at least 5.4×10^{-5} **[Tier X]**; such a low floor reflects soft modes of defected configurations, not a size effect;
- the short-wavelength, nearest-neighbour modes have the largest eigenvalue $\lambda_{\max}$.

An explicit method is stable only for step sizes $h \lesssim 2/\lambda_{\max}$, while relaxation along the softest mode needs a time $\sim 1/\lambda_{\min}$. The number of explicit steps therefore grows like the *stiffness ratio* $\lambda_{\max}/\lambda_{\min}$, which in a crystal grows with the system size (the softest phonon wavenumber scales like $1/L$) **[Tier C]**. We give no numerical value for the ratio because $\lambda_{\max}$ is not among the recorded results. BDF methods, whose stability regions contain most or all of the left half-plane, take steps limited by accuracy instead of stability, and this is why CVODE's BDF is the right tool for the quench.

Contrast the point vortices of Section 4.1.7: there the dynamics is Hamiltonian, the eigenvalues are near the imaginary axis, and explicit (or, better, symplectic) integrators are the right tool. The same library must serve both regimes.

4.5.2 BDF versus Adams in rusty-SUNDIALS

CVODE offers two families: BDF for stiff problems, Adams–Moulton for non-stiff ones. In rusty-SUNDIALS the Adams path does not work as an Adams method. The crate description of `qf-bao-distances` records it in one sentence:

```
... integrated with this repository's own CVODE (BDF; its Adams path was measured to be stuck
    at order 1) ...
```

(`crates/qf-bao-distances/Cargo.toml`). The README of `crates/sundials-mcp` gives the measured detail: `Method::Adams` never leaves order 1 because a coefficient routine (`compute_l`) is a placeholder, and its error on a smooth quadrature problem scales like $\sqrt{\text{rtol}}$ (it reports 2.1×10^{-4} relative at rtol $= 10^{-7}$). The same README records that many output times from one BDF run degrade accuracy compared with a fresh solve per output (4×10^{-6} against 6×10^{-7}). These are that README's measurements, not ours **[Tier X]**.

Implications. The relaxations of Chapter 3 are stiff and use BDF, so they are not affected by the Adams defect. A point-vortex or any other Hamiltonian problem would want Adams (or a dedicated symplectic method), and today it would silently get a first-order method if it asked for Adams. *Opportunity:* implement the variable-order Adams coefficients and test them against a problem with a known high-order convergence rate (Project P2 below).

4.5.3 Analytic versus finite-difference Jacobians

A BDF step solves a nonlinear system by Newton's method and needs the Jacobian $\partial F/\partial x$. For the gradient flow this is $-\mathcal{H}$. The two crates of Chapter 3 make different choices, as their source says:

```
crates/qf-yukawa-crystal/src/lib.rs:
/// Overdamped relaxation `dx/dt = F(x)` with CVODE (BDF, analytic Jacobian) until `max |F| <
    ftol` or `t > tmax`.
crates/qf-yukawa-crystal/src/jellium.rs:
    /// Overdamped relaxation with CVODE (BDF, finite-difference Jacobian) until `max |F| <
    ftol` or `t > tmax`.
```

For the short-range Yukawa potential with a cutoff, the Hessian is a sparse sum of 2×2 blocks and is cheap to write down; the Yukawa relaxation passes it to CVODE (`.jacobian(jac)` with entries $-\mathcal{H}$). For jellium, the Ewald sum makes an analytic Hessian laborious, and CVODE forms the Jacobian by finite differences of the force: the difference-quotient loop in `crates/cvode/src/solver.rs` makes one extra right-hand-side call per column, $J_{:,j} = (f(y + \varepsilon e_j) - f(y))/\varepsilon$, i.e. $2N$ Ewald force evaluations per Jacobian for N particles. The Hessian used for the stability analysis of jellium is likewise "central finite differences of the analytic force (symmetrised)" (source comment in `jellium.rs`).

Opportunities **[Tier C]**: an analytic Ewald Hessian (the real-space and reciprocal-space parts can both be differentiated twice in closed form); automatic differentiation; or a Jacobian-free Newton–Krylov solver in which only Hessian–vector products are needed, which a single extra force evaluation approximates.

4.5.4 Constraints and DAEs: what IDA would add

Several physical set-ups of this chapter are *constrained*: particles confined by a hard wall (rather than the soft wall $\varepsilon(r - R)^4$ of Chapter 3), particles on a curved interface, vortices in a bounded domain with explicit boundary conditions, or relaxation at a fixed density or fixed angular impulse. The natural formulation is a differential-algebraic system $F(t, y, \dot{y}) = 0$, which is what SUNDIALS' IDA solves. In rusty-SUNDIALS, `crates/ida` is not that solver yet; its own header says so:

```
/// A fixed-step Backward Euler prototype for DAE residuals, NOT a general IDA solver.
///
/// It does one fixed-point iteration per step (no Newton-Krylov, no Nordsieck array, no
    variable
/// order/step size, no error control beyond the fixed-point residual). It approximates `dF/dy
    ~ 1`
/// (see the fixed-point loop in `solve`), which is only valid for mildly stiff residuals.
    Treat it
/// as a placeholder for a real IDA port, not as a replacement for it; `CvodeSolver` in the `
    cvode`
/// crate is the tested, variable-order solver in this repository.
```

(`crates/ida/src/solver.rs`, lines 3–9). It was not used in Chapter 3. *Opportunity:* a real IDA port (variable order and step, Newton iteration with the residual Jacobian, error control), validated on index-1 problems with closed-form solutions before any physics is run on it.

4.5.5 KINSOL: solving $F(x) = 0$ directly

The gradient flow finds *minima*: almost every initial condition flows away from saddle points. But some of the most informative configurations are saddles. The square lattice is one (Chapter 3 found it a saddle with 72 negative Hessian eigenvalues at $\kappa = 2$ and 64 at $\kappa = 1$) **[Tier X]**; transition states between defect configurations are others; and rigidly rotating vortex configurations, the critical points of H_Ω in (4.7), are generally saddles too. All of these are zeros of a force, $F(x) = 0$. SUNDIALS solves such systems with KINSOL (Newton and Newton–Krylov with globalisation). rusty-SUNDIALS has no KINSOL: no crate of that name exists, and a search of the crates' sources and manifests for "kinsol" finds nothing. *Opportunity* **[Tier C]**: a KINSOL port, tested on the square-lattice saddle (whose existence and index Chapter 3 already measured) and on the regular vortex ring of Section 4.1.7 in its rotating frame.

4.5.6 The sundials-mcp server: solvers for AI agents

`crates/sundials-mcp` is a Model Context Protocol server (JSON-RPC over standard input and output) through which AI agents can run rusty-SUNDIALS solvers. Its README lists the tools `about`, `list_problems`, `solve` (named CVODE problems: `exponential`, `robertson`, `vanderpol`, `lorenz`, `domain_exit`), `pgpe_run`, `cmb_bound` and `bao_distances`. Two of its design rules are worth copying in any scientific tool that an agent will call:

- every result carries a field `ran`, and `ran: false` means that nothing was computed—never a pass;
- a solver failure is reported as a tool error with the solver's own message and *no* partial trajectory; each known-answer check names an external reference.

The server also runs every solve in a worker subprocess, because the CVODE code prints diagnostics to standard output on some failure paths, which would corrupt the protocol stream. `pgpe_run` runs a projected Gross–Pitaevskii field model, in which quantised vortices can occur, but no tool simulates point particles or point vortices, and we did not check whether either GPE crate (`qf-pgpe`, `qf-gpe2d`) supports a rotating frame. *Opportunity* **[Tier C]**: a `relax_crystal` tool (Yukawa or jellium relaxation with the Chapter 3 controls built in) and a `point_vortices` tool (with the ring known-answer test and the conservation report of Section 4.1.7), each returning its controls alongside its result.

4.5.7 Student projects

P1. Point vortices with CVODE. Port `rhs` of Section 4.1.7 to Rust and integrate it with rusty-SUNDIALS' CVODE (BDF, the only reliable method today). Reproduce the ring

rotation rate and the conservation table. Compare with the SciPy output above; explain any difference in the drift of H. Add the $1/r$-kernel control. (No CVODE run of this problem exists yet; this is a proposal.)

P2. Fix Adams. Implement the variable-order Adams–Moulton coefficients in the CVODE crate and show order > 1 on a smooth problem (e.g. $y' = -y$), by plotting error against tolerance. Then rerun P1 with Adams.

P3. Vortex crystals in a rotating frame. Minimise H_Ω of (4.7) for N equal vortices by gradient flow with CVODE (this is a stiff relaxation, like Chapter 3), measure $|\psi_6|$ in the interior, and check that the interior density approaches $2\Omega/\Gamma$ of (4.8).

P4. A Newton solver for saddles. Write a small Newton iteration (a "KINSOL-lite") for $F(x) = 0$ and converge onto the square-lattice saddle of the Yukawa crystal from a slightly perturbed square start; count the negative Hessian eigenvalues and compare with Chapter 3.

P5. An analytic Ewald Hessian. Derive and code it, test it against the existing finite-difference Hessian, and compare the CVODE cost of the jellium relaxation with both Jacobians.

P6. Hydrodynamic interactions. Add a Rotne–Prager mobility (see the literature for its form) to the Yukawa gradient flow and verify numerically that the relaxed configurations and their energies do not change while the relaxation paths do.

P7. An MCP tool. Add a `point_vortices` tool to `sundials-mcp` that follows its honesty conventions, with the ring as the known-answer check.

4.6 Other domains, briefly

Ion crystals. Ions held in Penning traps can form single-plane crystals. The ions interact by the three-dimensional Coulomb law (Riesz $s = 1$ in the plane), and the trap together with the plasma's rotation provides a harmonic radial confinement, which, as in (4.7), plays the role of a neutralising background. These planar ion crystals are the $s = 1$ analogue of the rotating vortex lattice; triangular order is expected for them **[Tier C]** (general knowledge; we cite no source here).

Abrikosov lattices in type-II superconductors. In a type-II superconductor in a magnetic field between the two critical fields, flux enters as quantised vortices, which arrange themselves in a lattice. Near the upper critical field, Ginzburg–Landau theory reduces the choice of lattice to minimising a single number (the Abrikosov parameter), and the triangular lattice is the minimiser; the square lattice that was first proposed is higher **[Tier C]** (standard physics, no citation here). The mathematics behind such statements is close to this book's: Gaussian lattice sums. Montgomery proved that among lattices of fixed covolume the triangular lattice minimises the theta function $\sum_a e^{-\pi\alpha|a|^2}$ for every $\alpha > 0$ [7] **[Tier L]**, and Rankin and Cassels studied the analogous problem for the Epstein zeta function [5, 6] **[Tier L]**. These are *lattice* results: they compare lattices with lattices. The theorem of Chapter 2 compares the triangular lattice with *every* configuration of density one, periodic or not. Which lattice sum, exactly, expresses the Abrikosov parameter, and how the result relates to Montgomery's theorem, we leave to the literature **[Tier C]**.

Foams and honeycombs are a different problem. Soap froths in a plane and bees' honeycombs show hexagonal cells, and it is tempting to file them with the triangular lattice, whose Voronoi cells are regular hexagons. But the underlying optimisation is different: one minimises the total *perimeter* of a partition of the plane into cells of equal area. There are no point particles and no pair potential. The "honeycomb conjecture" that the regular hexagonal partition is optimal was proved by Hales **[Tier C]** (general knowledge; no citation here). Neither that theorem nor the universal optimality theorem implies the other; they share an answer, not a problem.

Summary

- Planar point vortices interact through the 2D Green's function, i.e. logarithmically: the case $s = 0$, which lies outside the finite-energy class and needs renormalisation (Chapter 5). In a rotating frame the vortex energy becomes a logarithmic jellium with background density $2\Omega/\Gamma$ **[Tier C]**.
- Vortex crystals in electron columns [14] and vortex lattices in rotating condensates [15] are observations **[Tier L]**; their connection with energy minimisation is an analogy **[Tier C]**.
- Superparamagnetic colloids [10] realise a Riesz $s = 3$ law, inside the theorem's class **[Tier A]** for the potential, **[Tier C]** for the modelling. Their overdamped motion is the gradient flow of Chapter 3.
- Dusty plasmas [11] are Yukawa (Tier C admissibility); electrons on helium [13] are jellium only when the electrodes are far, and dipolar otherwise.
- rusty-SUNDIALS handles stiff relaxation with BDF well; its Adams path is stuck at order 1, its IDA is a prototype, it has no KINSOL, and its MCP server has no particle tools. Each gap is a student project.

Exercises

Exercise 4.1. Repeat the proof of Proposition 4.1 in $\mathbb{R}^3$ and show that $G(x) = 1/(4\pi|x|)$. In $\mathbb{R}^d$, $d \geq 3$, show that $G(x) = c_d|x|^{2-d}$, and find c_d in terms of the surface area of the unit sphere. Which Riesz exponent s is the Green-function ("Coulomb") exponent in dimension d? Conclude that the *planar* Coulomb interaction, the Green function of the plane, is the case $s = 0$ (logarithm), whereas the $1/r$ law that this book calls 'Coulomb' (the three-dimensional law restricted to the plane) is Riesz $s = 1$.

Exercise 4.2. Show that n equal vortices of circulation Γ at the vertices of a regular n-gon of radius R rotate rigidly at $\Omega = (n-1)\Gamma/(4\pi R^2)$. (Hint: by symmetry compute the velocity of one vortex; use $\sum_{k=1}^{n-1} \frac{1}{1-e^{2\pi ik/n}} = \frac{n-1}{2}$.) Check the case $n = 2$ directly.

Exercise 4.3. Let $\dot{x}_i = \sum_{j\neq i} \Gamma_j\,(x_i - x_j)^{\perp} f(r_{ij})$ with $(a,b)^{\perp} = (-b,a)$ and any function f. Prove that P, Q and I are conserved. Show that for $N = 2$ the distance r_{12}, hence the logarithmic H, is conserved for *every* f. Then explain why, for $N = 8$, the negative-control line of the point-vortex output shows a drift of H but not of P, Q, I. What would a negative control with $N = 2$ have shown?

Exercise 4.4. Let $g(t) = -\frac{1}{2}\log t$. Show that $(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 1$ and $t > 0$, but that $g(t) < 0$ for $t > 1$. Which clause of `AdmissiblePotential` fails? Show that adding a constant cannot repair it. Show also that $\sum_{a\in A\setminus 0} g(|a|^2)$ diverges to $-\infty$ (all $|a|^2 > 1$), and that the Lean `latticeEnergy`, which applies `ENNReal.ofReal` termwise, is 0.

Exercise 4.5. Derive (4.7). Then show that for equal vortices the confining term $\frac{\Omega\Gamma}{2}|x|^2$ equals, up to a constant, the interaction energy of a vortex with a uniform disk of vorticity $-\Gamma n_b$ (containing it), with $n_b = 2\Omega/\Gamma$. Use Proposition 4.1 and the radial form of the Laplacian.

Exercise 4.6. Show that the $E \times B$ drift velocity in a uniform field $B\hat{z}$ is divergence-free when ϕ is smooth, and that the electron density is advected by it. Write the analogue of (4.4) for the density. Which boundary condition does a grounded cylindrical wall impose, and what does it do to a single off-centre vortex (method of images)?

Exercise 4.7. Using $\gamma = 6\pi\eta a$, estimate the momentum relaxation time $\tau_m = m/\gamma$ of a silica sphere of radius $1\,\mu$m in water, and the time it takes to diffuse a distance equal to its own radius (a^2/D, $D = k_B T/\gamma$) at room temperature. Look up the material constants yourself and state your sources. Justify the overdamped limit.

Exercise 4.8. Show that along $\dot{x} = -M\nabla E$, with M symmetric positive definite, E is non-increasing, and that the linear stability of a critical point (its number of unstable directions) is the number of negative eigenvalues of the Hessian, independently of M. (Hint: $M^{1/2}\mathcal{H}M^{1/2}$ and $\mathcal{H}$ have the same inertia.)

Exercise 4.9. Prove that $g(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$ is completely monotone on $(0,\infty)$ for $\kappa \geq 0$. Use: (i) if φ is completely monotone and h is positive with a completely monotone derivative, then $\varphi \circ h$ is completely monotone; (ii) products of completely monotone functions are completely monotone. Prove (ii) by Leibniz's rule. Why is this not enough to put Yukawa crystals at Tier A in this book?

Exercise 4.10. Derive the image-charge interaction for two electrons at height d above a grounded plane, and its expansion $\frac{2d^2}{r^3}$ (in units of $e^2/4\pi\varepsilon_0$) for $r \gg d$. Compute the next term. Which Riesz exponent governs the large-distance behaviour, and is it in the finite-energy class?

Exercise 4.11. For the Riesz energy per particle $e_s(\rho) = c_s\rho^{s/2}$, show that a relative density error $\delta\rho/\rho$ gives a relative energy error $\frac{s}{2}\frac{\delta\rho}{\rho}$ to first order. Evaluate it for $s = 3$ and explain why this matters for the L1 kill criterion.

Exercise 4.12. For the gradient flow $\dot{x} = -\mathcal{H}x$ with $\mathcal{H}$ symmetric positive semidefinite, show that forward Euler is stable iff $h < 2/\lambda_{\max}$, and that backward Euler is stable for all $h > 0$. Estimate the number of forward-Euler steps needed to reduce the softest non-zero mode by a factor e. Why do the zero (translation) modes not matter?

Exercise 4.13. Modify the point-vortex script: (a) replace DOP853 by RK45 and tighten or loosen `rtol`; plot the relative drift of H against `rtol`. (b) Give SciPy's BDF the analytic Jacobian of (4.5) and compare `nfev`. (c) Change the seed until two opposite vortices come very close, and explain the effect on the drift of H. Report your actual output.

Exercise 4.14. Compute, to six digits, the lattice sums $\sum_{a\in A\setminus 0}|a|^{-3}$ for the triangular lattice and $\sum_{a\in\mathbb{Z}^2\setminus 0}|a|^{-3}$ for the square lattice, both of covolume one. (Sum over a large disk and correct the tail by an integral; test your tail correction by doubling the radius.) Which is smaller, and what does the theorem predict? Use `norm_sq_triangularPoint` for the triangular norms.

Chapter 5

Open problems, research directions and industry

The first four chapters followed one theorem: for every completely monotone g, the triangular lattice A of covolume one has the least lower energy among locally finite planar configurations of density one [1]. This chapter is about the edges of that theorem. Some are places where the theorem is silent and something else must be said: divergent interactions, the strong Cohn–Kumar conjecture, the endpoint $s = 2$. One is a detail of the definitions that has now been settled (open versus closed disks). The rest are questions the theorem does not ask: uniqueness, temperature, other dimensions, engineering.

Recall the status of the main theorem itself: it is AI-generated (an OpenAI release), kernel-checked in Lean and accepted by Comparator in this project, and not peer reviewed [1]. Every statement below carries a tier. Most of this chapter is **[Tier C]** (argument, conjecture, proposal) or **[Tier L]** (cited literature). That is the honest state of an open-problems chapter, and it should not be read as a weakness. Table 5.1 summarises where things stand.

5.1 The divergent regime $0 < s \le 2$ and the logarithm

5.1.1 Why the raw sums diverge

Write $N(r) = \#\{a \in A : 0 < |a| \le r\}$. The lattice has covolume one, so $N(r)$ is close to the area πr^2.

Lemma 5.1 (Lattice point count)**.** *There is a constant C such that $|N(r) - \pi r^2| \le C(r+1)$ for all $r \ge 0$.*

Human proof. **[Tier C]** Let P be a fundamental parallelogram of A with $0 \in P$, of area 1 and diameter δ. The translates $a + P$, $a \in A$, tile the plane. Every translate with $|a| \le r$ lies in the disk of radius $r + \delta$, and every point of the disk of radius $r - \delta$ lies in a translate with $|a| \le r$. Comparing areas,

$$\pi(r-\delta)^2 \le N(r) + 1 \le \pi(r+\delta)^2 \qquad (r \ge \delta),$$

so $|N(r) - \pi r^2| \le 2\pi\delta r + \pi\delta^2 + 1$. For $r < \delta$ both $N(r)$ and πr^2 are bounded. □

Proposition 5.2 (Divergence rates)**.** *Let $r_0 = \min_{a \ne 0} |a|$. As $R \to \infty$,*

$$\sum_{0<|a|\le R} |a|^{-2} = 2\pi \log R + O(1),$$

$$\sum_{0<|a|\le R} |a|^{-s} = \frac{2\pi}{2-s} R^{2-s} + O(R^{1-s} + \log R + 1) \qquad (0 < s < 2).$$

Table 5.1: Status of the questions treated in this chapter. "Lean" means a kernel-checked statement in the toolchain of Chapter 2, with axiom footprint inside {`propext`, `Classical.choice`, `Quot.sound`}.

Question	Mathematical status	Lean status
Riesz $s > 2$, Gaussians, every completely monotone g	proved upstream [1] (AI-generated, not peer reviewed)	kernel-checked (upstream + our D1, D1b, D2)
Raw energies for $0 < s \le 2$	infinite for every competitor (§5.1)	divergence of the lattice sum not formalised
Renormalised / jellium, $0 \le s < 2$	proved in upstream's manuscript, §8 [1]; implication from Petrache–Serfaty [4]	not formalised anywhere
Endpoint $s = 2$	open; outside upstream's statement	—
Strong (per-potential) Cohn–Kumar	open	—
Open vs closed disks	resolved in this project (D4)	see §5.3
Uniqueness among periodic configurations	open	—
Non-uniqueness among all configurations	proved (D3)	kernel-checked
Positive temperature	open	—
Dimension 3	no universal optimum known	—
Dimensions 8, 24	proved [2]	not formalised here

Human proof. **[Tier C]** Summation by parts (Stieltjes integration against the step function N) gives, for $s > 0$,

$$\sum_{0<|a|\le R} |a|^{-s} = \int_{r_0^-}^{R} r^{-s}\,\mathrm{d}N(r) = N(R)R^{-s} + s\int_{r_0}^{R} N(r)\,r^{-s-1}\,\mathrm{d}r.$$

Insert $N(r) = \pi r^2 + E(r)$ with $|E(r)| \le C(r+1)$ from Lemma 5.1. For $s = 2$:

$$N(R)R^{-2} = \pi + O(R^{-1}), \qquad 2\int_{r_0}^{R} \pi r^2\, r^{-3}\,\mathrm{d}r = 2\pi\log R - 2\pi\log r_0,$$
$$2\int_{r_0}^{R} |E(r)|\,r^{-3}\,\mathrm{d}r \le 2C\int_{r_0}^{\infty} (r+1)r^{-3}\,\mathrm{d}r < \infty.$$

The error integral converges absolutely, so the sum is $2\pi\log R + O(1)$. For $0 < s < 2$ the main term is $\pi R^{2-s} + s\pi\int_{r_0}^{R} r^{1-s}\,\mathrm{d}r = \pi R^{2-s} + \frac{s\pi}{2-s}(R^{2-s} - r_0^{2-s}) = \frac{2\pi}{2-s}R^{2-s} + O(1)$, and the error is $s\int_{r_0}^{R} C(r+1)r^{-s-1}\,\mathrm{d}r$, which is $O(R^{1-s})$ for $s < 1$, $O(\log R)$ for $s = 1$ and $O(1)$ for $s > 1$. □

The heuristic behind the proof is worth keeping: a lattice of density one looks, from far away, like a uniform area density, and $\int_{1\le|x|\le R} |x|^{-s}\,\mathrm{d}x = 2\pi\int_1^R r^{1-s}\,\mathrm{d}r$, which is $2\pi\log R$ for $s = 2$ and grows like R^{2-s} for $s < 2$. The sum diverges at infinity exactly when this integral does, namely when $s \le d = 2$.

Numerics (illustration of Proposition 5.2)

[Tier X] The script below sums $|a|^{-2}$ over $0 < |a| \le R$ for the covolume-one triangular lattice $p(j,k) = b^{-1/2}(j + k/2,\ kb)$, $b = \sqrt{3}/2$, the parametrisation of Chapter 1. It was run as `taskset -c 2,3 .venv/bin/python -I ch5_checks.py`.

```python
def tri_s2_partial(R: float) -> float:
    b = math.sqrt(3) / 2
    sc = 1 / math.sqrt(b)
    K = int(R * 1.2 / (sc * b)) + 3
    j, k = np.meshgrid(np.arange(-K, K + 1), np.arange(-K, K + 1), indexing="ij")
    x = sc * (j + k / 2.0)
    y = sc * (k * b)
    r2 = x * x + y * y
    m = (r2 > 0) & (r2 <= R * R)
    return float(np.sum(1.0 / r2[m]))
```

Output:

```
R=   50  S=27.101261  S-2pi logR=2.521295
R=  100  S=31.457323  S-2pi logR=2.522185
R=  200  S=35.808069  S-2pi logR=2.517760
R=  400  S=40.163973  S-2pi logR=2.518491
R=  800  S=44.519131  S-2pi logR=2.518477
```

The partial sums grow by about $2\pi \log 2 \approx 4.36$ per doubling of R, and $S(R) - 2\pi \log R$ stays near 2.52 over this range. The fluctuations come from the lattice-point error $E(r)$. We claim no more digits of the constant and no closed form for it.

5.1.2 What the Lean theorem says in this regime

Upstream's Lean theorem (Chapter 2) states, for admissible g,

$$\texttt{latticeEnergy g} \le \texttt{energy g C} \wedge \texttt{latticeEnergy g} = \texttt{energy g A},$$

with both sides in $[0, \infty]$. Our `riesz_admissible` shows that $t \mapsto t^{-s/2}$ is admissible for every $s > 0$ **[Tier A]**. For $0 < s \le 2$, Proposition 5.2 shows that the lattice sum is $+\infty$ (the sum of nonnegative terms dominates every partial sum). The Lean inequality then forces `energy (riesz s) C` $= \top$ for every locally finite density-one competitor C. That is a true statement, but the comparison it contains is $\infty \le \infty$, so it carries no information about which configuration is better. This deduction is **[Tier C]**: our D1b proves `latticeEnergy_riesz_lt_top` for $s > 2$ **[Tier A]**, but the complementary statement `latticeEnergy (riesz s)` $= \top$ for $0 < s \le 2$ has not been formalised (it is Project M2 in §5.6).

The logarithm $g(t) = -\frac{1}{2}\log t$ (that is, $-\log|x|$) is not even in the class: it is negative for $t > 1$, and `AdmissiblePotential` requires $g \ge 0$. So the theorem says nothing at all about the true two-dimensional Coulomb interaction. Yet the cases $s = 1$ (the electron gas, §5.5) and $s = 0$ (the one-component plasma, vortices) are the physically central ones. To say something there, the energy itself has to be redefined.

5.1.3 Local integrability and the jellium energy

The standard cure is to neutralise the points against a uniform background of opposite charge and the same density, so that the divergent far-field contributions cancel. This needs the background to have a finite self-interaction, and that is a local question:

$$\int_{|x|<1} |x|^{-s}\,\mathrm{d}x = 2\pi \int_0^1 r^{1-s}\,\mathrm{d}r = \begin{cases} \dfrac{2\pi}{2-s}, & s < 2, \\ +\infty, & s \ge 2. \end{cases} \tag{5.1}$$

So $|x|^{-s}$ is locally integrable in $\mathbb{R}^2$ if and only if $s < 2$, and the logarithm is locally integrable as well. Put $g_s(x) = |x|^{-s}$ for $0 < s < 2$ and $g_0(x) = -\log|x|$. Upstream [1, §8, eq. (finite-ordered-jellium)] defines, for N distinct points $a_1, \dots, a_N$ and the square $K_R = [-R/2, R/2]^2$

with $R = \sqrt{N}$, the *finite ordered-pair jellium energy*

$$J_{s,R}(a_1, \dots, a_N) = \sum_{i \neq j} g_s(a_i - a_j) - 2 \sum_{i=1}^{N} \int_{K_R} g_s(a_i - x)\, \mathrm{d}x + \int_{K_R} \int_{K_R} g_s(x - x')\, \mathrm{d}x\, \mathrm{d}x', \quad (5.2)$$

with $J = +\infty$ at collisions, and the density-one thermodynamic minimum

$$e_{\mathrm{Jel}}^{(s)} = \lim_{N \to \infty} \frac{1}{N} \min_{a_1, \dots, a_N \in K_{\sqrt{N}}} J_{s,\sqrt{N}}(a_1, \dots, a_N).$$

The three terms are point–point, point–background and background–background. Each of the last two is finite precisely by (5.1). Note the convention: the sum is over *ordered* pairs, so each pair is counted twice. Physics papers often count each pair once; a factor 2 between conventions must be checked before comparing numbers. The existence of the limit is part of upstream's argument; it relies on scalar thermodynamic results of Lewin–Lieb–Seiringer ($0 < s < 2$) and Lauritsen ($s = 0$), both cited in [1, §8].

5.1.4 Upstream's renormalised corollary

Upstream also uses the *renormalised energy* W_s of Petrache and Serfaty. It is defined through a field (electric-field) formulation: for $0 < s < 2$ one passes to $\mathbb{R}^2 \times \mathbb{R}$ with the weight $|y|^\gamma$, $\gamma = s - 1$, so that g_s becomes the fundamental solution of $-\operatorname{div}(|y|^\gamma \nabla \cdot)$ with constant $\kappa_s = 4\pi$ ($\kappa_0 = 2\pi$ for the logarithm). The fields are taken in L^p_{loc} with

$$1 < p < \min\Big(2, \frac{2}{s}, \frac{3}{s+1}\Big) \qquad (0 < s < 2),$$

the energy is truncated at a small radius η, averaged over large squares and the truncation removed; $W_s(C)$ is the infimum over compatible fields. We do not reproduce the full definition; it follows Petrache–Serfaty's 2017 field construction as cited in [1, §8]. On a torus $T = \mathbb{R}^2/L$ of area N with distinct points $a_1, \dots, a_N$, the periodic value $W_{s,L}$ has the closed form

$$W_{s,L}(a_1, \dots, a_N) = \kappa_s^2 \Big[\frac{1}{N} \sum_{i \neq j} \mathcal{G}_L(a_i - a_j, 0) + \lim_{x \to 0} \Big(\mathcal{G}_L(x, 0) - \frac{g_s(x)}{\kappa_s} \Big) \Big],$$

where $\mathcal{G}_L$ is the mean-zero unit-source Green function of the extension problem [1, §8]. For a periodic configuration $C = \bigcup_{i=1}^{N} (a_i + L)$ of density one write $V_s(C) = W_{s,L}(a_1, \dots, a_N)$.

Corollary 5.3 (Planar renormalised and jellium energies; upstream, §8)**.** ***[Tier L]*** *Let $0 \leq s < 2$. For every positive integer n put $N = n^2$ and enumerate $A/(nA)$ by $a_1^{\triangle}, \dots, a_N^{\triangle}$. Every configuration of N distinct points in $\mathbb{R}^2/(nA)$ satisfies*

$$W_{s,nA}(a_1, \dots, a_N) \geq W_{s,nA}(a_1^{\triangle}, \dots, a_N^{\triangle}).$$

Moreover, A minimises W_s among density-one planar configurations, and $e_{\mathrm{Jel}}^{(s)} = \kappa_s^{-1} W_s(A)$.

The engine of the proof is an identity that compares two periodic configurations *with different period lattices* through heat kernels.

Lemma 5.4 (Heat comparison across period lattices; upstream, §8)**.** ***[Tier L]*** *Let $0 \leq s < 2$ and let C, D be simple periodic configurations of density one. Put*

$$q_0 = 2\pi, \qquad q_s = \frac{2^{2-s} \pi\, \Gamma(1 - s/2)}{\Gamma(s/2)} \ (0 < s < 2), \qquad \Psi_t(x) = (4\pi t)^{-1} e^{-|x|^2/(4t)}.$$

If $C = \bigcup_{i=1}^{N}(a_i + L)$ *with* $|\mathbb{R}^2/L| = N$*, define*

$$P_{t,L}(x) = \sum_{v \in L} \Psi_t(x - v), \qquad H_C(t) = \frac{1}{N} \sum_{i,j=1}^{N} P_{t,L}(a_i - a_j) - \Psi_t(0),$$

and define H_D *with its own period lattice and point count. Then the following integral converges absolutely and*

$$V_s(C) - V_s(D) = \frac{\kappa_s q_s}{\Gamma(1 - s/2)} \int_0^\infty \big(H_C(t) - H_D(t)\big)\, t^{-s/2}\, \mathrm{d}t.$$

How the Gaussian theorem enters. [Tier L] The subtraction of $\Psi_t(0)$ removes only the coincident point. Counting periodic images in large disks gives [1, §8]

$$H_C(t) = \frac{1}{4\pi t}\, E_{g_t}(C), \qquad g_t(u) = e^{-u/(4t)},$$

the Gaussian energy with parameter $\alpha = 1/(4\pi t)$ in the normalisation $e^{-\pi\alpha u}$. The Gaussian case of the main theorem gives $E_{g_t}(C) \ge E_{g_t}(A)$, hence $H_C(t) \ge H_A(t)$ for every $t > 0$. The prefactor is positive: $\kappa_s q_s/\Gamma(1 - s/2) = \kappa_s 2^{2-s}\pi/\Gamma(s/2) > 0$, and $t^{-s/2} > 0$. Lemma 5.4 with $D = A$ then gives $V_s(C) \ge V_s(A)$ for every periodic C. Upstream passes to the infinite system by a square-periodic approximation of the field infimum, which is why the lemma must allow *different* period lattices: the approximants are square-periodic while the minimiser is triangular. The identification of e_{Jel} uses the scalar thermodynamic inputs cited above and a confinement argument based on a weighted strong minimum principle (Fabes–Kenig–Serapioni, cited in [1, §8]), which needs $-1 < \gamma < 1$.

The structure is the same as in the Bernstein step of Chapter 2: a nonnegative superposition of Gaussian inequalities. Here the weight is $t^{-s/2}\,\mathrm{d}t$, and the superposition is applied to *differences*, so the divergent parts cancel before anything is integrated. This is why the renormalised problem follows from the Gaussian theorem even though the raw energies are infinite.

Petrache and Serfaty had earlier shown [Tier L] that universal optimality of the triangular lattice, in the form conjectured by Cohn and Kumar, implies the crystallisation results for the Coulomb and Riesz renormalised energies [4, Theorem 2]. Upstream's §8 carries out that transfer for its stated field normalisation and order of limits, starting from its own Gaussian theorem rather than from the conjecture.

Caution (what is not formalised)

[Tier C] *Nothing* in this section is in Lean. No Lean file in openai/math mentions jellium or renormalised energy (checked by grep on the pinned tree, as recorded in the extension plan of this project). Our own Lean work (Chapter 2) covers only summable interactions. Corollary 5.3 and Lemma 5.4 are [Tier L]: they are proved in an AI-generated manuscript that has not been peer reviewed [1]. The field formulation, the thermodynamic limits and the confinement argument would each be a substantial formalisation project.

5.1.5 A Lean roadmap

The extension plan of this project (`docs/EXTENSION_RENORMALIZED_JELLIUM_2026-10-11.md`) splits off the pieces that look reachable. Its priors are subjective, recorded in the plan before any attempt; they are not results.

#	Target	Prior	Kill criterion
E3	For every periodic density-one C and $t > 0$: $H_C(t) \geq H_A(t)$, derived from upstream's Gaussian theorem	$\sim 30\%$	lower energy of a periodic C cannot be identified with its periodic Gaussian sum
E4	$t^{-s/2} = \Gamma(s/2)^{-1} \int_0^\infty u^{s/2-1} e^{-tu}\,\mathrm{d}u$, and the positivity transfer: if $H_C \geq H_A$ pointwise then the $t^{-s/2}$-weighted integrals compare	$\sim 70\%$	Mathlib's Gamma-integral API does not reach the needed form
E5	Definition of $J_{s,R}$ in (5.2), with a non-vacuity check (finite at distinct points)	$\sim 60\%$	the double integral needs an integrability lemma Mathlib lacks for s near 2
E6	Endpoint $s = 2$	—	outside upstream's statement; not attempted

None of E3–E5 had been started when this chapter was written. If E3 and E4 both succeed, they give a kernel-checked version of the second half of the heat-comparison argument (the inequality $V_s(C) \geq V_s(A)$, *given* the identity of Lemma 5.4). They would not give the field formulation, the identity itself or the thermodynamic limit. Even the easiest step, E4, is low value on its own; its point is to be the first brick.

5.1.6 A numerical check at $s = 1$

Numerics (2D Wigner crystal, $s = 1$ jellium)

[Tier X] On the branch `physics/jellium-wigner-2026-10-11` of the rusty-SUNDIALS crate `qf-yukawa-crystal` [18] (preregistration e706f85, code 3bc55fa), the $s = 1$ jellium energy per particle at density one was computed by Ewald summation. The triangular value is $e_{\mathrm{tri}} = -1.960515789320$, against the literature value -1.960515789825 credited to Alves et al. [9] (relative difference 2.6×10^{-10}). The square lattice gives $e_{\mathrm{sq}} = -1.950132460$, which agrees with the corresponding literature value for the square lattice to relative 4.3×10^{-10}, so $e_{\mathrm{tri}} < e_{\mathrm{sq}}$, in agreement with the classical lattice comparison of Bonsall and Maradudin [8]. The square lattice is a saddle (64 unstable modes in the tested cell), and the result is invariant under the Ewald splitting parameter to better than 10^{-10} over a factor-4 range. The crate counts each pair once (Chapter 3), so its energies per particle are one half of those in the ordered-pair convention of (5.2); the *ordering* triangular below square does not depend on the factor.
Status of the main relaxation study, the study E1 (prediction J1: no relaxed periodic configuration below e_{tri}): the main jellium relaxation study (prediction J1) was still running when this book was assembled.
Even a perfect outcome is Tier X: it would be a consistency check of Corollary 5.3 in one case, not evidence for it.

5.1.7 The endpoint $s = 2$

The endpoint $s = d = 2$ is open, and it lies outside upstream's statement. Here is precisely what breaks.

1. **The background is not integrable.** By (5.1), $\int_{K_R} |a - x|^{-2}\,\mathrm{d}x = +\infty$ for every $a \in K_R$, and so is the background self-energy. In (5.2) the second and third terms are both infinite and enter with opposite signs: $J_{2,R}$ is of the form $\infty - \infty$ and has no definition. This is the basic obstruction; the next three are what it looks like inside upstream's proof.

2. **The field class is empty.** The admissible range $1 < p < \min(2, 2/s, 3/(s+1))$ becomes $1 < p < 1$ at $s = 2$.

3. **The weight leaves the good class.** The extension weight $|y|^\gamma$ has $\gamma = s - 1 = 1$, outside the range $-1 < \gamma < 1$ used for the weighted minimum principle in the confinement step.

4. **The fractional Laplacian degenerates.** In the proof of Lemma 5.4 the operator is $(-\Delta)^{1-s/2}$; at $s = 2$ it is the identity, which has no Green function, and q_s has a pole from $\Gamma(1 - s/2)$.

Items 2–4 are **[Tier C]** observations read off the formulas of [1, §8], not theorems about every possible approach. They do not say that $s = 2$ is meaningless. They say that a different renormalisation would be needed.

Remark 5.5 (a possible periodic quantity at $s = 2$). **[Tier C]** (unverified sketch). The bounds that make the integral in Lemma 5.4 converge, $|H_C(t) - H_D(t)| \le Ct^{-1}e^{-c/t}$ as $t \downarrow 0$ and $\le Ce^{-\lambda t}$ as $t \to \infty$, do not involve s. The prefactor $\kappa_s q_s/\Gamma(1 - s/2) = \kappa_s 2^{2-s}\pi/\Gamma(s/2)$ also has a finite limit at $s = 2$. So for periodic configurations the right-hand side $\int_0^\infty (H_C - H_A)(t)\, t^{-1}\, dt$ is a finite number, and it is nonnegative by the Gaussian theorem. What is missing is a left-hand side: an independently defined $s = 2$ renormalised energy whose differences this integral computes, and an infinite-system theory that reduces to it. We have not checked whether such a quantity appears in the literature. This is Project P2 in §5.6, and it should not be quoted as a result.

5.2 The strong (per-potential) Cohn–Kumar conjecture

5.2.1 Statement

Cohn and Kumar conjectured that the triangular, E_8 and Leech lattices are universally optimal [3, Conjecture 9.4]. The form of the conjecture is stronger than an energy inequality: it asks for a *certificate* for each potential.

Definition 5.6 (Sharp auxiliary function)**.** Let g be completely monotone on $(0, \infty)$ with sufficient decay. A *sharp auxiliary function* for g and A is a radial function $f : \mathbb{R}^2 \to \mathbb{R}$ in an admissible class (integrable, with integrable Fourier transform and enough decay of both for Poisson summation over A and A^*; see [3, Conjecture 9.4] and its context for the exact class) such that

$$\begin{aligned} f(x) &\le g(|x|^2) \quad (x \neq 0), & \hat{f}(\xi) &\ge 0 \quad (\xi \in \mathbb{R}^2), \\ f(a) &= g(|a|^2) \quad (a \in A \setminus \{0\}), & \hat{f}(w) &= 0 \quad (w \in A^* \setminus \{0\}). \end{aligned}$$

The *strong Cohn–Kumar conjecture* in the plane is the statement: *for each* completely monotone g with sufficient decay, a sharp auxiliary function exists. **[Tier L]** In upstream's words, the conjecture "asks for a sharp auxiliary function for each sufficiently decaying completely monotone squared-distance interaction: a Fourier lower bound that attains the lattice energy and therefore proves the comparison for periodic competitors" [1, §1].

Why the conditions certify optimality is the Cohn–Elkies-type argument of Chapter 2: the two inequalities give $E_g(C) \ge \hat{f}(0) - f(0)$ for every competitor of density one, and Poisson summation on A with both contact conditions gives

$$f(0) + \sum_{a \neq 0} g(|a|^2) = \sum_{a \in A} f(a) = \sum_{w \in A^*} \hat{f}(w) = \hat{f}(0),$$

so the bound equals the lattice energy.

Caution (status)

The strong per-potential conjecture is **open** in the plane. Upstream proves the energy inequality for every completely monotone g, and constructs sharp auxiliary functions f_α for every Gaussian $G_\alpha(x) = e^{-\pi\alpha|x|^2}$. It states **[Tier L]**: "Integrating the resulting energy inequalities does not construct a sharp auxiliary for each mixed potential in the class required by Cohn and Kumar's conjecture. That stronger per-potential quantifier is not claimed here." [1, §1]

In dimensions 8 and 24, Cohn, Kumar, Miller, Radchenko and Viazovska proved universal optimality and constructed sharp Gaussian bounds and Fourier interpolation formulas [2, Theorems 1.4, 1.7 and 1.9]. We say no more about the per-potential form there than that.

5.2.2 What integrating the Gaussians does and does not give

It is tempting to think that the per-potential statement follows at once. Write a completely monotone g (with no constant part: by upstream's lemma `energy_one`, a constant has infinite energy at density one) as

$$g(t) = \int_{(0,\infty)} e^{-\pi\alpha t}\,\mathrm{d}\mu(\alpha),$$

with μ a positive measure, and try

$$f(x) = \int f_\alpha(x)\,\mathrm{d}\mu(\alpha). \tag{5.3}$$

Formally, every condition of Definition 5.6 holds by linearity: $f \le g$ integrates $f_\alpha \le G_\alpha$; $\hat{f} = \int \hat{f}_\alpha\,\mathrm{d}\mu \ge 0$; at $a \in A \setminus \{0\}$, $f(a) = \int G_\alpha(a)\,\mathrm{d}\mu = g(|a|^2)$; at $w \in A^* \setminus \{0\}$, $\hat{f}(w) = 0$. So the issue is not that mixing destroys the contact conditions or the signs. It does not.

What upstream actually integrates is different. It integrates the *scalar* inequalities $E_{G_\alpha}(C) \ge \vartheta_A(\alpha) := \theta_A(\alpha) - 1 = \sum_{a\neq 0} e^{-\pi\alpha|a|^2}$ (here θ_A is the full theta function of Chapter 1, which includes $a = 0$, and ϑ_A omits it), which are explicit and measurable in α on both sides; Tonelli and the lower-limit argument of upstream's §5 then give the energy inequality for g. No function is ever mixed. To turn the formal computation (5.3) into a proof, one would need at least the following **[Tier C]**:

1. **Measurability.** The construction is carried out for each α separately, with finite reference lists and certified inequalities chosen per range of parameters. One needs a choice $\alpha \mapsto f_\alpha$ for which $(\alpha, x) \mapsto f_\alpha(x)$ is jointly measurable.

2. **Absolute integrability.** To exchange Fourier transform and μ-integral (Fubini), one needs $\int \|f_\alpha\|_{L^1}\,\mathrm{d}\mu(\alpha) < \infty$, and similarly for $\hat{f}_\alpha$.

3. **The admissible class.** The mixture must again have the decay that Poisson summation and the class of [3, Conjecture 9.4] require. This needs bounds on the decay constants of f_α and $\hat{f}_\alpha$ that are uniform, or at least μ-integrable, in α.

4. **Finite values at the origin.** The quantities $f(0) = \int f_\alpha(0)\,\mathrm{d}\mu$ and $\hat{f}(0) = \int \hat{f}_\alpha(0)\,\mathrm{d}\mu$ must be finite separately, not only their difference.

Upstream gives none of these estimates, and neither does this book.

A Riesz example shows where each end matters. **[Tier C]** For $g(t) = t^{-s/2}$, the Gamma integral gives $t^{-s/2} = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^\infty \alpha^{s/2-1} e^{-\pi\alpha t}\, \mathrm{d}\alpha$, so $\mathrm{d}\mu \propto \alpha^{s/2-1}\, \mathrm{d}\alpha$.

- As $\alpha \to 0$: Poisson summation gives $\vartheta_A(\alpha) = \alpha^{-1} \sum_{w \in A^*} e^{-\pi|w|^2/\alpha} - 1 \sim \alpha^{-1}$, so $\int_0 \vartheta_A \, \mathrm{d}\mu < \infty$ iff $s > 2$. This is the finiteness of the lattice energy (D1b), and it is the same at the level of functions, since $\hat{f}_\alpha(0) - f_\alpha(0) = \vartheta_A(\alpha)$.
- As $\alpha \to \infty$: μ has infinite mass ($\int^\infty \alpha^{s/2-1} \mathrm{d}\alpha = \infty$ for $s > 0$). The constraints give only $-\vartheta_A(\alpha) \le f_\alpha(0) \le 1$: the upper bound from $f_\alpha \le G_\alpha$, the lower from $\hat{f}_\alpha(0) = f_\alpha(0) + \vartheta_A(\alpha) \ge 0$. So $f(0)$ is finite only if $f_\alpha(0)$ decays fast enough as $\alpha \to \infty$. Upstream gives no such estimate. Note also that $g(|x|^2) \to \infty$ at $x = 0$, so $f \le g$ does not bound $f(0)$.

For a g whose measure μ is finite and supported in a compact interval $[\alpha_0, \alpha_1] \subset (0, \infty)$, items 2–4 reduce to bounds that are locally uniform in α. In that case the obstruction may well be technical, but we have not checked that the construction satisfies such bounds, so we do not claim it. For Riesz potentials, the large-α end is a genuine quantitative question about the construction, not a formality.

Why one would want the certificate anyway. **[Tier C]** The energy inequality is already proved, so a per-potential certificate would not change the minimum value. It would give an explicit object that can be checked numerically for each g. It would match the form of the conjecture as Cohn and Kumar posed it. It could also carry information that the integrated inequality loses: the gap $g(|x|^2) - f(x)$ and the function $\hat{f}$ measure how far a competitor is from optimal, and that is the kind of input that stability and uniqueness questions (§5.4) need.

Small research projects. (i) Take $g(t) = \int_1^2 e^{-\pi\alpha t}\, \mathrm{d}\alpha$ and try to establish items 1–4 for upstream's construction on $[1, 2]$. (ii) Measure numerically how $f_\alpha(0)$ behaves for large α, using upstream's verification scripts. (iii) Build a sharp auxiliary function directly, by linear programming, for one non-Gaussian g (Project M4). Each of these is a step, not a solution.

5.3 Open versus closed disks: resolved (D4)

Upstream measures density and energy in *closed* disks $\overline{B}(0, R)$. A reviewer asked whether open disks would change the theorem. They do not.

Proposition 5.7 (D4). ***[Tier C]*** *(human proof; Lean status below). For every locally finite $C \subset \mathbb{R}^2$: C has density one in closed disks if and only if it has density one in open disks; and for every g, the lower energy computed with open disks equals the lower energy computed with closed disks. Hence upstream's theorem holds verbatim with open disks.*

Human proof. Local finiteness makes $C \cap \overline{B}(0, R)$ finite, so only finitely many norms $|x|$, $x \in C$, lie in $[0, R]$. Hence there is $\delta \in (0, 1)$ with no point of C of norm in $[R-\delta, R)$, and then the open-disk point set at radius R equals the closed-disk point set at radius $R - \delta$. Likewise the closed-disk set at R equals the open-disk set at some $R+\delta'$ with $\delta' \in (0, 1)$. So each disk-energy function takes the values of the other at radii within distance 1: $E^{\mathrm{open}}(R) = E^{\mathrm{closed}}(R - \delta_R) \ge \inf_{r \ge R-1} E^{\mathrm{closed}}(r)$. Letting $R \to \infty$ gives $\liminf E^{\mathrm{open}} \ge \liminf E^{\mathrm{closed}}$, and the symmetric argument gives the reverse. For the counts, $N_{\mathrm{closed}}(R-1) \le N_{\mathrm{open}}(R) \le N_{\mathrm{closed}}(R)$, and $(R-1)^2/R^2 \to 1$, so the two densities have the same limit. The full proof is in Chapter 2. □

Lean status: **[Tier A]** kernel-checked for density one and for the lower energy, independently re-verified; other densities **[Tier C]** (details in the Editor's note).

What D4 does *not* touch is uniqueness. Even with both disk conventions settled, whether A is the unique minimiser among *periodic* configurations of density one remains open in the plane.

5.4 Uniqueness, temperature and other dimensions

5.4.1 Uniqueness

Upstream is explicit **[Tier L]**: its theorem "identifies the minimum value, not all minimizers" [1]. Among all locally finite configurations, uniqueness is in fact false. Our D3 **[Tier A]** proves that the configuration $A \setminus \{0\}$ (`vacancy`) has density one and the same lower energy as A for every admissible g (`vacancy_energy`, `vacancy_universally_optimal`), and that it is not an isometric image of A (`vacancy_not_isometric_image`); hence `universal_optimum_not_unique`. A single vacancy changes a density-one average by nothing. The meaningful uniqueness questions are therefore of the following kinds **[Tier C]**:

- **Periodic uniqueness.** Is A (up to isometry) the only periodic density-one minimiser for, say, every Gaussian? Open in the plane.
- **Uniqueness up to negligible sets.** Is every minimiser equal to an isometric image of A outside a set of density zero? Even formulating this needs care with lower energies.
- **Local versions.** Several restricted results are cited in [1, §1]: Faulhuber et al. (honeycomb, products of one-dimensional configurations, unions of two translates), Hardin–Tenpas (four-point and six-point periodic classes, under a rapid-decay assumption) and Leblé (sufficiently small bounded displacements cannot lower the energy). We cite them only through [1].

5.4.2 Positive temperature

The theorem is a statement at zero temperature: it compares energies, not free energies. Whether particles with a completely monotone repulsion crystallise into a triangular arrangement at small positive temperature, and in what sense of order, is not addressed by any result in this book's references. Physicists expect that two-dimensional positional order at positive temperature is weaker than at zero temperature; we do not cite that literature, so treat it as a pointer, not a claim **[Tier C]**.

The numerics of Chapter 3 show the gap between "is the minimum" and "is reached" **[Tier X]**. In the Yukawa study, annealing reached the perfect lattice in only 1 of 20 runs per initial-condition family at $\kappa = 2$, and in 0 of 10 and 2 of 10 runs at $\kappa = 1$ (prediction P2 failed there); zero-temperature quenches at $\kappa = 2$ ended defected in 15 of 20 and 14 of 20 runs. Energetic optimality and kinetic accessibility are different questions.

5.4.3 Other dimensions

[Tier L] Per CKMRV, Ventevogel and Nijboer proved universal optimality of $\mathbb{Z}$ in dimension 1, and Cohn and Kumar gave another proof; before CKMRV no universal optimum was known in any dimension above 1, and the case of A_2 "in full generality is an important open problem" [2, §1.2]. CKMRV settled dimensions 8 and 24 (E_8 and Leech) [2]; upstream claims dimension 2 [1]. Cohn and Kumar's conjecture singled out the triangular, E_8 and Leech lattices [3, 1]. None of this is formalised here.

Dimension 3: fcc against bcc. **[Tier C]** Scale the face-centred cubic lattice $D_3 = \{x \in \mathbb{Z}^3 : x_1 + x_2 + x_3 \text{ even}\}$ by $2^{-1/3}$ and the body-centred cubic lattice $\mathbb{Z}^3 \cup (\mathbb{Z}^3 + \frac{1}{2}(1,1,1))$ by $2^{1/3}$, so both have covolume one. With these scalings the two lattices are exactly dual to each other, since $D_3^* = \mathbb{Z}^3 \cup (\mathbb{Z}^3 + \frac{1}{2}(1,1,1))$ and $(c\Lambda)^* = c^{-1}\Lambda^*$. Poisson summation for a covolume-one lattice $\Lambda \subset \mathbb{R}^3$ gives $\sum_{x\in\Lambda} e^{-\pi\alpha|x|^2} = \alpha^{-3/2} \sum_{w\in\Lambda^*} e^{-\pi|w|^2/\alpha}$, hence

$$\Theta_{\mathrm{fcc}}(\alpha) = \alpha^{-3/2}\, \Theta_{\mathrm{bcc}}(1/\alpha), \qquad \Theta(\alpha) := \sum_{x\in\Lambda} e^{-\pi\alpha|x|^2},$$

and at $\alpha = 1$ the two Gaussian energies are exactly equal. Consequently, if fcc has strictly lower Gaussian energy than bcc at some $\alpha_* > 1$, then bcc has strictly lower Gaussian energy than fcc at $1/\alpha_*$ (because $\Theta - 1$ is the energy and the identity transfers the strict inequality of Θ). Neither lattice can then be universally optimal.

Numerics (fcc against bcc at density one)

[Tier X] Same script (`ch5_checks.py`), summing $e^{-\pi\alpha|x|^2}$ over $x \neq 0$ in the box $\{-12, \dots, 12\}^3$ before scaling:

```
def theta3(kind: str, alpha: float, K: int = 12) -> float:
    rng = np.arange(-K, K + 1)
    X, Y, Zc = np.meshgrid(rng, rng, rng, indexing="ij")
    pts = np.stack([X.ravel(), Y.ravel(), Zc.ravel()], axis=1).astype(float)
    if kind == "fcc":
        pts = pts[(pts.sum(axis=1) % 2) == 0]   # D3, covolume 2
        pts = pts * 2.0 ** (-1.0 / 3.0)
    else:
        pts = np.vstack([pts, pts + 0.5])   # bcc, covolume 1/2
        pts = pts * 2.0 ** (1.0 / 3.0)
    r2 = (pts ** 2).sum(axis=1)
    r2 = r2[r2 > 0]
    return float(np.sum(np.exp(-math.pi * alpha * r2)))
```

Output:

```
alpha= 0.50  fcc=1.8419799351  bcc=1.8408091753  fcc-bcc=+1.171e-03
alpha= 0.80  fcc=0.5182946954  bcc=0.5170433085  fcc-bcc=+1.251e-03
alpha= 1.00  fcc=0.2315362662  bcc=0.2315362662  fcc-bcc=-2.776e-17
alpha= 1.25  fcc=0.0855078280  bcc=0.0864032476  fcc-bcc=-8.954e-04
alpha= 2.00  fcc=0.0043777160  bcc=0.0047916420  fcc-bcc=-4.139e-04
```

fcc is lower for $\alpha > 1$ (short range) and bcc for $\alpha < 1$ (long range), with equality at $\alpha = 1$ to rounding error, as the duality predicts. The crossing is a floating-point observation and supports no claim by itself; a student can make it rigorous with interval arithmetic (Project U2).

If the Tier X inequality at $\alpha = 2$ is certified (Project U2), then neither fcc nor bcc is universally optimal; that is all it would show. It does not show that no configuration in $\mathbb{R}^3$ is universally optimal; that stronger statement remains conjectural, and in this book it is **[Tier C]**.

5.5 Industry and engineering implications

Everything in this section is **[Tier C]**. We give no performance numbers, because none of the references in this book measures an engineering benefit of the theorem, and we will not invent one.

Caution (what the theorem does and does not say)

- It is about **energy minimisation** for pair potentials that are completely monotone functions of squared distance, at fixed density, at zero temperature. It says the triangular lattice achieves the least lower energy.
- It does *not* say that a physical system reaches that state (Chapter 3: kinetics, defects), that it is the unique ground state (D3), or anything about non-pair, anisotropic, attractive or three-body interactions.

- Packing, quantisation, covering and sampling optimisation are *different functionals*. The theorem settles none of them directly.

5.5.1 Which interactions are covered

A physical system is in scope if its effective pair potential is $V(r) = g(r^2)$ with g completely monotone. Three families matter.

- **Screened Coulomb (Yukawa)** $V(r) = e^{-\kappa r}/r$. In $t = r^2$ this is $t^{-1/2}e^{-\kappa\sqrt{t}}$. Here $\sqrt{t}$ is a Bernstein function, so $e^{-\kappa\sqrt{t}}$ is completely monotone; $t^{-1/2}$ is completely monotone; and products of completely monotone functions are completely monotone. So Yukawa is covered directly. This is the model of Chapter 3 (shifted-force cutoffs in the simulations are a modelling choice and are not covered).
- **Inverse powers** r^{-s} with $s > 2$: covered (our D1, D1b). Dipolar repulsion r^{-3} is the case $s = 3$.
- **Unscreened Coulomb** r^{-1} ($s = 1$) and the logarithm ($s = 0$). Here 'Coulomb' means the three-dimensional $1/r$ law restricted to the plane (Riesz $s = 1$); the planar Coulomb (logarithmic, $s = 0$) interaction is distinguished where it matters. Both are *not* covered by the main theorem; only by the renormalised Corollary 5.3 (**[Tier L]**, not formalised), which assumes a neutralising uniform background.

5.5.2 Colloidal self-assembly and coatings

Two-dimensional colloidal crystals are studied by video microscopy [10] and in confined geometries [12]. Where the effective interaction is a screened Coulomb repulsion or, for magnetic colloids in a field, a dipolar repulsion (check the interaction model in each paper before relying on it), the theorem identifies the triangular lattice as the minimum-energy arrangement at the given density. For monolayer coatings and colloidal masks, this is a useful *target statement*: an assembly process that ends in a non-triangular, defect-free monolayer of purely repulsive, isotropic particles is not at its energy minimum, and something (kinetics, the substrate, anisotropy, attraction) is responsible. The Chapter 3 numerics are a warning in the other direction: defects and incomplete annealing are typical, so the theorem says what the target is, not that it is reached.

5.5.3 Photonic and phononic crystals

Many photonic and phononic devices use triangular (hexagonal) arrays of holes or rods. The design criterion there is a band-structure property (a band gap, a dispersion), which depends on the geometry through a wave equation, not through a pair energy. The theorem does not imply that the triangular lattice gives the best band gap. Its relevance is indirect: if such arrays are fabricated by self-assembly of repulsive particles, the triangular arrangement is the energetically favoured one, so the fabrication route and the design target agree. Any stronger statement needs its own argument.

5.5.4 Wigner crystals: electrons on helium, ion traps, two-dimensional materials

Electrons on the surface of liquid helium form a two-dimensional crystal [13]; dusty plasmas form "plasma crystals" [11]. Bare electrons interact by r^{-1}, so the relevant mathematics is the $s = 1$ jellium problem, covered only by Corollary 5.3; our numerical check above agrees with the classical lattice comparison [8, 9]. Dusty plasmas are commonly modelled with screened (Yukawa)

interactions, which are in scope of the main theorem. Trapped-ion crystals and electron crystals in two-dimensional materials are quantum, often confined by a non-uniform trap, and sometimes not purely two-dimensional. The theorem is classical and assumes uniform density, so it gives a reference arrangement for the bulk of a large, nearly uniform layer at best; edge structure and quantum fluctuations are outside it. For device engineers the value is a clean benchmark: simulation codes for such crystals can be tested against the exact minimum (Chapter 3 used exactly that control).

5.5.5 Hexagonal sampling, quantisation and sensor arrays

Hexagonal grids are used in image sensors, antenna arrays and sampling of two-dimensional fields. The reasons there are of three kinds, and it matters which.

- **Packing.** As $\alpha \to \infty$, a Gaussian energy is dominated by the shortest distances. A heuristic argument therefore relates universal optimality to optimal packing: a density-one configuration with larger minimal distance than A would beat it for large α. In the plane the optimal packing is classical and is not new information.
- **Sampling.** The efficiency of a sampling lattice for an isotropic band-limited field is a packing-type statement about the *dual* lattice in frequency space. Grating lobes of an antenna array are likewise governed by the dual lattice. The duality $A \leftrightarrow A^*$ appears in the theorem's proof too, but the theorem is not a sampling theorem.
- **Quantisation.** The mean-squared quantisation error of a lattice is the normalised second moment of its Voronoi cell: a single-cell integral, not a pair energy. The theorem says nothing about it.

The honest summary: the triangular lattice is optimal for several unrelated planar criteria, and universal optimality adds one more, the strongest one for pair energies. It does not imply the others, and an engineer choosing a hexagonal array should justify it by the criterion that actually matters for the device.

5.5.6 Verified numerical software for safety-critical simulation

The most transferable lesson of this project is methodological. Its pipeline has four layers.

1. A theorem is checked by the Lean kernel against stated definitions, with an axiom audit. The audit flags planted controls (Chapter 2). This is Tier A for the formal statement only. Statement adequacy, meaning whether the Lean definitions say what the prose means, is a separate human audit. In this project it has not been done: 6 ledger flags are open.
2. Inequalities used by the proof are certified with ball arithmetic (upstream's verification scripts use python-flint).
3. Floating-point simulations (CVODE in rusty-SUNDIALS [18]) are preregistered and carry positive and negative controls: the perfect lattice, the square-lattice saddle, incommensurate boxes (Chapter 3).
4. Each claim carries a tier, so that a reader can see what supports it [17].

For safety-critical simulation (structural, nuclear, aerospace, medical), the same separation is useful. Prove the properties that can be proved. Certify the finite computations that can be certified. Label the rest as evidence of a weaker kind, and test the simulation against exactly known solutions, of which a universally optimal lattice is a clean example. We make no claim that this pipeline is ready for certification regimes; that would need its own assessment.

5.5.7 Fluid dynamics: vortex arrays, mixing and superfluids

Point vortices in two-dimensional ideal flow interact through the logarithm. Experiments have observed relaxation of two-dimensional turbulence in magnetised electron columns to "vortex crystals" [14], and triangular vortex lattices in rotating Bose–Einstein condensates [15]. The mathematics here is the $s = 0$ case: in a rotating frame, the uniform rotation plays a role analogous to the neutralising background, and the relevant statement is the logarithmic part of Corollary 5.3 (**[Tier L]**, not formalised). The analogy is a modelling statement **[Tier C]**: real systems are finite, have boundaries, and (for the electron-column experiments) do not form a uniform-density lattice. For superfluid devices such as rotation sensors, the theorem gives the reference bulk arrangement of a large vortex array; for mixing applications, where vortex arrays are a transient structure, it is a statement about the end state of a relaxation, not about the mixing rate.

5.6 Student projects

Each project has a level, a concrete first step and a deliverable that can fail. "U" is undergraduate, "M" master's level, "P" doctoral. Respect the book's rules: tier every claim, run a positive and a negative control before reporting a number, and never cite a Lean name you have not checked exists.

1. **U1. An explicit lattice-point bound.** *First step:* compute the diameter δ of a fundamental parallelogram of A and make Lemma 5.1 explicit. *Deliverable:* a constant C with $|N(r) - \pi r^2| \le C(r+1)$ for all $r \ge 0$, checked exactly (integer arithmetic on $j^2 + jk + k^2$) for all $r \le 1000$. It fails if one value violates it.

2. **U2. The fcc/bcc crossing, rigorously.** *First step:* prove the duality $\Theta_{\mathrm{fcc}}(\alpha) = \alpha^{-3/2}\Theta_{\mathrm{bcc}}(1/\alpha)$ with the scalings above. *Deliverable:* an interval-arithmetic certificate that $\Theta_{\mathrm{fcc}}(2) < \Theta_{\mathrm{bcc}}(2)$, including a rigorous bound on the truncated tail. It fails if the intervals overlap.

3. **U3 / M. The $s = 0$ jellium (logarithmic) Ewald energy.** *First step:* implement the mean-zero periodic Green function of $-\Delta$ on a torus by Ewald splitting, and check that the energy does not depend on the splitting parameter (positive control). Choose the pair-counting convention of (5.2) and say so. *Deliverable:* the log-jellium energies per particle of the triangular and square lattices at density one, with an independent second method (for example a direct Fourier-side formula you derive) agreeing to stated precision, and the sign of $e_{\mathrm{tri}} - e_{\mathrm{sq}}$. Corollary 5.3 predicts triangular lower. The project fails, and the failure must be reported, if the two methods disagree or the sign is reversed.

4. **M1. Formalise the Laplace-transform step (E4).** *First step:* in the pinned toolchain, find Mathlib's Gamma-integral lemmas. In the Mathlib checkout used by this repository's `formal/` directory, `Real.integral_rpow_mul_exp_neg_mul_Ioi` states $\int_{t>0} t^{a-1}e^{-rt}\,\mathrm{d}t = (1/r)^a\,\Gamma(a)$ for $a, r > 0$. Check that it exists in Mathlib d13f23b7 before relying on it. *Deliverable:* a kernel-checked proof that $t^{-s/2} = \Gamma(s/2)^{-1}\int_0^\infty u^{s/2-1}e^{-tu}\,\mathrm{d}u$ for $t, s > 0$, and the positivity transfer: if $h \ge 0$ is integrable against $t^{-s/2}$, the integral is ≥ 0. Gate it as in Chapter 2: axiom audit, and a planted `sorry` control that the audit flags.

5. **M2. Divergence in Lean.** *First step:* state `latticeEnergy (riesz s)` $= \top$ for $0 < s \le 2$, the complement of D1b. *Deliverable:* a kernel-checked proof, for example by bounding the sum below by a constant times $\sum_{n\ge1} n \cdot n^{-s}$ using shells of A. A negative control is $s = 3$, where the proof must fail because D1b proves finiteness.

6. **M3. The jellium energy as a Lean definition (E5).** *First step:* define $J_{s,R}$ of (5.2) for $0 < s < 2$ using the Bochner integral. *Deliverable:* a kernel-checked non-vacuity lemma, namely that $J_{s,R}$ is finite at distinct points. This requires the local integrability (5.1) in Lean. It fails if the integrability lemma cannot be established.

7. **M4. A sharp auxiliary function for one non-Gaussian g.** *First step:* take $g(t) = \int_1^2 e^{-\pi\alpha t}\,\mathrm{d}\alpha = (e^{-\pi t} - e^{-2\pi t})/(\pi t)$, a radial basis such as Gaussians times Laguerre polynomials (whose Fourier transforms are explicit), and set up a linear program: impose $f \le g$ and $\hat{f} \ge 0$ on fine radial grids, impose the contact conditions on the first shells of A and A^*, and maximise $\hat{f}(0) - f(0)$. Positive control: run it for a Gaussian, where the optimum must reach $\vartheta_A(\alpha)$. *Deliverable:* the gap between the LP optimum and $\sum_{a\neq 0} g(|a|^2)$ as the basis grows, and a report of where it fails: which constraint is active off the grid, and which shell's contact condition cannot be met. This is evidence, not proof (Tier X), unless the final function is certified with ball arithmetic.

8. **M5. Jellium relaxation at other s.** *First step:* reproduce the $s = 1$ Ewald controls of §5.1 (triangular against literature, square-lattice saddle). *Deliverable:* the same relaxation test (no periodic configuration below triangular) at $s = 1/2$ or $s = 3/2$, preregistered, with the outcome reported whatever it is.

9. **P1. The strong conjecture for compactly supported mixtures.** *First step:* read upstream's construction and decide whether $\alpha \mapsto f_\alpha$ can be chosen measurably on $[\alpha_0, \alpha_1]$. *Deliverable:* either a proof of items 1–4 of §5.2 for every μ supported in a compact subinterval of $(0, \infty)$, hence a sharp auxiliary function for each such g, or an explicit point where the construction's bounds are not uniform in α.

10. **P2. A periodic $s = 2$ energy.** *First step:* make Remark 5.5 rigorous, that is, prove the convergence of $\int_0^\infty (H_C - H_A)(t)\, t^{-1}\,\mathrm{d}t$ for periodic C. *Deliverable:* an identification of this integral with the difference of an independently defined periodic quantity (for example a finite part of the divergent lattice sums of Proposition 5.2), or a proof that no such identification holds.

11. **P3. The heat-trace comparison in Lean (E3).** *First step:* for a periodic C, prove in Lean that upstream's lower energy `energy g C` for a Gaussian g equals the periodic sum $\frac{1}{N}\sum_{i,j}\sum'_{v\in L}$ that defines H_C. *Deliverable:* the kernel-checked inequality $H_C(t) \ge H_A(t)$ for all $t > 0$, derived from the upstream theorem. The prior recorded before any attempt is about 30%.

12. **P4. Periodic uniqueness for Gaussians.** *First step:* use the Gaussian auxiliary functions and their equality cases: a periodic minimiser must have all its difference vectors where $f_\alpha = G_\alpha$. *Deliverable:* a proof that every periodic density-one minimiser for one fixed Gaussian is an isometric image of A, or a description of what the contact sets of the known f_α allow. It fails if the equality set of f_α is larger than $A \setminus \{0\}$.

Exercises

Exercise 5.1. Show that the sum $\sum_{0<|a|\le R} |a|^{-s}$ over $\mathbb{Z}^2$ also behaves like $2\pi \log R$ for $s = 2$. Explain why the leading term does not depend on the lattice, and why the constant term can.

Exercise 5.2. Compute $\int_{K_R} |a - x|^{-s}\,\mathrm{d}x$ for a at the centre of K_R, show that it is finite for $s < 2$, and find its behaviour as $s \uparrow 2$.

Exercise 5.3. Verify that $q_s\kappa_s/\Gamma(1 - s/2) = \kappa_s 2^{2-s}\pi/\Gamma(s/2)$, and use the Gamma integral to show $\int_0^\infty (4\pi t)^{-1} e^{-r^2/(4t)} t^{-s/2}\,\mathrm{d}t = (4\pi)^{-1}\Gamma(s/2)\, 4^{s/2} r^{-s}$ for $r, s > 0$.

Exercise 5.4. Prove that $t \mapsto t^{-1/2} e^{-\kappa\sqrt{t}}$ is completely monotone on $(0, \infty)$ directly from the definition, for $\kappa > 0$.

Exercise 5.5. In the proof of Proposition 5.7, where exactly is local finiteness used? Describe what goes wrong in the choice of δ if infinitely many points of C accumulate at the circle $|x| = R$ from inside, and recall how upstream's `diskPoints` treats an infinite intersection.

Summary

For $0 < s \leq 2$ the raw energies are infinite: at $s = 2$ the partial sums grow like $2\pi \log R$. The renormalised and jellium energies for $0 \leq s < 2$ are minimised by A according to upstream's §8 **[Tier L]**, and none of it is in Lean. At $s = 2$ even the jellium energy is undefined, because $|x|^{-2}$ is not locally integrable in the plane. The strong per-potential Cohn–Kumar conjecture is open. Mixing the Gaussian certificates satisfies its conditions formally, but needs measurability, integrability and decay estimates that nobody has supplied. Open versus closed disks is settled (D4). Uniqueness of the minimiser fails among all locally finite configurations (D3) and is open among periodic ones. In engineering the theorem gives a clean target and a clean benchmark for repulsive, isotropic pair interactions, and nothing more.

Editor's note: status at assembly

The five chapters were written independently, each from the same fact sheet (`FACTS.md`), and combined afterwards. This note records what changed while they were being written.

D4: open versus closed disks [Tier A]. The theorem of Section 2.7 (open-disk density and lower energy coincide with the closed-disk ones) is kernel-checked in `formal_cert/Triangular/OpenDisk.lean` for density one and for the lower energy. The density statement at other densities $\rho \neq 1$ is **[Tier C]**: the same argument applies, but it is not formalised. The Lean theorems are:

- `energy_eq_openEnergy`, `densityOne_iff_open` and `universal_energy_minimum_open`, which is upstream's theorem verbatim with open disks;
- `A_densityOneOpen`, the non-vacuity check.

It was preregistered before the proof was written. An independent verification run accepted it with corrections. That run found that the two identities need no local-finiteness hypothesis at all, and kernel-checked this in `OpenDiskAny.lean`. It also added a control showing that the identity is not definitional: for a single point on the unit circle, the open and closed counts differ at $R = 1$. Every footprint is within {propext, Classical.choice, Quot.sound}.

Scope: this settles the open/closed question for upstream's observables only (centred disks, $N/(\pi R^2)$ density, lim inf energy in $[0, \infty]$). It says nothing about other window shapes, lim sup energies or physical boundary conditions.

Jellium ([Tier X]). Status: the main jellium relaxation study (prediction J1) was still running when this book was assembled.

Erratum carried from the published paper (found while writing Chapter 3). The numerical study uses a shifted-force truncation of the Yukawa potential at $r_c = 8$. The quantity reported in the paper as the truncation size is $V(r_c)$, which is *not* the change in the energy per particle. Summing the unshifted potential out to $r = 60$ changes the absolute energies by about 9.5×10^{-6} at $\kappa = 2$ and 1.8×10^{-2} at $\kappa = 1$; at $\kappa = 1$ that exceeds $e_{\mathrm{sq}} - e_{\mathrm{tri}}$. The square–triangular gap itself is almost unchanged, and every comparison in the study uses the same truncated potential, so the P1 observation stands for that potential. But the truncated potential is not completely monotone, so P1 concerns an approximation of the Yukawa interaction, not the interaction the theorem covers. The next version of the paper will state this.

Scripts. The Python scripts quoted in Chapters 3–5 are in `code/`. Chapter 1 prints its short checks inline. The scratch checks quoted in Chapter 3's footnotes are not shipped.

Bibliography

[1] OpenAI, Universal optimality of the triangular lattice, OpenAI Math Release preprint, 2026-09-23, `https://github.com/openai/math`. AI-generated; not peer reviewed.

[2] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, Universal optimality of the E_8 and Leech lattices and interpolation formulas, Ann. of Math. 196 (2022) 983–1082, doi:10.4007/annals.2022.196.3.3.

[3] H. Cohn, A. Kumar, Universally optimal distribution of points on spheres, J. Amer. Math. Soc. 20 (2007) 99–148, doi:10.1090/S0894-0347-06-00546-7.

[4] M. Petrache, S. Serfaty, Crystallization for Coulomb and Riesz interactions as a consequence of the Cohn–Kumar conjecture, Proc. AMS 148 (2020) 3047–3057, doi:10.1090/proc/15003.

[5] R. A. Rankin, A minimum problem for the Epstein zeta-function, Proc. Glasgow Math. Assoc. 1 (1953) 149–158, doi:10.1017/S2040618500035668.

[6] J. W. S. Cassels, On a problem of Rankin about the Epstein zeta-function, Proc. Glasgow Math. Assoc. 4 (1959) 73–80, doi:10.1017/S2040618500033906.

[7] H. L. Montgomery, Minimal theta functions, Glasgow Math. J. 30 (1988) 75–85, doi:10.1017/S0017089500007047.

[8] L. Bonsall, A. A. Maradudin, Some static and dynamical properties of a two-dimensional Wigner crystal, Phys. Rev. B 15 (1977) 1959, doi:10.1103/PhysRevB.15.1959.

[9] E. Alves, G. L. Bendazzoli, S. Evangelisti, J. A. Berger, Accurate ground-state energies of Wigner crystals from a simple real-space approach, Phys. Rev. B 103 (2021) 245125, doi:10.1103/PhysRevB.103.245125.

[10] K. Zahn, A. Wille, G. Maret, S. Sengupta, P. Nielaba, Elastic properties of 2D colloidal crystals from video microscopy, Phys. Rev. Lett. 90 (2003) 155506, doi:10.1103/PhysRevLett.90.155506.

[11] H. Thomas, G. E. Morfill, V. Demmel, J. Goree, B. Feuerbacher, D. Möhlmann, Plasma crystal: Coulomb crystallization in a dusty plasma, Phys. Rev. Lett. 73 (1994) 652, doi:10.1103/PhysRevLett.73.652.

[12] R. Bubeck, S. Neser, C. Bechinger, P. Leiderer, Structure and dynamics of two-dimensional colloidal crystals in confined geometry, Prog. Colloid Polym. Sci. (1998) 41–45, doi:10.1007/BFb0118046.

[13] C. C. Grimes, G. Adams, Evidence for a liquid-to-crystal phase transition in a classical, two-dimensional sheet of electrons, Phys. Rev. Lett. 42 (1979) 795, doi:10.1103/PhysRevLett.42.795.

[14] K. S. Fine, A. C. Cass, W. G. Flynn, C. F. Driscoll, Relaxation of 2D turbulence to vortex crystals, Phys. Rev. Lett. 75 (1995) 3277–3280, doi:10.1103/PhysRevLett.75.3277.

[15] J. R. Abo-Shaeer, C. Raman, J. M. Vogels, W. Ketterle, Observation of vortex lattices in Bose–Einstein condensates, Science 292 (2001) 476–479, doi:10.1126/science.1060182.

[16] L. Chanussot et al., OC20 IS2RES val id, ColabFit (2024), doi:10.60732/b4005525.

[17] X. Callens, SocrateAI-Scientific-Elenchus, `https://github.com/xaviercallens/SocrateAI-Scientific-Elenchus`.

[18] X. Callens, rusty-SUNDIALS, `https://github.com/xaviercallens/rusty-SUNDIALS`.

[19] X. Callens, Riesz-energy, finiteness and every-density corollaries of the openai/-math planar universal-optimality theorem, kernel-checked in Lean 4, Zenodo (2026), doi:10.5281/zenodo.23285631.

[20] X. Callens, Planar universal optimality, tiered (v2), Zenodo (2026), doi:10.5281/zenodo.23293429.